

with a few thousandths of an inch radial clearance, is frequently used. Grooves, sometimes in the form of a thread, are machined in the ring; the relative motion of the shaft with respect to the ring creates the sealing action.

Limiting Speeds

Most catalogs list limiting speeds for each bearing. Exceeding these limits may result in excessively high operating temperatures due to friction between the cages supporting the rolling elements. Generally, the limiting speed is lower for larger bearings than for smaller bearings. Also, a given bearing will have a lower limiting speed as loads increase. With special care, either in the fabrication of the bearing cage or in the lubrication of the bearing, bearings can be operated at higher speeds than those listed in the catalog. The manufacturer should be consulted in such applications. The use of ceramic rolling elements with their lower mass can result in higher limiting speeds.

Standards

Several groups are involved in standard setting for the bearing industry. Some of them are as follows:

American Bearing Manufacturers Association	(ABMA) (Internet site 11)
Annular Bearing Engineers Committee	(ABEC)
American National Standards Institute	(ANSI)
International Standards Organization	(ISO)

Many standards are listed by the ANSI and ABMA organizations including:

Load Ratings and Fatigue Life for Ball Bearings,
ABMA 9

Load Ratings and Fatigue Life for Roller Bearings,
ABMA 11

Tolerances

Several different tolerance classes are recognized in the bearing industry to accommodate the needs of the wide variety of equipment using rolling contact bearings. In general, of course, all bearings are precision machine elements and should be treated as such. As noted before, the general range of tolerances is of the order of a few thousandths of an inch. The standard tolerance classes are defined by ABEC, as identified below.

- ABEC 1: Standard radial ball and roller bearings
- ABEC 3: Semiprecision instrument ball bearings
- ABEC 5: Precision radial ball and roller bearings
- ABEC 5P: Precision instrument ball bearings
- ABEC 7: High-precision radial ball bearings
- ABEC 7P: High-precision instrument ball bearings

Most machine applications would use ABEC 1 tolerances, the data for which are usually provided in the catalogs. Machine tool spindles requiring extra smooth and accurate running would use ABEC 5 or ABEC 7 classes.

14-15 IMPORTANCE OF OIL FILM THICKNESS IN BEARINGS

In bearings operated at heavy loads or high speeds, it is critical to maintain a film of lubricating oil at the surface of the rolling elements. A steady supply of clean lubricant with an adequate viscosity is required. With careful analysis of the geometry of the bearing, the speed of rotation, and the properties of the lubricant, it is possible to estimate the thickness of the oil film between the rolling elements and the races. Although the film thickness may be only a few microinches, it has been shown that oil starvation of the contact area is a chief cause of premature failure of rolling contact bearings. Conversely, if a film thickness significantly greater than the height of surface roughness features can be maintained, the expected life will be several times greater than published data from catalogs. (See References 4, 6, and 12.)

The nature of the lubrication at the interface between rolling elements and the races is called *elastohydrodynamic lubrication* because it depends on the specific elastic deformation of the mating surfaces under the influence of high contact stresses and the creation of a pressurized film of lubricant by the dynamic action of the rolling elements.

The data needed to evaluate film thickness for ball bearings are given in the following lists:

Bearing Geometry Factors

- Ball diameter
- Number of balls
- Radius of curvature of the inner race groove in both the circumferential and the axial directions
- Pitch diameter of the bearing; the average of the bore diameter and the outside diameter
- Surface roughness of the balls and the races
- Contact angle for angular contact bearings

Bearing Material Factors

- Modulus of elasticity of balls and races
- Poisson's ratio

Lubricant Factors

- Dynamic viscosity at the operating temperature *within* the bearing
- Pressure coefficient of viscosity; the change of viscosity with pressure

Operational Factors

- Rotational speed of both the inner and the outer races
- Radial load
- Thrust load

The details of the analysis are given in Reference 6.

The results of the analysis include the following:

The minimum film thickness of the lubricant, h_o

The composite roughness of the balls and the races, S

The ratio $\Lambda = h_o/S$

The service life of the bearing is dependent on the value of Λ :

- If $\Lambda < 0.90$, a life less than the manufacturer's rated life is to be expected because of surface damage due to inadequate lubricant film.
- If Λ is in the range of 0.90 to 1.50, rated service life can be expected.
- If Λ is in the range of 1.50 to 3.0, an increase of life of up to three times the rated life is possible.
- If Λ is greater than 3.0, a life of up to six times the rated life is possible.

General Recommendations for Achieving Long Life for Bearings

1. Choose a bearing with an adequate rated life using the procedures outlined in this chapter.
2. Ensure that the bearing has a fine surface finish and is not damaged by rough handling, poor installation practices, corrosion, vibration, or exposure to electrical current flow.
3. Ensure that operating loads are within the design values.
4. Supply the bearing with a copious flow of clean lubricant at an adequate viscosity at the operating temperature within the bearing according to the bearing manufacturer's recommendations. Provide external cooling for the lubricant if necessary. For an existing system, this is the factor over which you have the most control without a significant redesign of the system itself.
5. If redesign is possible, design the system to operate at as low a speed as possible.

14-16 LIFE PREDICTION UNDER VARYING LOADS

The design and analysis procedures used so far in this book assumed that the bearing would operate at a single design load throughout its life. It is possible to

predict the life of the bearing under such conditions fairly accurately using manufacturers' data published in catalogs. If loads vary with time, a modified procedure is required.

One procedure that is recommended by bearing manufacturers is called the *Palmgren-Miner rule*, or sometimes simply *Miner's rule*. References 9 and 10 describe their work, and Reference 8 discusses a modified approach more tailored to bearings.

The basis of Miner's rule is that if a given bearing is subjected to a series of loads of different magnitudes for known lengths of time, each load contributes to the eventual failure of the bearing in proportion to the ratio of the load to the expected life that the bearing would have under that load. Then the cumulative effect of the series of loads must account for all such contributions to failure.

A similar approach, described in Reference 6, introduces the concept of *mean effective load*, F_m :

Mean Effective Load under Varying Loads

$$F_m = \left(\frac{\sum_i (F_i)^p N_i}{N} \right)^{1/p} \quad (14-10)$$

where

F_i = individual load among a series of i loads

N_i = number of revolutions at which F_i operates

N = total number of revolutions in a complete cycle

p = exponent on the load/life relationship;
 $p = 3$ for ball bearings, and $p = 10/3$ for rollers

Alternatively, if the bearing is rotating at a constant speed, and because the number of revolutions is proportional to the time of operation, N_i can be the number of minutes of operation at F_i , and N is the sum of the number of minutes in the total cycle. That is,

$$N = N_1 + N_2 + \cdots + N_i$$

Then the total expected life, in millions of revolutions of the bearing, would be

$$L = \left(\frac{C}{F_m} \right)^p \quad (14-11)$$

Example Problem 14-6

A single-row, deep-groove ball bearing number 6208 is subjected to the following set of loads for the given times:

Condition	F_i	Time
1	650 lb	30 min
2	750 lb	10 min
3	250 lb	20 min

This cycle of 60 min is repeated continuously throughout the life of the bearing. The shaft carried by the bearing rotates at 600 rpm. Estimate the total life of the bearing.

Solution Using Equation (14–10), we have

$$F_m = \left(\frac{\sum_i (F_i)^p N_i}{N} \right)^{1/p} \quad (14-10a)$$

$$F_m = \left(\frac{30(650)^3 + 10(750)^3 + 20(250)^3}{30 + 10 + 20} \right)^{1/3} = 597 \text{ lb}$$

Now use Equation (14–11):

$$L = \left(\frac{C}{F_m} \right)^p \quad (14-11a)$$

From Table 14–3, for the 6208 bearing, we find that $C = 6902 \text{ lb}$. Then

$$L = \left(\frac{6902}{597} \right)^3 = 1545 \text{ million rev}$$

At a rotational speed of 600 rpm, the number of hours of life would be

$$L = \frac{1545 \times 10^6 \text{ rev}}{1} \cdot \frac{\text{min}}{600 \text{ rev}} \cdot \frac{\text{h}}{60 \text{ min}} = 42\,917 \text{ h}$$

Note that this is the same bearing used in Example Problem 14–3 which was selected to operate at least 30 000 hr when carrying a steady load of 650 lb.

14-17 BEARING DESIGNATION SERIES

An example of a bearing designation system has been shown in Table 14–3, using the 6000 series for single-row, deep-groove ball bearings. The following list shows the series designations for other types of bearings. Note that different manufacturers may use modifications of these series designations.

Series	Bearing type	Example for 30 mm bore, light (206)
1000	Double row self-aligning ball bearing	1206
2000	Tapered roller bearing	2206
6000	Single row, deep groove ball bearing	6206
7000	Single row angular contact ball bearing	7206
21 000	Spherical roller bearing	21 206
51 000	Ball thrust bearing	51 206
81 000	Cylindrical roller thrust bearing	81 206
NU	Single row cylindrical roller	NU206
Needle bearings use a modified system in which the last two digits indicate the bore size in mm.		
NK	Cylindrical needle roller bearing	NK30

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INTERNET SITES RELATED TO ROLLING CONTACT BEARINGS

1. **Boca Bearings Company.** Manufacturer of many styles of rolling contact bearings made from steel, ceramic, and ceramic hybrids in both inch and metric sizes. Special capabilities in miniature bearings as small as 0.040 in and 0.60 mm bore. Applications to precision industrial machinery, semiconductor industry, flowmeters, medical devices, robotics, optics, cryogenic pumps, food processing, bicycles, RC cars, and planes. Also markets specialized lubricants for bearings.
2. **SKFUSA, Inc.** Manufacturer of rolling contact bearings under the SKF brand. Online catalog.
3. **FAG Bearings.** Manufacturer of rolling contact bearings under the FAG and INA brands. Online catalog. A subsidiary of the Schaeffler Group.
4. **RBC Bearings.** Manufacturer of many types of rolling contact bearings, cam followers, and specialized integrated assemblies containing bearings for the automotive, agricultural, lawn equipment, semiconductor, material handling, and other similar markets.
5. **NSK Corporation.** Manufacturer of rolling contact bearings under the NSK brand. Online catalog.
6. **Timken Company.** Manufacturer of rolling contact bearings under the Timken brand. Online catalog.
7. **Baldor/Dodge.** Manufacturer of numerous power transmission products, including mounted rolling contact bearings along with many other types of bearings. Online catalog.
8. **Regal Beloit.** Manufacturer of numerous power transmission products, including bearings from their Browning, McGill, Rollway, and Sealmaster units. Online catalog.
9. **Power Transmission Engineering.** A site that lists numerous manufacturers of power transmission components including rolling contact bearings, gears, gear drives, clutches, couplings, motors, and more. Descriptive information about the listed companies is provided along with links to their websites.
10. **Machinery Lubrication.** On online service of *Machinery Lubrication Magazine* providing technical articles and information on lubrication of industrial machinery, including bearings.
11. **American Bearing Manufacturers Association.** A non-profit association of manufacturers of anti-friction bearings. ABMA defines U. S. national standards for bearings.
12. **High Temp Metals, Inc.** Distributor of high temperature, corrosion resistant grades of nickel and cobalt alloys.
13. **Igus, Inc.** Producer of plastic ball bearings using their proprietary polymers xiros® and iglide®.
14. **KMS Bearings, Inc.** Producer of mounted bearings made from combinations of plastic housings and stainless steel balls.

PROBLEMS

1. A radial ball bearing has a basic dynamic load rating of 2350 lb for a rated (L_{10}) life of 1 million rev. What would its L_{10} life be when operating at a load of 1675 lb?
2. Determine the required basic dynamic load rating for a bearing to carry 1250 lb from a shaft rotating at 880 rpm if the design life is to be 20 000 h.
3. A catalog lists the basic dynamic load rating for a ball bearing to be 3150 lb for a rated life of 1 million rev. What would be the L_{10} life of the bearing if it were subjected to a load of (a) 2200 lb and (b) 4500 lb?
4. Compute the required basic dynamic load rating, C , for a ball bearing to carry a radial load of 1450 lb at a shaft speed of 1150 rpm for an industrial fan.
5. Specify suitable bearings for the shaft of Example Problem 12–1. Note the data contained in Figures 12–1, 12–2, 12–11, and 12–12.
6. Specify suitable bearings for the shaft of Example Problem 12–2. Note the data contained in Figures 12–13, 12–14, 12–15, and 12–16.
7. Specify suitable bearings for the shaft of Example Problem 12–3. Note the data contained in Figures 12–17, 12–18.
8. For any of the bearings specified in Problem 2–7, make a scale drawing of the shaft, the bearings, and the part of the housing to support the outer races of the bearings. Be sure to consider fillet radii and axial location of the bearings.
9. A bearing is to carry a radial load of 455 lb and no thrust load. Specify a suitable bearing from Table 14–3 if the shaft rotates at 1150 rpm and the design life is 20 000 hr.

For each of the problems in Table 14–9, repeat Problem 9 with the new data.

TABLE 14–9

Problem number	Radial load	Thrust load	rpm	Design life, h
10.	857 lb	0	450	30 000
11.	1265 lb	645 lb	210	5000
12.	235 lb	88 lb	1750	20 000
13.	2875 lb	1350 lb	600	15 000
14.	3.8 kN	0	3450	15 000
15.	5.6 kN	2.8 kN	450	2000
16.	10.5 kN	0	1150	20 000
17.	1.2 kN	0.85 kN	860	20 000

18. In Chapter 12, Figures P12–1 through P12–40 showed shaft design exercises related to the problems at the end of the chapter. For each bearing of each shaft, specify a suitable bearing from Table 14–3. If the shaft design has already been completed to the point that the minimum acceptable diameter of the shaft at the bearing seat is known, consider that diameter when specifying the bearing. Note the Chapter 12 problem statements for the speed of the shaft and the loading data.

19. Bearing number 6324 from Table 14–3 is carrying the set of loads in Table P14–19 while rotating at 600 rpm. Compute the expected L_{10} life for the bearing under these conditions if the cycle repeats continuously.

TABLE P14–19

Condition	Load, F_i	Time, N_i
1	4500 lb	25 min
2	2500 lb	15 min

20. Bearing number 6314 from Table 14–3 is carrying the set of loads in Table P14–20 while rotating at 600 rpm. Compute the expected L_{10} life for the bearing under these conditions if the cycle repeats continuously.

TABLE P14–20

Condition	Load, F_i	Time, N_i
1	2500 lb	25 min
2	1500 lb	15 min

21. Bearing number 6209 from Table 14–3 is carrying the set of loads in Table P14–21 while rotating at 1700 rpm. Compute the expected L_{10} life for the bearing under these conditions if the cycle repeats continuously.

TABLE P14–21

Condition	Load, F_i	Time, N_i
1	600 lb	480 min
2	200 lb	115 min
3	100 lb	45 min

22. Bearing number 6209 from Table 14–3 is carrying the set of loads in Table P14–22 while rotating at 1700 rpm. Compute the expected L_{10} life for the bearing under these conditions if the cycle repeats continuously.

TABLE P14–22

Condition	Load, F_i	Time, N_i
1	450 lb	480 min
2	180 lb	115 min
3	50 lb	45 min

23. Bearing number 6205 from Table 14–3 is carrying the set of loads in Table P14–23 while rotating at 101 rpm for one 8-h shift. Compute the expected L_{10} life for the bearing under these conditions if the cycle repeats continuously. If the machine operates two shifts per day, six days per week, in how many weeks would you expect to have to replace the bearing?

TABLE P14–23

Condition	Load, F_i	Time, N_i
1	500 lb	6.75 h
2	800 lb	0.40 h
3	100 lb	0.85 h

24. Bearing number 6211 from Table 14–3 is carrying the set of loads in Table P14–24 while rotating at 101 rpm for one 8-h shift. Compute the expected L_{10} life for the bearing under these conditions if the cycle repeats continuously. If the machine operates two shifts per day, six days per week, in how many weeks would you expect to have to replace the bearing?

TABLE P14–24

Condition	Radial load	Thrust load	Time, N_i
1	1750 lb	350 lb	6.75 h
2	600 lb	250 lb	0.40 h
3	280 lb	110 lb	0.85 h

25. Compute the basic dynamic load rating, C , for a ball bearing to carry a radial load of 1450 lb at a shaft speed of 1150 rpm for 15 000 hours. Use a reliability of 95%.
26. Compute the basic dynamic load rating, C , for a ball bearing to carry a radial load of 509 lb at a shaft speed of 101 rpm for 20 000 hours. Use a reliability of 99%.
27. Compute the basic dynamic load rating, C , for a ball bearing to carry a radial load of 436 lb at a shaft speed of 1700 rpm for 5000 hours. Use a reliability of 97%.
28. Compute the basic dynamic load rating, C , for a ball bearing to carry a radial load of 1250 lb at a shaft speed of 880 rpm for a design life of 20 000 hours. Use a reliability of 95%.

COMPLETION OF THE DESIGN OF A POWER TRANSMISSION

The Big Picture

- 15-1 Objectives of This Chapter
- 15-2 Description of the Power Transmission to Be Designed
- 15-3 Design Alternatives and Selection of the Design Approach
- 15-4 Design Alternatives for the Gear-Type Reducer
- 15-5 General Layout and Design Details of the Reducer
- 15-6 Final Design Details for the Shafts
- 15-7 Assembly Drawing

THE BIG PICTURE

Completion of the Design of a Power Transmission

Discussion Map

- Now we bring together the concepts and design procedures from chapters 7 to 14 to complete the design of the power transmission.

This chapter presents a summary of the steps that you should take to complete the design of the power transmission. Some procedures are quite detailed. You should be able to apply this experience to any design you are responsible for in the future.

This is where all of the work of Part II of this book comes together. In Chapters 7–14, you learned important concepts and design procedures for many kinds of machine elements that could all be part of a given power transmission. In each case, we emphasized the importance of how the elements need to work together. Now we show the approach to completing the design of a power transmission, which demonstrates an integrated approach and illustrates the title of this book, *Machine Elements in Mechanical Design*. The emphasis is on the whole design.

Discover

Consider how all of the machine elements that you have studied in Chapters 7–14 fit together.

Think also about the entire life cycle of the transmission, from its design to its disposal.

The lesson of this chapter is that you as a designer must constantly keep in mind how the part you are currently working on fits with other parts and how its design can affect the design of other parts. You must also consider how the part is to be manufactured, how it may be serviced and repaired as necessary, and how it will eventually be taken out of service. What will happen to the materials in the product when they have served their useful life as part of your current project?

Although we are using a power transmission in this example, the skills and insights that you will acquire should be transferable to the design of almost any other mechanical device or system.

15-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Bring together the individual components of a mechanical, gear-type power transmission into a unified, complete system.
2. Resolve the interface questions where two components fit together.
3. Establish reasonable tolerances and limit dimensions on key dimensions of components, especially where assembly and operation of the components are critical.
4. Verify that the final design is safe and suitable for its intended purpose.
5. Add details to some of the components that were not considered in earlier analyses.

15-2 DESCRIPTION OF THE POWER TRANSMISSION TO BE DESIGNED

The project to be completed in this chapter is the design of a single-reduction speed reducer that uses spur gears. We will use the data from Example Problems 9–3 through 9–6 in which the gears for the drive for an industrial saw were designed. You should review those problems now.

We will also use the elements of the mechanical design process that were first outlined in Section 1–4 and, state the functions and design requirements for the power transmission, establish a set of criteria for evaluating design decisions, and implement the design tasks that were outlined in Section 1–5. References 3, 6, 8, and 10–13 in this chapter provide additional approaches that you may find useful for other design projects.

Basic Statement of the Problem

We will design a power transmission for an industrial saw that will be used to cut tubing for vehicle exhaust pipes to length prior to the forming processes. The saw will receive 25 hp from the shaft of an electric motor rotating at 1750 rpm. The drive shaft for the saw should rotate at approximately 500 rpm.

Functions, Design Requirements, and Selection Criteria for the Power Transmission

Functions. The functions of the power transmission are as follows:

1. To receive power from an electric motor through a rotating shaft
2. To transmit the power through machine elements that reduce the rotational speed to a desired value
3. To deliver the power at the lower speed to an output shaft, which ultimately drives the saw

Design Requirements. Additional information is presented here for the specific case of the industrial saw. You would normally be responsible for acquiring the necessary information and for making design decisions at this point in the design process. You would be involved in the design of the saw and would be able to discuss its desirable features with colleagues in marketing, sales, manufacturing planning, and production management and field service, and, perhaps, with customers. The kinds of information that you should seek are provided in the following list:

1. The reducer must transmit 25 hp.
2. The input is from an electric motor whose shaft rotates at a full-load speed of 1750 rpm. It has been proposed to use a NEMA frame 284T motor having a shaft diameter of 1.875 in and a keyway to accommodate a $1/2 \times 1/2$ in key. See Figure 21–18 and Table 21–3 for more data on the dimensions of the motor. Sections 21–3 to 21–8 describe industrial AC motors.
3. The output of the reducer delivers power to the saw through a shaft that rotates at 500 rpm. The speed reduction ratio will then be 3.50.
4. A mechanical efficiency of greater than 95% is desirable.
5. The minimum torque delivered to the saw should be 2950 lb·in.
6. The saw is a band saw. The cutting operation is generally smooth, but moderate shock may be encountered as the saw blade engages the tubes and if there is any binding of the blade in the cut.
7. The speed reducer will be mounted on a rigid plate that is part of the base of the saw. The means of mounting the reducer should be specified.
8. The design for the shaft for the band-saw drive has not yet been completed. It is likely that its diameter will be the same as that of the output shaft of the reducer.
9. Whereas a small, compact size for the reducer is desirable, space in the machine base should be able to accommodate most reasonable designs.
10. The saw is expected to operate 16 h per day, 5 days per week, with a design life of 5 years. This is approximately 20 000 h of operation.
11. The machine base will be enclosed and will prevent any casual contact with the reducer. However, the functional components of the reducer should be enclosed in their own rigid housing to protect them from contaminants and to provide for the safety of those who work with the equipment.
12. The saw will operate in a factory environment and should be capable of operation in the temperature range of 50°F to 100°F.

13. The saw is expected to be produced in quantities of 5000 units per year.
14. A moderate cost is critical to the marketing success of the saw.

Selection Criteria. The list of criteria should be produced by an interdisciplinary team composed of people having broad experience with the market for and use of such equipment. The details will vary according to the specific design. As an illustration of the process, the following criteria are suggested for the present design:

1. **Safety:** The speed reducer should operate safely and provide a safe environment for people near the machine.
2. **Cost:** Low cost is desirable so that the saw appeals to a large set of customers.
3. **Small size.**
4. **High reliability.**
5. **Low maintenance.**
6. **Smooth operation, low noise, and low vibration.**

15-3 DESIGN ALTERNATIVES AND SELECTION OF THE DESIGN APPROACH

There are many ways that the speed reduction for the saw can be accomplished. Figure 15-1 shows four possibilities: (a) belt drive, (b) chain drive, (c) gear-type drive connected through flexible couplings, and (d) gear-type drive with a belt drive on the input side and connected to the saw with a flexible coupling.

Reference 9 includes a more extensive set of alternatives and a more detailed analysis. Internet site 2 provides numerous examples of commercially available speed reducers.

Selection of the Basic Design Approach

Table 15-1 shows an example of the rating that could be done to select the type of design to be produced for the speed reducer for the saw. A 10-point scale is used, with 10 being the highest rating. Of course, with more information about the actual application, a different design approach could be selected. Also, it may be desirable to proceed with more than one design to determine

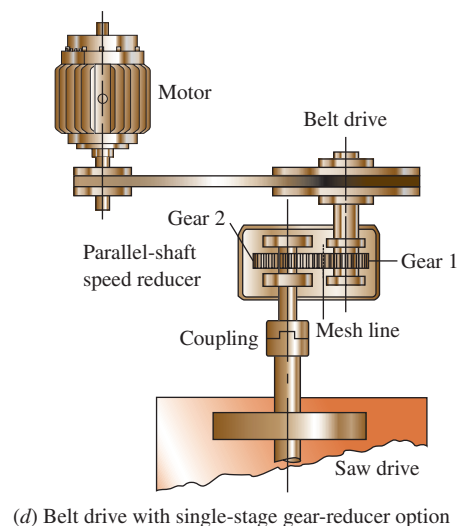
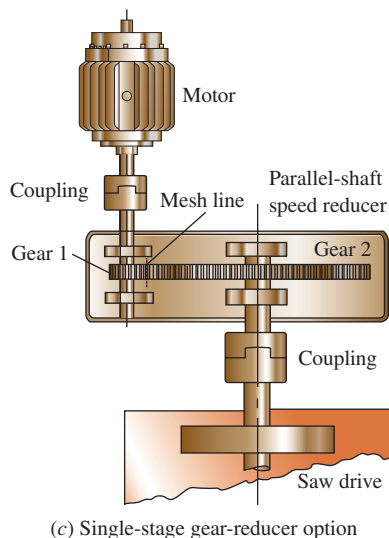
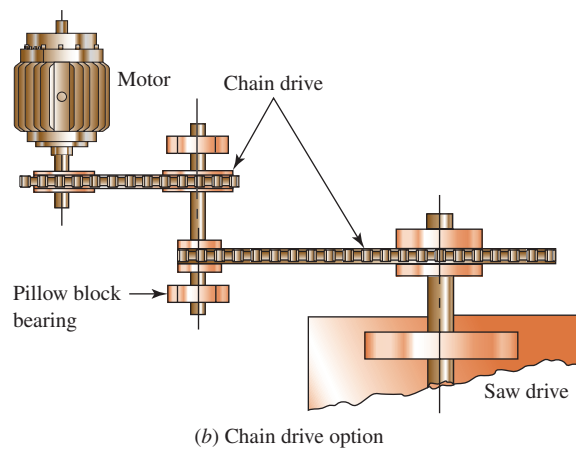
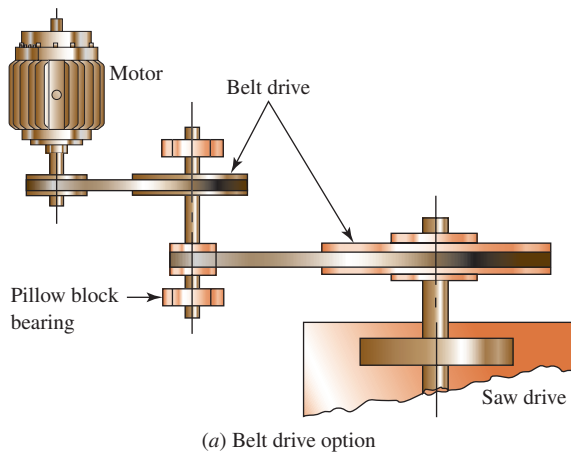


FIGURE 15-1 Options for speed reduction for saw drive design project

TABLE 15-1 Decision Analysis Chart

Criteria	Alternatives			
	(a) Belt	(b) Chain	(c) Gear with flexible couplings	(d) Gear with input belt drive
1. Safety	6	6	9	7
2. Cost	9	8	7	6
3. Size	5	6	9	6
4. Reliability	7	6	10	7
5. Maintenance	6	5	9	6
6. Smoothness	8	6	9	8
Total:	41	31	53	40

more of the details, thus allowing a more rational decision. A modification of the design decision matrix calls for weighting factors to be assigned to each criterion to reflect its relative importance. See References 5 and 10 and Internet site 1 for extensive discussions of rational decision analysis techniques.

On the basis of this decision analysis, let's proceed with (c), the design of a *gear-type speed reducer using flexible couplings* to connect with the drive motor and the driven shaft of the saw. It is considered to have a higher level of safety for operators and maintenance people because its rotating components are enclosed. The input and output shafts and the couplings can be covered at the time of installation. Reliability is expected to be higher because precision metallic parts are used and the drive is enclosed in a sealed housing. The flexing of belts and the significant number of moving parts in a chain drive are judged to provide lower reliability. Initial cost may be higher than for belt or chain drives. However, it is expected that maintenance will be somewhat less, leading to lower cost overall. The space taken by the design should be small, simplifying the design of other parts of the saw. Design alternative (d) is attractive if there is some interest in providing a variable-speed operation in the future. By using different belt drive ratios, we can achieve different cutting speeds for the saw. A further alternative would be to consider a variable-speed electric drive motor, either to replace the need for a reducer at all or to be used in conjunction with the gear-type reducer.

15-4 DESIGN ALTERNATIVES FOR THE GEAR-TYPE REDUCER

Now that we have selected the gear-type reducer, we need to decide which type to use. Here are some alternatives:

1. **Single-reduction spur gears:** The nominal ratio of 3.50:1 is reasonable for a single pair of gears. Spur gears produce only radial loads which simplify selection of the bearings that support the shafts. Efficiency should be greater than 95% with reasonable precision of the gears, bearings, and seals. Spur gears are relatively inexpensive to produce. Shafts would be parallel and should be fairly easy to align with the motor and the drive shaft for the saw.
2. **Single-reduction helical gears:** These gears are equally practical as spur gears. Shaft alignment is similar. A smaller size should be possible because of the greater capacity of helical gears. However, axial thrust loads would be created which must be accommodated by the bearings and the housing. Cost is likely to be somewhat higher.
3. **Bevel gears:** These gears produce a right-angle drive which may be desirable, but not necessary in the present design. They are also somewhat more difficult to design and assemble to achieve adequate precision.
4. **Worm and wormgear drive:** This drive also produces a right-angle drive. It is typically used to achieve a higher reduction ratio than 3.50:1. Efficiency is usually much lower than the 95% called for in the design requirements. Heat generation could be a problem with 25 hp and the lower efficiency. A larger motor could be required to overcome the loss of power and still provide the required torque at the output shaft.

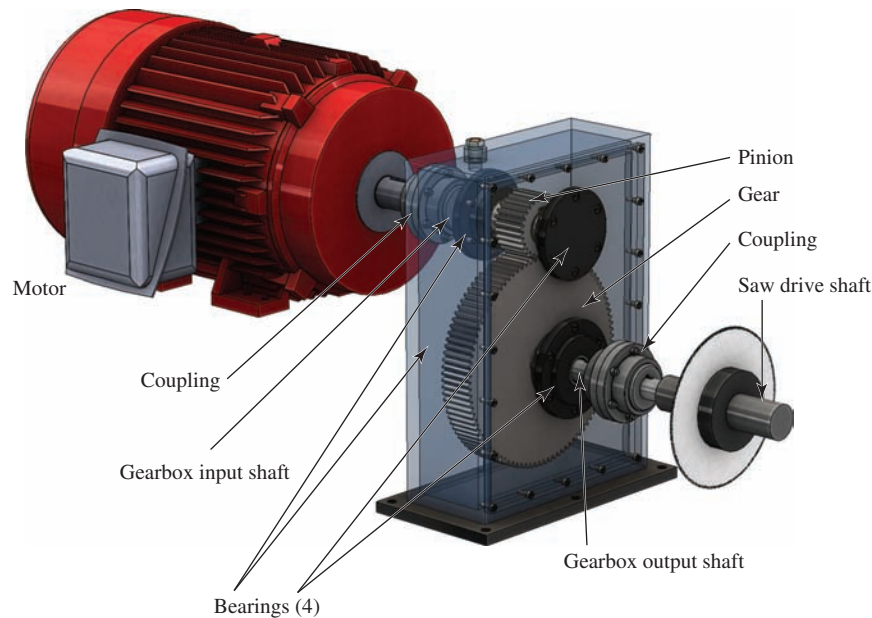
Design Decision for the Gear Type Reducer

For the present design, we choose the *single-reduction spur gear reducer*. Its simplicity is desirable, and the final cost is likely to be lower than that of the other proposed designs. The smaller size of the helical reducer is not considered to be of high priority.

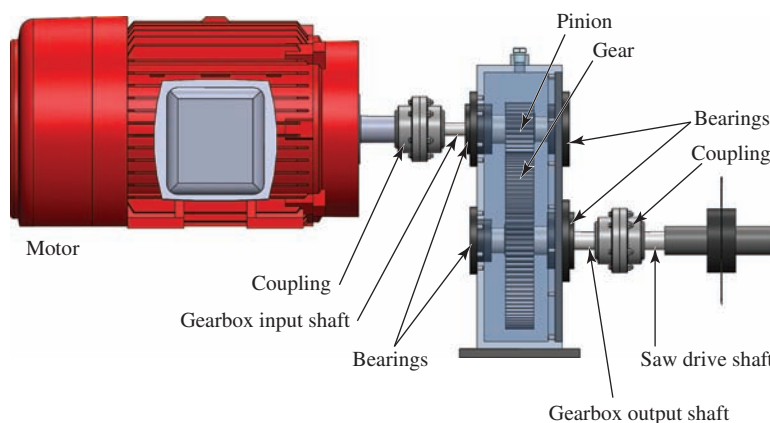
15-5 GENERAL LAYOUT AND DESIGN DETAILS OF THE REDUCER

Figure 15-2 shows the proposed arrangement of the components for the single-reduction spur gear-type speed reducer. Note that the illustration in Figure 15-2(b) is the side view. The design involves the following tasks:

1. Design a pinion and gear to transmit 25 hp with a pinion speed of 1750 rpm and a gear speed of 500 rpm. The ratio is 3.50:1. Design for both strength and pitting resistance to achieve approximately 20 000 h of life and a reliability of at least 0.999.
2. Design two shafts, one for the pinion and one for the gear. Provide positive axial location for the gears on the shaft. The input shaft must be designed to extend beyond the housing to enable the motor shaft to be coupled to it. The output shaft must accommodate a coupling that mates with the drive shaft of the saw. Use a design reliability of 0.999.
3. Design six keys: one for each gear; one for the motor; one for the input shaft at the coupling; one for the output shaft at the coupling; and one for the drive shaft for the saw.
4. Specify two flexible couplings: one for the input shaft and one for the output shaft.
5. Specify four commercially available rolling contact bearings, two for each shaft. The L_{10} design life should be 20 000 h.
6. Design a housing to enclose the gears and the bearings and to support them rigidly.
7. Provide a means of lubricating the gears within the housing.



(a) Pictorial view



(b) Side view

FIGURE 15-2 General layout of saw drive with a single-stage gear reducer

8. Provide seals for the input and output shafts at the place where they pass through the housing. We will not specify the particular seals because of lack of data in this book. However, refer to Chapter 11 for suggestions for the types of seals that may be suitable.

Gear Design

The conditions described above are similar to those in Example Problems 9–3 to 9–6. We propose modest changes to design decisions to achieve a gear pair that would operate more smoothly and quietly with more and smaller teeth. Namely, we select the following:

- Diametral pitch: $P_d = 8$ instead of 6 as was used before
- Number of pinion teeth = 28 and number of gear teeth = 98 instead of 20 and 70 as was used before

The modified design is shown in Figure 15–3 in the form of the output from the gear design spreadsheet developed in Chapter 9. The primary results that affect the completion of the design for the reducer are summarized as follows:

- Diametral pitch: $P_d = 8$; 20° , full-depth, involute teeth
- Number of teeth in the pinion: $N_p = 28$
- Number of teeth in the gear: $N_G = 98$
- Diameter of the pinion: $D_p = 3.500$ in
- Diameter of the gear: $D_G = 12.250$ in
- Center distance: $C = 7.875$ in
- Face width: $F = 2.00$ in
- Quality number: $A_v = 9$
- Tangential force: $W_t = 514$ lb
- Required bending stress number for pinion: $s_{at} = 20\,915$ psi
- Required contact stress number for pinion: $s_{ac} = 153\,363$ psi; requires 386 HB steel
- Material specified: SAE 4140 OQT 800; 429 HB; $s_u = 210$ ksi; 16% elongation

Shaft Design

1. **Forces:** Figure 15–4(a) shows the proposed configuration for the input shaft that carries the pinion and connects to the motor shaft through a flexible coupling. Figure 15–4(b) shows the output shaft, similarly configured. The only active forces on the shafts are the tangential force and the radial force from the gear teeth. The flexible couplings at the ends of the shafts allow torque transmission, but no radial or axial forces are transmitted when the alignment of the shafts is within the recommended limits for the coupling. (See Chapter 11.) Without the flexible couplings, it is very likely that significant radial loads would be produced, requiring somewhat larger diameters for the shaft and larger bearings. (See Chapter 12.)

The spreadsheet analysis for the gears gives the tangential force as $W_t = 514$ lb. It acts downward

in the vertical plane on the pinion and upward on the gear. The radial force is

$$W_r = W_t \tan \phi = (514 \text{ lb}) \tan (20^\circ) = 187 \text{ lb}$$

The radial force acts horizontally toward the left on the pinion, tending to separate the pinion from the gear. It acts toward the right on the gear.

2. **Torque values:** The torque on the input shaft is

$$\begin{aligned} T_1 &= (63\,000)(P)/n_p = (63\,000)(25)/1750 \\ T_1 &= 900 \text{ lb} \cdot \text{in} \end{aligned}$$

This value acts from the coupling at the left end of the shaft to the pinion where the power is delivered through the key to the pinion and thus to the mating gear.

The torque on the output shaft is computed next, assuming that no power is lost. The resulting value of torque is conservative for use in the shaft design:

$$\begin{aligned} T_2 &= (63\,000)(P)/n_G = (63\,000)(25)/500 \\ T_2 &= 3150 \text{ lb} \cdot \text{in} \end{aligned}$$

The torque acts in the output shaft from the gear to the coupling at the right end of the shaft. Assuming that the system is 95% efficient, the actual output torque is approximately

$$T_o = T_2(0.95) = 2992 \text{ lb} \cdot \text{in}$$

This value is within the required range, as indicated in the design requirements, item 5.

3. **Shearing force and bending moment diagrams:** Figure 15–4 shows the shearing force and bending moment diagrams for the two shafts. Because the active loading occurs only at the gears, the form of each diagram is the same in the vertical and horizontal directions. The first number given is the value for load, the shearing force, or the bending moment in the vertical plane. The second number in parentheses is the value in the horizontal plane. The maximum bending moment in each shaft occurs where the gears are mounted. The values are

$$M_y = 643 \text{ lb} \cdot \text{in} \quad M_x = 234 \text{ lb} \cdot \text{in}$$

The resultant moment is $M_{\max} = 684 \text{ lb} \cdot \text{in}$.

The bending moment is zero at the bearings and in the extensions for the input and output shafts.

4. **Support reactions—bearing forces:** The reactions at all bearings are the same for this example because of the simplicity of the loading pattern and the symmetry of the design. The horizontal and vertical components are

$$F_y = 257 \text{ lb} \quad F_x = 93.5 \text{ lb}$$

The resultant force is the radial force that must be carried by the bearings: $F_r = 274$ lb. This value also produces vertical shearing stress in the shaft at the bearings.

5. **Material selection for shafts:** Each shaft will have a series of different diameters, shoulders with fillets, keyseats, and a ring groove, as shown in

APPLICATION:	Industrial saw + electric motor		
Data for design project in Chapter 15			
Initial Input Data:			
Overload Factor:	$K_o =$	1.50 Table 9–1	
Input Power:	$P =$	25 hp	
Design Power:	$P_{des} =$	37.5	
Input Speed:	$n_P =$	1750 rpm	
Diametral Pitch:	$P_d =$	8	
Number of Pinion Teeth:	$N_P =$	28	
Desired Output Speed:	$n_G =$	500 rpm	
Computed Number of Gear Teeth:		98.0	
Enter: Chosen No. of Gear Teeth:	$N_G =$	98	
Computed Data:			
Actual Output Speed:	$n_G =$	500.0 rpm	
Gear Ratio:	$m_G =$	3.50	
Pitch Diameter—Pinion:	$D_P =$	3.500 in	
Pitch Diameter—Gear:	$D_G =$	12.250 in	
Center Distance:	$C =$	7.875 in	
Pitch Line Speed:	$v_t =$	1604 ft/min	
Transmitted Load:	$W_t =$	514 lb	
Secondary Input Data:			
Face Width Guidelines (in):	Min	Nom	Max
	1.000	1.500	2.000
Enter: Face Width:	$F =$	2.000 in	
Ratio: Face Width/Pinion Diameter:	$F/D_P =$	0.57	
Recommended Range of Ratio:	$0.50 < F/D_P$	<2.00	
Enter: Elastic Coefficient:	$C_p = 2300$	Table 9–7	
Enter: Quality Number:	$A_v = 9$	Table 9–5	
Dynamic Factor:	$K_v = 1.33$	Table 9–6	
[Factors for computing K_v :]	$B = 0.630$	$C = 70.71$	
Reference:	$N_P = 28$	$N_G = 98$	
Bending Geometry Factor—Pinion:	$J_P = 0.380$	Figure 9–10	
Bending Geometry Factor—Gear:	$J_G = 0.440$	Figure 9–10	
Reference:	$m_G = 3.50$		
Enter: Pitting Geometry Factor:	$I = 0.115$	Figure 9–17	

Factors in Design Analysis:			
Alignment Factor, $K_m = 1.0 + C_{pf} + C_{ma}$	If $F < 1.0$		If $F > 1.0$
Pinion Proportion Factor, $C_{pf} =$	0.032	0.045	$[0.50 < F/D_p < 2.00]$
Enter: $C_{pf} =$	Figure 9–12		
Type of Gearing:	Open	Commer.	Precision
Mesh Alignment Factor, $C_{ma} =$	0.280	0.158	0.093
Enter: $C_{ma} =$	Figure 9–13		
Alignment Factor: $K_m =$	1.20	[Computed]	
Size Factor: $K_s =$	1.00	Table 9–2: Use 1.00 if $P_d \geq 5$	
Pinion Rim Thickness Factor: $K_{BP} =$	1.00	Figure 9–14: Use 1.00 if solid blank	
Gear Rim Thickness Factor: $K_{BG} =$	1.00	Figure 9–14: Use 1.00 if solid blank	
Service Factor: $SF =$	1.00	Use 1.00 if no unusual conditions	
Reliability Factor: $K_R =$	1.50	Table 9–11: Use 1.00 for $R = 0.99$	
Enter: Design Life:	20 000 hours	See Table 9–12	
Pinion—Number of Load Cycles: $N_P =$	2.1E + 09	Guidelines: Y_N, Z_N	
Gear—Number of Load Cycles: $N_G =$	6.0E + 08	10^7 cycles	$> 10^7$ $< 10^7$
Bending Stress Cycle Factor: $Y_{NP} =$	0.93	1.00	0.93 Figure 9–21
Bending Stress Cycle Factor: $Y_{NG} =$	0.95	1.00	0.95 Figure 9–21
Pitting Stress Cycle Factor: $Z_{NP} =$	0.88	1.00	0.88 Figure 9–22
Pitting Stress Cycle Factor: $Z_{NG} =$	0.91	1.00	0.91 Figure 9–22
Stress Analysis: Bending			
Pinion: Required $s_{at} =$	20 912 psi	See Figure 9–18 or	
Gear: Required $s_{at} =$	17 680 psi	Table 9–9 or 9–10	
Stress Analysis: Pitting			
Pinion: Required $s_{ac} =$	153 353 psi	See Figure 9–19 or	
Gear: Required $s_{ac} =$	148 298 psi	Table 9–9 or 9–10	
Required Hardness of Pinion HB:	386	Equations in Figure 9–19-Grade 1	
Required Hardness of Gear HB:	370	Equations in Figure 9–19-Grade 1	
Specify materials, alloy and heat treatment, for most severe requirement.			
One possible material specification:			
Pinion requires HB 386; SAE 4140 OQT 800; HB 429, 16% Elongation			
Gear requires HB 370; SAE 4140 OQT 800; HB 429, 16% Elongation			
Comments:			
It is reasonable to specify the same material for both the pinion and the gear because the required hardness is similar.			

FIGURE 15-3 Spreadsheet data used for final design project for power transmission in Chapter 15. Input data from Example Problems 9-3 to 9-6

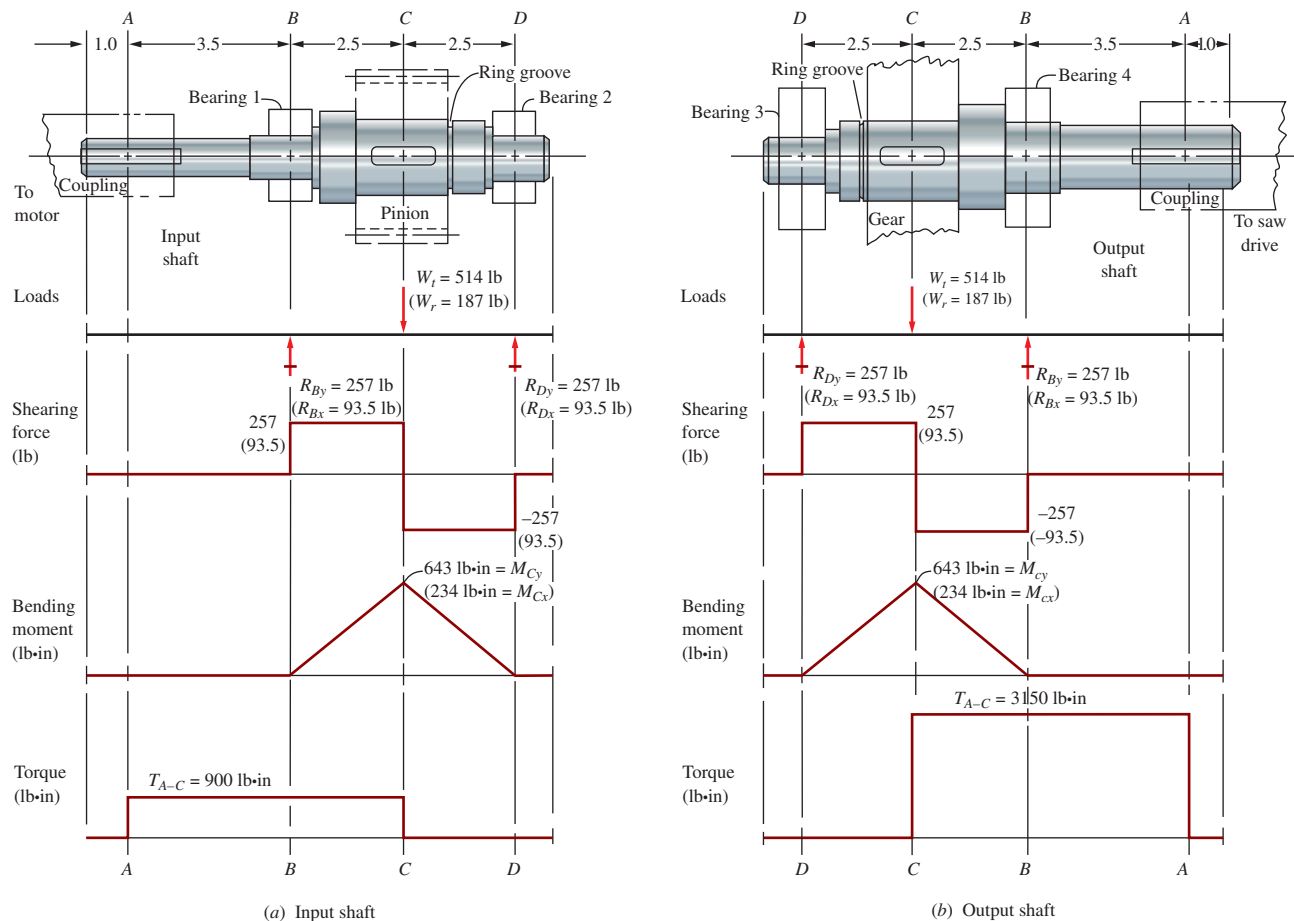


FIGURE 15-4 Diagrams of shaft configurations, shearing force, bending moment, and torque

Figure 15-4. Thus, much machining will be required. The shafts will be subjected to a combination of steady torque and reversed, repeated bending during normal use as the saw cuts steel tubing for vehicular exhaust systems. Occasional moderate shock loading is expected as the saw engages the tubing and when binding occurs in the cut due to dullness of the blade or unusually hard steel in the tubing.

These conditions call for a steel that has moderately high strength, good fatigue resistance, good ductility, and good machinability. Such shafts are typically made from a medium-carbon-alloy steel (0.30% to 0.60% carbon) in either the cold-drawn or the oil-quenched and tempered condition. Good machinability is obtained from a steel having moderately high sulfur content, a characteristic of the 1100 series. Where good hardenability is also desired, higher manganese content is used.

An example of such an alloy is SAE 1144, which has 0.40% to 0.48% carbon, 1.35% to 1.65% manganese, and 0.24% to 0.33% sulfur. It is called a *resulfurized, free-machining* grade of steel. Figure A4-2 shows the range of properties available for this material when it is oil quenched and tempered. We select a tempering temperature of 1000°F, which produces good balance between strength and ductility.

In summary, the material specified is,

SAE 1144 OQT 1000 steel: $s_u = 118\,000 \text{ psi}$
 $s_y = 83\,000 \text{ psi}$
 20% elongation

The endurance strength of the material can be estimated using the method outlined in Chapters 5 and 12: Basic endurance strength: $s_n = 43\,000 \text{ psi}$ (from Figure 5-11 for a machined surface)

Size factor: $C_s = 0.81$ (from Figure 5-12 with an estimate of 2.0 in diameter)

Reliability factor: $C_R = 0.75$ (desired reliability of 0.999)

Modified endurance strength:

$$s'_n = s_n(C_s)(C_R) = (43\,000 \text{ psi})(0.81)(0.75)$$

$$s'_n = 26\,100 \text{ psi}$$

- Design factor, N :** The choice of a design factor, N , should consider the many factors discussed in Chapter 5 where a nominal value of $N = 2$ was suggested for general machine design. With the expectation of moderate shock and impact loading, let's specify $N = 4$ for extra safety.
- Minimum allowable shaft diameters:** The minimum allowable shaft diameters are now computed

at several sections along the shaft using Equation (12–24) if there is any combination of torsion or bending loads at the section of interest. For those sections subjected only to vertical shearing loads, such as at the bearings labeled *D* in Figure 15–4, Equation (12–16) is used. Table 15–2 summarizes the data used in these equations for each section and reports the computed minimum diameter. The spreadsheet from Section 12–9 was used to complete the analysis.

The last column in Table 15–2 also lists some *preliminary* design decisions for convenient diameters at the given locations. These will be reevaluated and refined as the design is completed. The diameters suggested for the shaft extensions at *A* on both the input and the output shafts have been set to standard values available for the bores of flexible couplings. The actual flexible couplings are discussed later.

Note that the diameters for the bearing seats at sections *B* and *D* have not been given. The reason is that the next task in the design project is to specify commercially available rolling contact bearings to carry the

radial loads with suitable life. The diameters of the shafts must be specified according to the limit dimensions recommended by the bearing manufacturer. Therefore, we will leave Table 15–2 as it is for now and revisit it after completing the bearing selection process.

Another check advisable at this stage of the design process is at section *C* where the pinion is mounted on the shaft to ensure that adequate material is above the keyseat in the pinion hub to the root of the pinion teeth. This issue was discussed in Chapter 9 in relation to the *rim thickness factor*, K_B , and shown in Figures 9–14 and 9–15. The dimension, t_R , is defined as the radial distance from the top of the key to the root of the teeth and it is recommended that $t_R/h_t > 1.2$, where h_t is the whole depth of the teeth. For the proposed shaft design with $D_C = 1.75$ in, a 3/8 in square key is used, as shown in Table 11–1. It can be shown that the radial distance to the top of the key is 1.044 in. We need to calculate the pinion root diameter and then the root radius. Recalling that the dedendum, b , is the distance from the pitch line of

TABLE 15–2 Summary of the Results of Shaft Diameter Calculations for the Preliminary Design Sizing for the Shaft

A. Input shaft

Section	Diameter (and related component)	Torque (lb · in)	Bending moments		Shearing forces		K_t	Feature	Diameter (in)	
			M_x (lb · in)	M_y (lb · in)	V_x (lb)	V_y (lb)			Minimum	Design
A	D_1 (coupling)	900	0	0	0	0	1.60	S.R. keyseat	0.73	0.875
B (to right)	D_2 (bearing)	900	0	0	0	0	2.50	Sharp fillet	0.73	*
Note: D_3 must be greater than D_2 or D_4 to provide shoulders for bearing and gear.										2.000
C	D_4 (gear)	900	234	643	94	257	2.00	Profile keyseat	1.29	1.750
C (to right)	D_4 (gear)	0	234	643	94	257	3.00	Ring groove	1.47	1.750
D	D_5 (bearing)	0	0	0	94	257	2.50	Sharp fillet	0.56	*

B. Output shaft

Section	Diameter (and related component)	Torque (lb · in)	Bending moments		Shearing forces		K_t	Feature	Diameter (in)	
			M_x (lb · in)	M_y (lb · in)	V_x (lb)	V_y (lb)			Minimum	Design
A	D_1 (coupling)	3150	0	0	0	0	1.60	S.R. keyseat	1.10	1.250
B (to left)	D_2 (bearing)	3150	0	0	0	0	2.50	Sharp fillet	1.10	*
Note: D_3 must be greater than D_2 or D_4 to provide shoulders for bearing and gear.										2.000
C	D_4 (gear)	3150	234	643	94	257	2.00	Profile keyseat	1.36	1.750
C (to left)	D_4 (gear)	0	234	643	94	257	3.00	Ring groove	1.47	1.750
D	D_5 (bearing)	0	0	0	94	257	2.50	Sharp fillet	0.56	*

Note: Bearing seat diameters denoted by * are to be specified. S.R. = sled runner

the gear to the bottom of the tooth space, we find the radius to the root to be

$$\begin{aligned}\text{Root radius} &= R_R = D/2 - b = D/2 - 1.25/P_d \\ &= 3.500/2 - 1.25/8 = 1.594 \text{ in}\end{aligned}$$

Then,

$$t_R = R_R - 1.044 \text{ in} = 1.594 - 1.044 = 0.550 \text{ in}$$

The whole depth of the gear teeth is as follows:

$$\begin{aligned}h_t &= a + b = 1/P_d + 1.25/P_d \\ &= 2.25/P_d = 2.25/8 = 0.281 \text{ in} \\ t_R/h_t &= 0.550/0.281 = 1.95 > 1.2\end{aligned}$$

Therefore, the space is adequate and it is acceptable to mount the pinion on the shaft. If this check failed, it is recommended that the design be changed—possibly using a smaller shaft diameter or machining the pinion directly into the shaft and eliminating the need for a key.

Bearing Selection

We will use the method outlined in Section 14–9 to select commercially available, single-row, deep-groove ball bearings from the data given in Table 14–3. The design load is equal to the radial load, and the value can be found from the shaft analysis shown in Figure 15–4. The reactions at the supports for each shaft are, in fact, the radial loads to which the bearings are subjected.

Because of the symmetry of the design of this system, and because there are no radial loads on the shaft except those produced by the action of the **gear teeth**, the radial loads on each of the four bearings in this design are the same. Earlier in this chapter, in the shaft design section, we determined that the bearing load is 274 lb.

Recall that design life for the bearings, L_d , is the total number of revolutions expected in service. Therefore, it is dependent on both the rotational speed of the shaft and the design life in hours. We are using a design

life of 20 000 h for all bearings. Shaft 1, the input shaft, rotates at 1750 rpm, resulting in a total number of revolutions of

$$L_d = (20\,000 \text{ h})(1750 \text{ rev/min})(60 \text{ min/h}) = 2.10 \times 10^9 \text{ rev}$$

Shaft 2, the output shaft, rotates at 500 rpm. Then its design life is

$$L_d = (20\,000 \text{ h})(500 \text{ rev/min})(60 \text{ min/h}) = 6.0 \times 10^8 \text{ rev}$$

Data for the bearings in Table 14–3 are for a life of 1.0 million rev (10^6 rev).

Now we will use Equation (14–3) with $k = 3$ to compute the required basic dynamic load rating C for each ball bearing. For the bearings on shaft 1,

$$C = P_d(L_d/10^6)^{1/k} = (274 \text{ lb})(2.10 \times 10^9/10^6)^{1/3} = 3510 \text{ lb}$$

Similarly, for the bearings on shaft 2,

$$C = P_d(L_d/10^6)^{1/k} = (274 \text{ lb})(6.0 \times 10^8/10^6)^{1/3} = 2310 \text{ lb}$$

Listed in Table 15–3 are candidate bearings for each shaft having basic dynamic load ratings at least as high as those just computed. We also need to refer to Table 15–2 to determine the minimum acceptable diameters for the shafts at each bearing seat to ensure that the inner race diameter for the bearing is compatible.

We have selected the smallest bearing for each location on shaft 1 that had a suitable value for the basic dynamic load rating. For shaft 2, we decided that the diameter of the shaft extension should be 1.25 in and that the bearing bore must be larger. Bearing 6207 provides a suitable bore and an additional safety factor on the load rating.

Note that the dimensions for the bearings are actually listed in mm by the manufacturer as indicated in Table 14–3. The decimal-inch equivalents are somewhat

TABLE 15–3 Candidate Bearings for Shafts 1 and 2

A. For shaft 1: Minimum bore = 0.73 in at section B; 0.56 in at section D; $C_{\min} = 3516$ lb

Bearing no.	C (lb)	d (in)	D (in)	B (in)	$r_{\max}$ (in)	Comment
6206	4384	1.1811	2.4409	0.6299	0.039	Large for both B and D
6305	5058	0.9843	2.4409	0.6693	0.039	Specify for both B and D

B. For shaft 2: Minimum bore = 1.10 in at section B; 0.56 in at section D; $C_{\min} = 2310$ lb

Bearing no.	C (lb)	d (in)	D (in)	B (in)	$r_{\max}$ (in)	Comment
6205	3147	0.9843	2.0472	0.5906	0.039	Small for B ; specify for D
6207	5733	1.3780	2.8346	0.6693	0.039	Specify for B

inconvenient, but they must be used. The dimensions in mm are listed as follows:

Inch dimension	mm	Inch dimension	mm
0.5906	15	1.3780	35
0.6299	16	2.0472	52
0.6693	17	2.4409	62
0.9843	25	2.8346	72
1.1811	30	0.039	1.00-mm (fillet radius)

Bearing Mounting on the Shafts and in the Housing

With the specifications for the bearings, we can finalize the basic dimensions for the shaft diameters. Table 15–4 is an update of the data in Table 15–2 with the bearing bore dimensions given. A few other changes are included as well, which is typical of the iterative

nature of design. For example, the diameter of the input shaft at the coupling (section A) was made slightly smaller than the bearing seat diameter. This permits the bearing to be slid onto the shaft easily to the point where it is then pressed into position on its seat at section B and against the shoulder. Another check needs to be made at section D for both shafts where a 1.750-in diameter steps down to the bearing seat diameter of 0.984 in. There is the possibility that the step is too large and that it may interfere with the outer race of the bearing. That will be checked as we complete the details of mounting the bearings. If interference does occur, it should be a simple matter to provide another small step to make the bearing shoulder an acceptable height.

Mounting of ball and roller bearings on shafts and into housings requires very careful consideration of limit dimensions on all mating parts to ensure proper fits as defined by the bearing manufacturer. The total tolerances on shaft diameters are only a few ten-thousandths of an inch in sizes up to about 6.00 in. Total tolerances on housing bore diameters range from about 0.001 to 0.004 in

TABLE 15–4 Summary of the Results of the Shaft Diameter Calculations for the Revised Design Sizing for the Shaft

A. Input shaft

Section	Diameter (and related component)	Torque (lb · in)	Bending moments		Shearing forces			Feature	Diameter (in)	
			M_x (lb · in)	M_y (lb · in)	V_x (lb)	V_y (lb)	K_t		Minimum	Design
A	D_1 (coupling)	900	0	0	0	0	1.60	S.R. keyseat	0.73	0.875
B (to right)	D_2 (bearing)	900	0	0	0	0	2.50	Sharp fillet	0.73	0.984
Note: D_3 must be greater than D_2 or D_4 to provide shoulders for bearing and gear.										2.000
C	D_4 (gear)	900	234	643	94	257	2.00	Profile keyseat	1.29	1.750
C (to right)	D_4 (gear)	0	234	643	94	257	3.00	Ring groove	1.47	1.750
D	D_5 (bearing)	0	0	0	94	257	2.50	Sharp fillet	0.56	0.984

B. Output shaft

Section	Diameter (and related component)	Torque (lb · in)	Bending moments		Shearing forces			Feature	Diameter (in)	
			M_x (lb · in)	M_y (lb · in)	V_x (lb)	V_y (lb)	K_t		Minimum	Design
A	D_1 (coupling)	3150	0	0	0	0	1.60	S.R. keyseat	1.10	1.250
B (to left)	D_2 (bearing)	3150	0	0	0	0	2.50	Sharp fillet	1.10	1.378
Note: D_3 must be greater than D_2 or D_4 to provide shoulders for bearing and gear.										2.000
C	D_4 (gear)	3150	234	643	94	257	2.00	Profile keyseat	1.36	1.750
C (to left)	D_4 (gear)	0	234	643	94	257	3.00	Ring groove	1.47	1.750
D	D_5 (bearing)	0	0	0	94	257	2.50	Sharp fillet	0.56	0.984

Note: S.R. = sled runner

for sizes from about 1.00 in to over 16.0 in. Violation of the recommended fits will likely cause unsatisfactory performance and possibly early failure of the bearing.

The bore of a bearing is typically pressed onto the shaft seat with a light interference fit to ensure that the inner race rotates with the shaft. The OD of the bearing is typically a close sliding fit in the housing, with the minimum clearance being zero. This facilitates installation and allows some slight movement of the bearing as thermal deformation occurs during operation. Tighter fits than those recommended by the manufacturer may cause the rolling elements to bind between the inner and outer races, resulting in higher loads and higher friction. Looser fits may permit the outer race to rotate relative to the housing, a very undesirable situation.

Only one of the two bearings on a shaft should be located and held fixed axially in the housing to provide proper alignment of functional components such as the gears in this design. The second bearing should be installed in a way that allows some small axial movement during operation. If the second bearing is held fixed also, it is likely that extra axial loads will be developed for which the bearing has not been designed.

We first discuss the specification of shaft limit dimensions at the bearing seats.

Bearing Seat Diameters. Because most commercially available bearings are produced to metric dimensions, the fits are specified according to the tolerance system of the International Organization for Standardization (ISO). Only a sampling of the data are listed here to illustrate the process of specifying limit dimensions for shafts and housings to accommodate bearings. Manufacturers' catalogs include much more extensive data.

For bearings carrying moderate to heavy loads such as those in this example design, the following tolerance grades are recommended for the bearing seats on shafts and housing bore fits with the outer race:

Bearing bore diameter range	Tolerance grade
10–18 mm	j5
20–100 mm	k5
105–140 mm	m5
150–200 mm	m6
Housing bore (any)	H8

Table 15–5 shows representative data for the actual limit dimensions for these grades over the size ranges included for the bearings listed in Table 14–3. Note that the bearing bore and the bearing OD dimensions are those expected from the bearing manufacturers. You must control the shaft diameter and the housing bore to the specified minimum and maximum dimensions. The table also lists the minimum and maximum fits that result. The symbol L indicates that there is a net clearance (loose) fit; T indicates an interference (tight) fit. So bearings must

be pressed onto the shaft seat. Sometimes heating of the bearing and cooling of the shaft are used to produce a clearance to facilitate assembly. When the parts return to normal temperatures, the final fit is produced.

We now show the determination of the limit dimensions for the shaft at each bearing seat.

Shaft 1: Input Shaft. Both bearings 1 and 2 are number 6305.

Nominal bore = 25 mm (0.9843 in)

From Table 15–5: k5 ISO tolerance grade on the shaft seat; limits of 0.9847–0.9844 in

Resulting fit between bearing bore and shaft seat: 0.0001 in - tight to 0.0008 in - tight

OD of the outer race = 62 mm (2.4409 in)

From Table 15–5: H8 ISO tolerance grade on the housing bore; limits of 2.4409–2.4427 in

Resulting fit between outer race and housing bore: 0.0 to 0.0023 in - loose

Shaft 2: Output Shaft. Bearing 3 at *D* is number 6205.

Nominal bore = 25 mm (0.9843 in)

From Table 15–5: k5 ISO tolerance grade on the shaft seat; limits of 0.9847–0.9844 in

Resulting fit between bearing bore and shaft seat: 0.0001 in - tight to 0.0008 in - tight

OD of the outer race = 52 mm (2.0472 in)

From Table 15–5: H8 ISO tolerance grade on the housing bore; limits of 2.0472–2.0490 in

Resulting fit between outer race and housing bore: 0.0 to 0.0023 in - loose

Shaft 2: Output Shaft.

Bearing 4 at *B* is number 6207.

Nominal bore = 35 mm (1.3780 in)

From Table 15–5: k5 ISO tolerance grade on the shaft seat; limits of 1.3785–1.3781 in

Resulting fit between bearing bore and shaft seat: 0.0001 in - tight to 0.0010 in - tight

OD of the outer race = 72 mm (2.8346 in)

From Table 15–5: H8 ISO tolerance grade on the housing bore; limits of 2.8346–2.8364 in

Resulting fit between outer race and housing bore: 0.0 to 0.0023 in - loose.

Shaft and Housing Shoulder Diameters. Each of the bearings in this design is to be seated against a shoulder on one side of the bearing. The shaft shoulder must be sufficiently large to provide a solid, flat surface against which to seat the side of the inner race. But the shoulder must not be so high that it contacts the outer race because the inner race rotates at the shaft speed and the outer race is stationary.

TABLE 15-5 Shaft and Housing Fits for Bearings

A. Shaft fits

Bearing bore			ISO tolerance grade	Shaft diameter		Limits of fit	
Nominal (mm)	Maximum (in)	Minimum (in)		Maximum (in)	Minimum (in)	Minimum (in)	Maximum (in)
10	0.3937	0.3934	j5	0.3939	0.3936	0.0001L	0.0005T
12	0.4724	0.4721	j5	0.4726	0.4723	0.0001L	0.0005T
15	0.5906	0.5903	j5	0.5908	0.5905	0.0001L	0.0005T
17	0.6693	0.6690	j5	0.6695	0.6692	0.0001L	0.0005T
20	0.7874	0.7870	k5	0.7878	0.7875	0.0001T	0.0008T
25	0.9843	0.9839	k5	0.9847	0.9844	0.0001T	0.0008T
30	1.1811	1.1807	k5	1.1815	1.1812	0.0001T	0.0008T
35	1.3780	1.3775	k5	1.3785	1.3781	0.0001T	0.0010T
40	1.5748	1.5743	k5	1.5753	1.5749	0.0001T	0.0010T
45	1.7717	1.7712	k5	1.7722	1.7718	0.0001T	0.0010T
50	1.9685	1.9680	k5	1.9690	1.9686	0.0001T	0.0010T
55	2.1654	2.1648	k5	2.1660	2.1655	0.0001T	0.0012T
60	2.3622	2.3616	k5	2.3628	2.3623	0.0001T	0.0012T
65	2.5591	2.5585	k5	2.5597	2.5592	0.0001T	0.0012T
70	2.7559	2.7553	k5	2.7565	2.7560	0.0001T	0.0012T
75	2.9528	2.9522	k5	2.9534	2.9529	0.0001T	0.0012T
80	3.1496	3.1490	k5	3.1502	3.1497	0.0001T	0.0012T
85	3.3465	3.3457	k5	3.3472	3.3466	0.0001T	0.0015T
90	3.5433	3.5425	k5	3.5440	3.5434	0.0001T	0.0015T
95	3.7402	3.7394	k5	3.7409	3.7403	0.0001T	0.0015T
100	3.9370	3.9362	k5	3.9377	3.9371	0.0001T	0.0015T
105	4.1339	4.1331	m5	4.1350	4.1344	0.0005T	0.0019T
110	4.3307	4.3299	m5	4.3318	4.3312	0.0005T	0.0019T
115	4.5276	4.5268	m5	4.5287	4.5281	0.0005T	0.0019T
120	4.7244	4.7236	m5	4.7255	4.7249	0.0005T	0.0019T
125	4.9213	4.9203	m5	4.9226	4.9219	0.0006T	0.0023T
130	5.1181	5.1171	m5	5.1194	5.1187	0.0006T	0.0023T
140	5.5118	5.5108	m5	5.5131	5.5124	0.0006T	0.0023T
150	5.9055	5.9045	m6	5.9071	5.9061	0.0006T	0.0026T
160	6.2992	6.2982	m6	6.3008	6.2998	0.0006T	0.0026T
170	6.6929	6.6919	m6	6.6945	6.6935	0.0006T	0.0026T
180	7.0866	7.0856	m6	7.0882	7.0872	0.0006T	0.0026T
190	7.4803	7.4791	m6	7.4821	7.4810	0.0007T	0.0030T
200	7.8740	7.8728	m6	7.8758	7.8747	0.0007T	0.0030T

B. Housing fits

Bearing OD			ISO tolerance grade	Housing bore		Limits of fit	
Nominal (mm)	Maximum (in)	Minimum (in)		Maximum (in)	Minimum (in)	Minimum (in)	Maximum (in)
30	1.1811	1.1807	H8	1.1811	1.1824	0	0.0017L
32	1.2598	1.2594	H8	1.2598	1.2613	0	0.0019L
35	1.3780	1.3776	H8	1.3780	1.3795	0	0.0019L

(continued)

TABLE 15-5 Shaft and Housing Fits for Bearings (*continued*)

B. Housing fits

Bearing OD			ISO tolerance grade	Housing bore		Limits of fit	
Nominal (mm)	Maximum (in)	Minimum (in)		Maximum (in)	Minimum (in)	Minimum (in)	Maximum (in)
37	1.4567	1.4563	H8	1.4567	1.4582	0	0.0019L
40	1.5748	1.5744	H8	1.5748	1.5763	0	0.0019L
42	1.6535	1.6531	H8	1.6535	1.6550	0	0.0019L
47	1.8504	1.8500	H8	1.8504	1.8519	0	0.0019L
52	2.0472	2.0467	H8	2.0472	2.0490	0	0.0023L
62	2.4409	2.4404	H8	2.4409	2.4427	0	0.0023L
72	2.8346	2.8341	H8	2.8346	2.8364	0	0.0023L
80	3.1496	3.1491	H8	3.1496	3.1514	0	0.0023L
85	3.3465	3.3459	H8	3.3465	3.3486	0	0.0027L
90	3.5433	3.5427	H8	3.5433	3.5454	0	0.0027L
100	3.9370	3.9364	H8	3.9370	3.9391	0	0.0027L
110	4.3307	4.3301	H8	4.3307	4.3328	0	0.0027L
120	4.7244	4.7238	H8	4.7244	4.7265	0	0.0027L
125	4.9213	4.9206	H8	4.9213	4.9238	0	0.0032L
130	5.1181	5.1174	H8	5.1181	5.1206	0	0.0032L
140	5.5118	5.5111	H8	5.5118	5.5143	0	0.0032L
150	5.9055	5.9048	H8	5.9055	5.9080	0	0.0032L
160	6.2992	6.2982	H8	6.2992	6.3017	0	0.0035L
170	6.6929	6.6919	H8	6.6929	6.6954	0	0.0035L
180	7.0866	7.0856	H8	7.0866	7.0891	0	0.0035L
190	7.4803	7.4791	H8	7.4803	7.4831	0	0.0040L
200	7.8740	7.8728	H8	7.8740	7.8768	0	0.0040L
215	8.4646	8.4634	H8	8.4646	8.4674	0	0.0040L
225	8.8583	8.8571	H8	8.8583	8.8611	0	0.0040L
230	9.0551	9.0539	H8	9.0551	9.0579	0	0.0040L
240	9.4488	9.4476	H8	9.4488	9.4516	0	0.0040L
250	9.8425	9.8413	H8	9.8425	9.8453	0	0.0040L
260	10.2362	10.2348	H8	10.2362	10.2394	0	0.0046L
270	10.6299	10.6285	H8	10.6299	10.6331	0	0.0046L
280	11.0236	11.0222	H8	11.0236	11.0268	0	0.0046L
290	11.4173	11.4159	H8	11.4173	11.4205	0	0.0046L
300	11.8110	11.8096	H8	11.8110	11.8142	0	0.0046L
310	12.2047	12.2033	H8	12.2047	12.2079	0	0.0046L
320	12.5984	12.5968	H8	12.5984	12.6019	0	0.0051L
340	13.3858	13.3842	H8	13.3858	13.3893	0	0.0051L
360	14.1732	14.1716	H8	14.1732	14.1767	0	0.0051L
380	14.9606	14.9590	H8	14.9606	14.9641	0	0.0051L
400	15.7480	15.7464	H8	15.7480	15.7515	0	0.0051L
420	16.5354	16.5336	H8	16.5354	16.5392	0	0.0056L

Note: L = loose; and T = tight.

Similarly, a shoulder in the housing must provide for the solid location of the outer race but not be such that it contacts the inner race.

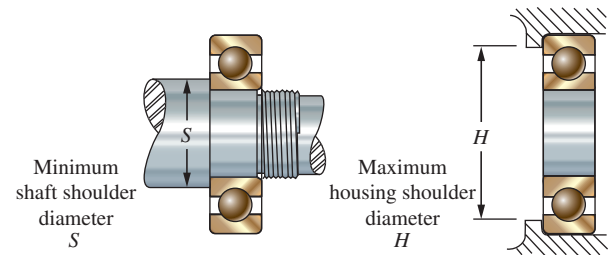
Bearing manufacturers' catalogs provide data such as those shown in Table 14–3 to guide you in specifying suitable shoulder heights. As shown in Figure 15–5, the value of S is the minimum shaft shoulder diameter. The nominal maximum diameter is the mean diameter for the bearing at the middle of the balls. The value of H is the maximum housing shoulder diameter, with the nominal minimum diameter being the mean diameter for the bearing.

For example, in the present design, the minimum shoulder diameter at each bearing on shaft 1 should be 1.260 in as indicated for the bearing number 305 in Table 14–3. (Note that the bearing number 6305 specified for the shaft is of the same *series* as the number 305, indicating that it would have similar dimensions.) The maximum housing shoulder diameter for the number 305 bearing is 2.165 in, where the outer race is to seat against a shoulder.

On shaft 2, the shoulder for bearing 6205 should also be at least 1.181 in, and the shoulder for bearing 6207 should be 1.654 in minimum. The maximum housing shoulder diameter for the 6205 bearing on shaft 2 is 1.850. For the 6207 bearing, the maximum housing shoulder diameter is 2.559 in.

Table 15–6 shows the pertinent data used to decide on the values for the shoulder diameters and, in the last two columns, gives the specified values. Where the specified shoulder diameter is less than the preliminary value shown in Table 15–4, another step in the shaft will be used to provide the proper shoulder for the bearing and for the gear. This can be seen in the drawings of the shafts given at the end of this chapter.

The use of a 1.75-in diameter for the shoulder at D on shaft 1 was specified because that is the diameter of the shaft chosen earlier. It is a bit higher than the mean



Notes:

S = minimum shaft shoulder diameter

Maximum diameter should not exceed the mean diameter of the bearing at the middle of the balls.

H = maximum housing shoulder diameter

Minimum diameter should not be less than the mean diameter of the bearing at the middle of the balls.

Refer to Table 14–3 in Chapter 14 for the values of S and H .

FIGURE 15–5 Shaft and housing shoulder diameters

diameter of the bearing, but it should still be lower than that of the outer race. More complete data in a manufacturer's catalog indicates the diameter of the inner surface of the outer race to be 2.00 in, so the 1.75 in diameter is acceptable.

Fillet Radii. Each of the bearings specified for the reducer calls for the maximum fillet radius at the shoulder that locates the bearing to be 0.039 in. (See Table 14–3.) Let's specify the limits on the radius to be 0.039 to 0.035. Before committing to the design, we will check the stress concentration factor at each shoulder.

Flexible Couplings. The use of flexible couplings on both the input and the output shafts has been taken into account in the shaft design and analysis. They allow the transmission of torque between two shafts but do not exert significant radial or axial forces on the shaft. In the present design, the use of flexible couplings made the

TABLE 15–6 Shaft and Housing Shoulder Diameters for Bearings in the Present Design

A. Shaft 1

Bearing no.	Bore	OD	Mean diameter	S Minimum	H Maximum	Specified shaft shoulder	Specified housing shoulder
6305 at B	0.9843	2.4409	1.713	1.260	2.165	1.50	2.00
6305 at D	0.9843	2.4409	1.713	1.260	2.165	1.75	2.00

B. Shaft 2

Bearing no.	Bore	OD	Mean diameter	S Minimum	H Maximum	Specified shaft shoulder	Specified housing shoulder
6207 at B	1.3780	2.8346	2.106	1.654	2.559	2.00	2.25
6205 at D	0.9843	2.0472	1.516	1.181	1.850	1.50	1.75

shaft design simpler and decreased the loads on bearings compared with having a device such as a belt sheave or a chain sprocket on the shaft.

Now we specify suitable couplings for the input and output shafts. Chapter 11 showed many examples for such couplings, and you should review them now. It is impractical to reproduce data for all couplings in this book. As you read this section, it would be good for you to examine the Internet site of one of the manufacturers of couplings and study their recommended selection procedures. (See the Internet sites for Chapter 11.)

We have selected couplings of the type pictured in Figure 11–22, called the *D-Flex Coupling* from the Baldor-Dodge Company. Elastomeric flex members accommodate parallel misalignment of the mating shafts up to 0.062 in, angular misalignment of $\pm 1^\circ$, and axial end float of the shafts of up to ± 0.125 in. It is important for you to design the drive system for the saw to provide this alignment of the input shaft to the drive motor and of the output shaft to the drive shaft of the saw.

The selection of a suitable coupling relies on the power transmission rating of the various sizes available. But the power rating must be correlated to the speed of rotation because the real variable is the torque to which the coupling is subjected. Both the input and the output couplings transmit nominally 25 hp in this design for the drive for the saw. But the input shaft rotates at 1750 rpm, and the output shaft rotates at 500 rpm. Because torque is inversely proportional to the speed of rotation, the torque experienced by the coupling on the output shaft is approximately 3.5 times higher than that on the input shaft. The coupling catalog data also call for the use of a *service factor* based on the kind of machine being driven, and some suggested values are included in the catalog. We judge that a service factor of 1.5 is suitable for the saw which will see mostly smooth power transmission with occasional moderate shock loading.

The service factor is applied to the nominal power being transmitted to compute a value for the *normal rating* for the couplings. Then,

$$\begin{aligned}\text{Normal rating} &= \text{power input} \times \text{service factor} \\ &= 25 \text{ hp}(1.5) = 37.5 \text{ hp}\end{aligned}$$

The catalog tables list coupling size 8 with a suitable normal rating at 1750 rpm for the input shaft and size 11 for the output shaft at 500 rpm.

We specify hubs for the couplings that have machined bores and keyways with a range of bores allowed. Each coupling half can have a different bore according to the shaft size on which it is to be mounted. For the input shaft, we have specified the diameter to be 0.875 in (7/8 in), and this will be the specification for the bore of that half for the coupling. The keyway is $3/16 \times 3/32$ to accept a 3/16-in square key.

The other half of the input shaft coupling mounts on the motor shaft. Recall that the design requirements listed at the beginning of this design process specified

a 25-hp motor with a NEMA Frame 284T. Table 21–3 gives the shaft diameter for this motor to be 1.875 in (1 7/8 in) with a $1/2 \times 1/4$ in keyway to accept a 1/2-in square key. This will be the bore specified for the motor half of the input shaft coupling. The reason for the large difference in the sizes of the shafts for the motor and for our reducer is that the general-purpose motor must be designed to carry a significant side load, and our shaft does not.

The output shaft of the reducer at the coupling has a diameter of 1.250 in, and the input shaft for the saw will have the same size. Therefore, both halves of the output shaft coupling will have that bore with a $1/4 \times 1/8$ in keyway to accept a 1/4-in square key.

Keys and Keyseats. A total of six keys need to be specified: two for each half of the flexible couplings on the input and output shafts, and one for each gear in the reducer. The methods of Chapter 11 are used to verify the suitability of the keys and to specify the required length using Equation (11–5). We will use standard key sizes made from SAE 1018 CD steel having a yield strength of 54 000 psi.

1. **Keys for the coupling on the input shaft:** First let's check the keys inside the couplings because their sizes have already been specified by the coupling manufacturer. The coupling half that mounts on the input shaft is critical because its bore diameter of 0.875 in is smaller, resulting in larger forces on the key when transmitting the torque of 900 lb·in which was computed earlier during the shaft design. The key is 3/16-in square (0.188 in). We use a design factor N of 4 as we did in the shaft design. Then, from Equation (11–5),

$$L = \frac{4TN}{DW_s_y} = \frac{4(900 \text{ lb} \cdot \text{in})(4)}{(0.875 \text{ in})(0.188 \text{ in})(54\,000 \text{ psi})} = 1.62 \text{ in}$$

We can specify a key length of 2.50 in for extra safety and to match the length of the hub of the coupling. The 1/2-in key for the motor shaft should be made to match the length of the coupling hub also, and it should be very safe because of the larger key size and the larger shaft size carrying the same torque.

2. **Keys for the output shaft coupling on the output shaft:** For the output shaft and the drive shaft for the saw,

$$T = 3150 \text{ lb} \cdot \text{in}$$

$$D = 1.25 \text{ in}$$

$$W = 0.250 \text{ in} \quad (\text{key width})$$

$$L = \frac{4TN}{DW_s_y} = \frac{4(3150 \text{ lb} \cdot \text{in})(4)}{(1.25 \text{ in})(0.250 \text{ in})(54\,000 \text{ psi})} = 2.99 \text{ in}$$

TABLE 15-7 Tolerances and Fits for Keys and Keyseats

Key size (width in inches)	Tolerance on key (all + 0.000)	Tolerance on keyseat (all - 0.000)	Fit range
Up to 1/2	-0.002	+0.002	0.000-0.004
Over 1/2 to 3/4	-0.002	+0.003	0.000-0.005
Over 3/4 to 1	-0.003	+0.003	0.000-0.006
Over 1 to 1 1/2	-0.003	+0.004	0.000-0.007
Over 1 1/2 to 2 1/2	-0.004	+0.004	0.000-0.008

We will make the key length 3.125 in (3 1/8 in), the full length of the hub of the coupling. The conservative design factor of 4 should make this length acceptable.

3. **Key for the pinion on shaft 1:** The bore of the pinion is to be nominally 1.75 in as determined in the shaft design and shown in Table 15-4. The key size for this diameter should be 3/8-in square according to Table 11-1. The torque being transmitted is 900 lb·in. Then Equation (11-5) gives

$$L = \frac{4TN}{DWs_y} = \frac{4(900 \text{ lb} \cdot \text{in})(4)}{(1.75 \text{ in})(0.375 \text{ in})(54\,000 \text{ psi})} = 0.406 \text{ in}$$

The face width of the gear is 2.00 in. Let's use a key length of 1.50 in and center the profile keyseat at section C on the shaft so that the keyseat does not significantly interact with the ring groove to the right or with the shoulder fillet to the left.

4. **Key for the gear on shaft 2:** The bore of the gear is to be nominally 1.75 in as determined in the shaft design and shown in Table 15-4. The key size for this diameter should be 3/8-in square according to Table 11-1. The torque being transmitted is 3150 lb·in. Then Equation (11-5) gives

$$L = \frac{4TN}{DWs_y} = \frac{4(3150 \text{ lb} \cdot \text{in})(4)}{(1.75 \text{ in})(0.375 \text{ in})(54\,000 \text{ psi})} = 1.42 \text{ in}$$

The face width of the gear is 2.00 in. Let's use a key length of 1.50 in for this key also.

The key designs are summarized in the following list:

Summary of Key Designs

Motor shaft: 1/2-in square key $\times$ 2.50 in long
 Input shaft of reducer at coupling: 3/16-in square key $\times$ 2.50 in long; sled runner keyseat

Input shaft at pinion: 3/8-in square key $\times$ 1.50 in long; profile shaft keyseat

Output shaft at gear: 3/8-in square key $\times$ 1.50 in long; profile shaft keyseat

Output shaft at coupling: 1/4-in square key $\times$ 3.125 in long; sled runner keyseat

Drive shaft for saw at coupling: 1/4-in square key $\times$ 3.125 in long; sled runner keyseat

Standard square bar stock in SAE 1018 steel is available to be used for keys. Typical tolerances are given in Table 15-7. Also given are recommended tolerances on the keyseat width dimension and the resulting fit between the key and the keyseat. A small clearance fit is desirable to permit easy assembly while not allowing the key to rock noticeably when installed.

Tolerances on Other Shaft Dimensions. You should review the discussion in Sections 13-5 and 13-9, on fits and tolerancing, respectively. See also References 1, 4, and 7 in this chapter, along with other comprehensive texts on technical drawing and interpretation of engineering drawings.

The fit of the bore of the gears on the shafts or the fit of the *ID* of the couplings on the ends of the shafts must be specified. Either a close sliding fit or a close locational fit is recommended for such components. Data are given in Table 13-3 for the RC2, RC5, and RC8 fits. More complete data are available in References 1-3, 5, 6, and 8 in Chapter 13. We will apply the RC5 fit to accurate mating parts that must assemble easily but where little perceptible play between the parts is desired. The RC fits use the *basic hole system*, as illustrated in Chapter 13.

15-6 FINAL DESIGN DETAILS FOR THE SHAFTS

Figures 15-6 and 15-7 show the final design for the input and output shafts, also called the *pinion shaft* and the *gear shaft*. Data from throughout this chapter have been used to specify pertinent dimensions. Where fillets are specified, a final check on the stress condition has

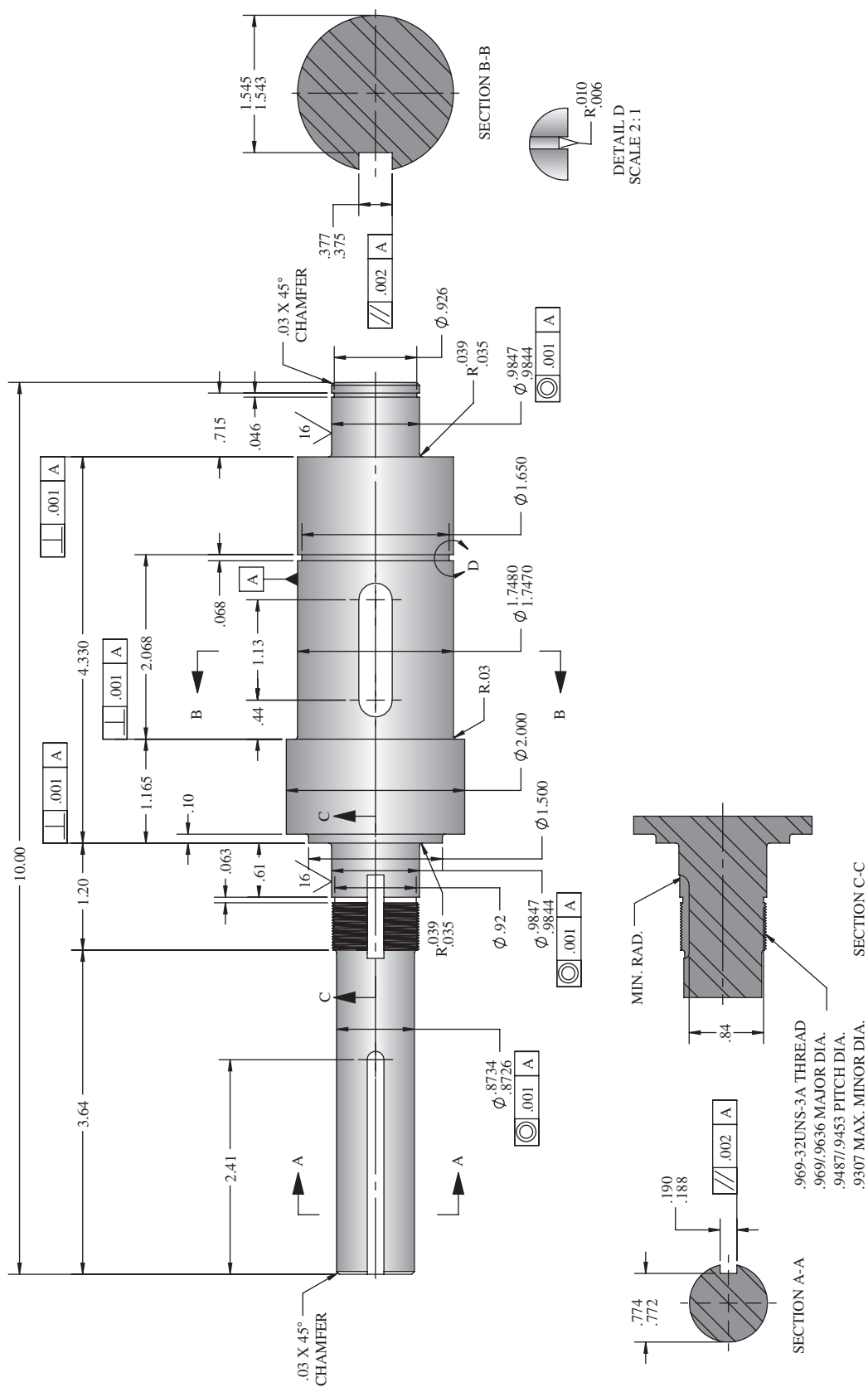


FIGURE 15-6 Final design of the pinion shaft

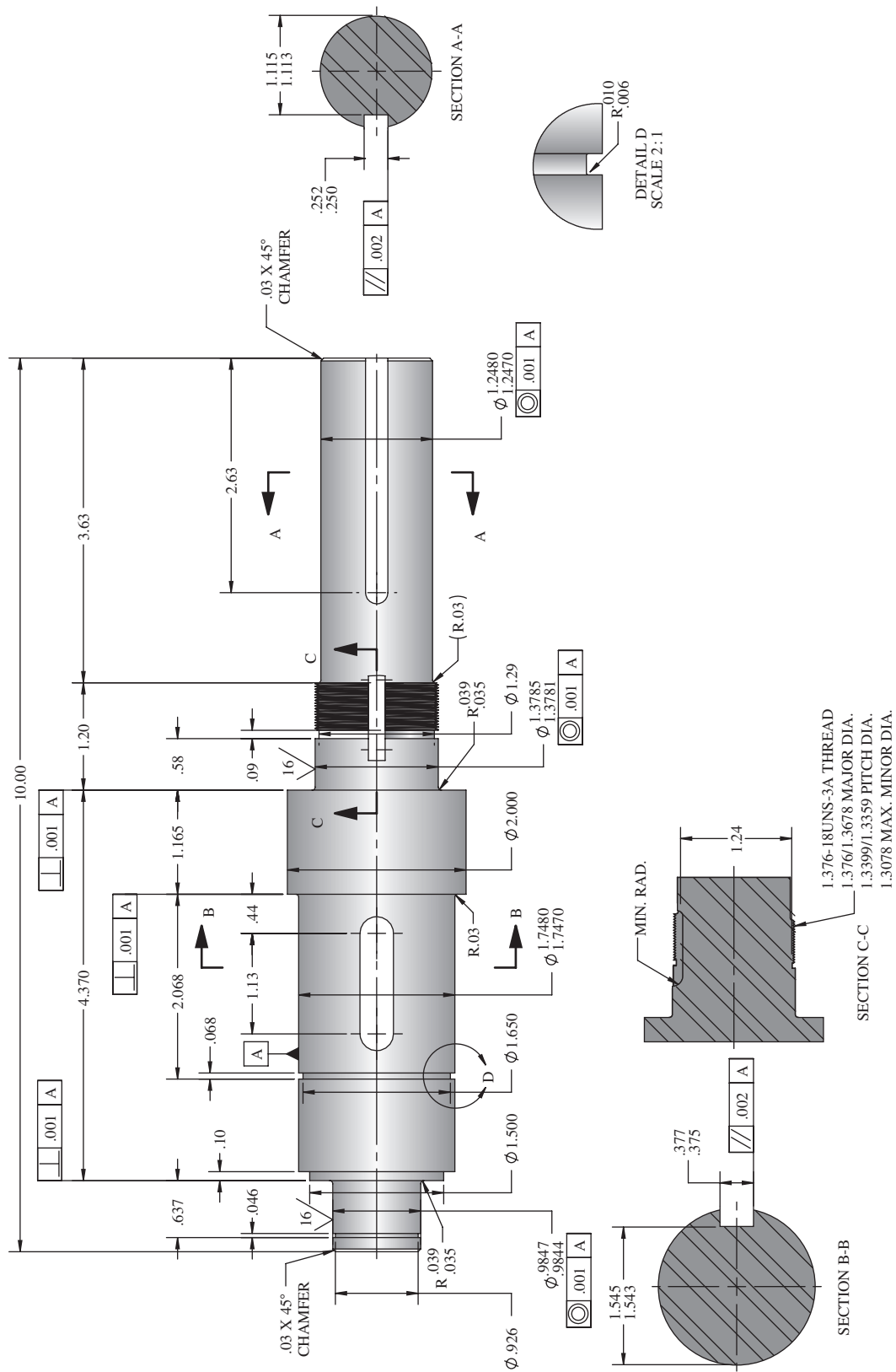


FIGURE 15-7 Final design of the gear shaft

TABLE 15-8 Calculations and Specifications for Fits of Elements on the Shafts of the Transmission

Location	Nominal diameter	Bore limit dimensions for external element	OD limit dimensions for shaft	Fit
Shaft 1:				
Input at coupling	0.8750	0.8762/0.8750	0.8734/0.8726	+0.0026/ +0.0036
Pinion	1.7500	1.7516/1.7500	1.7480/1.7470	+0.0020/ +0.0046
Shaft 2:				
Gear	1.7500	1.7516/1.7500	1.7480/1.7470	+0.0020/ +0.0046
Output at coupling	1.2500	1.2516/1.2500	1.2480/1.2470	+0.0020/ +0.0046

Note: RC5 fit is used for all locations; dimensions are in inches.

been made to ensure that the estimated stress concentration factors used in the earlier design analysis are satisfactory and that the final stress levels are safe.

Keyseat details have been shown in section views on the shaft drawings. See Chapter 11 for computing the vertical dimension from the bottom of the shaft to the bottom of the keyseat.

The retaining ring grooves are drawn to the dimensions specified for a basic external ring for a 1.75-in-diameter shaft as pictured in Figure 11-37(a). See Internet sites 11 and 22 in Chapter 11 for providers of retaining rings and application details.

Four shaft diameters are sized to the RC5 fit in Figures 15-6 and 15-7. On shaft 1 they are at the extension where the coupling mounts and at the pinion. On shaft 2 they are at the gear and at the coupling location on the output extension. Table 15-8 summarizes the data for the fits. You should verify these using the procedure shown in Chapter 13. Note that the limit dimensions for both the shaft diameter and the bore of the mating element are given and that the total tolerance on each dimension is small, less than 0.002 in for any dimension. Note also the small variation of the clearance on mating parts as indicated in the last column called “Fit.”

For the shafts in this project, we have specified geometric tolerances for concentricity of critical diameters for each shaft. The datum or reference diameter is specified at each gear. Then the diameters at the two bearing seats and at the end of the shaft where the coupling mounts are controlled with concentricity feature control blocks. Shoulders for locating bearings and gears are controlled for perpendicularity to the axis of the shaft as represented by the gear diameter. The keyseat is controlled for parallelism to the axis of the shaft.

15-7 ASSEMBLY DRAWING

Figure 15-8 is an assembly drawing for the reducer with all features drawn to scale. The housing has been shown as a rectangular box shape for simplicity. The assembly of all components into the housing is facilitated by

having the right side removable, and special care to control alignment of the cover is critical because it carries the right bearing for each shaft. Using locating dowel pins in the cover and the main housing is one way of ensuring proper location.

Shown are the right side view with the cover removed and a section through the two gears, along the vertical centerline. In the section view the cover is in place. Part numbers are shown and the parts list is included at the right side of the drawing.

The order of assembly could include the following:

1. Install the pinion and the gear onto their shafts, seating them against their shoulders and insert retaining rings to maintain axial location.
2. Press all four bearings onto the shafts, seating them against their shoulders.
3. Attach bearing locknuts for the left bearing on the pinion shaft and the right bearing on the gear shaft.
4. Attach the bearing caps to the left wall of the housing and to the cover for the right side, including the seals for the left pinion shaft and the right gear shaft.
5. Insert the two shaft assemblies into the retaining caps on the left side of the housing, sliding the outer races of the bearings into the caps.
6. Install the cover for the right side, ensuring proper location.
7. Attach the two bearing caps onto the cover carefully sliding them over the outer races of the bearings and securing them to the cover.

Note that the left bearing on the pinion shaft and the right bearing on the gear shaft are fixed. The other two bearings are free to float inside the bearing cap to accommodate thermal expansion and tolerance stack-up. Those bearings are held in axial position on their shaft seats with retaining rings.

Seals have been shown in the bearing retainers where the shafts penetrate the side walls of the housing. See Sections 11-10 and 11-11 and Internet sites 4, 12-16, 22, and 28 in Chapter 11 for more information on seals.

ITEM NO.	DESCRIPTION	QTY.
1	HOUSING GEARBOX	1
2	SHAFT PINION	1
3	SPUR GEAR, PINION DRIVING	1
4	SPUR GEAR, DRIVEN GEAR	1
5	EXTERNAL RETAINING RING #100-175	2
6	SHAFT GEAR	1
7	SKF BALL BEARING #6205	1
8	SKF BALL BEARING #6205	2
9	SKF BALL BEARING #6207	1
10	BEARING CAP FIXED BEARING PINION SHAFT	1
11	COVER, GEARBOX HOUSING	1
12	BEARING CAP-FLOATING BEARING PINION SHAFT	1
13	CR SEAL CIR A1 #19754	1
14	LOCKWASHER, BEARING W-65	1
15	LOCKNUT, BEARING #6-65	1
16	BEARING CAP-FLOATING BEARING GEAR SHAFT	1
17	LOCKWASHER, BEARING W-67	1
18	LOCKNUT, BEARING #6-67	1
19	BEARING CAP FIXED BEARING GEAR SHAFT	1
20	INTERNAL RETAINING RING #100-098	2
21	INTERNAL RETAINING RING #231	1
22	CR SEAL CIR A1 #19754	1
23	INTERNAL RETAINING RING #900-244	1
24	KEY, GEAR	2
25	BASEPLATE, GEARBOX	1
26	FILLER / BREATHER VENT	1

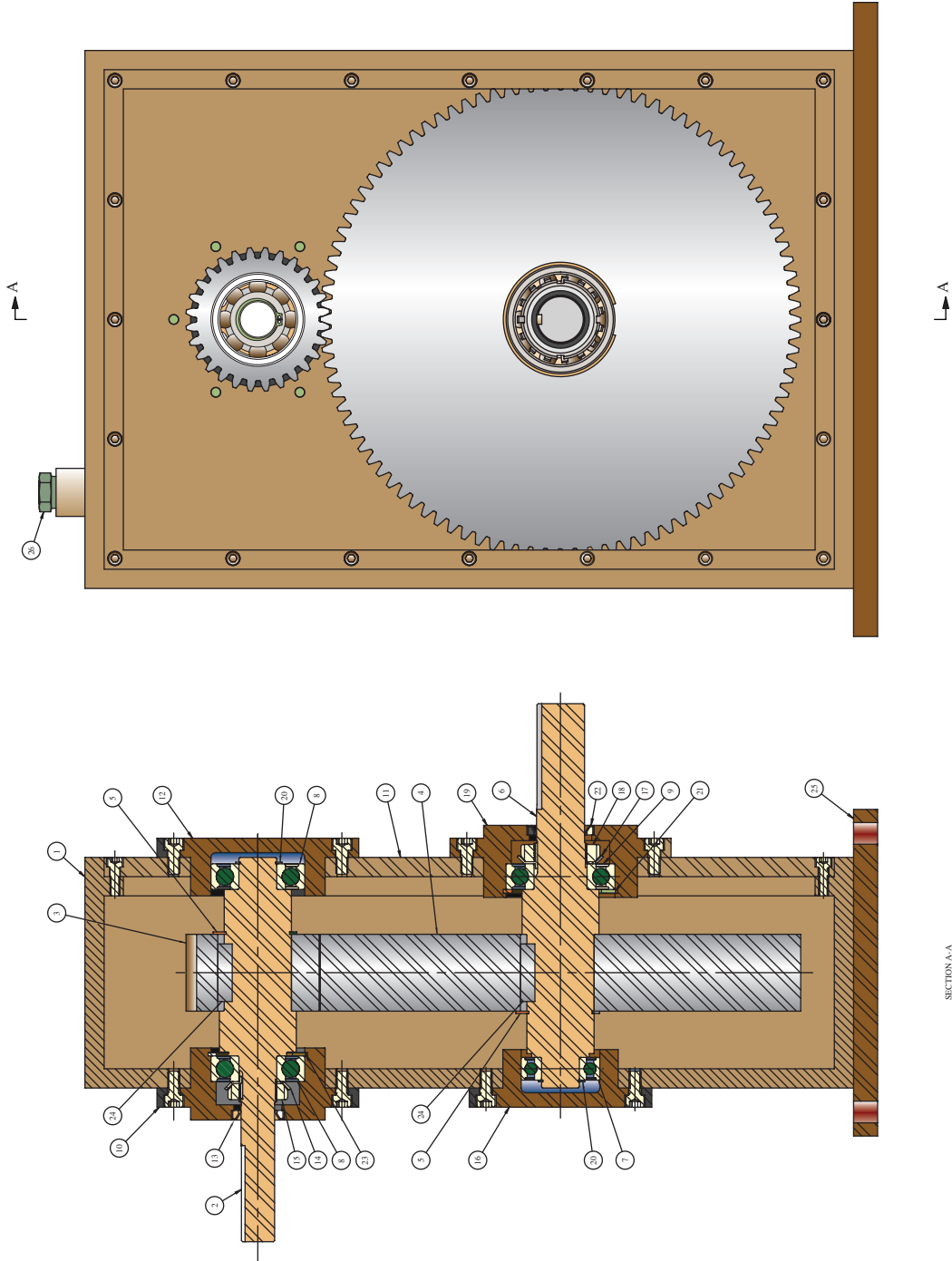


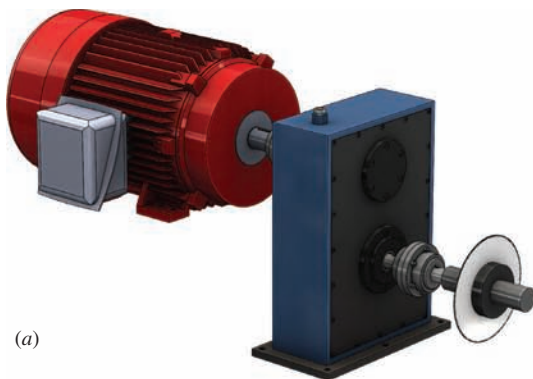
FIGURE 15-8 Assembly drawing for the reducer

Figure 15–9 shows additional details of parts of the gearbox. Comments on each part are as follows:

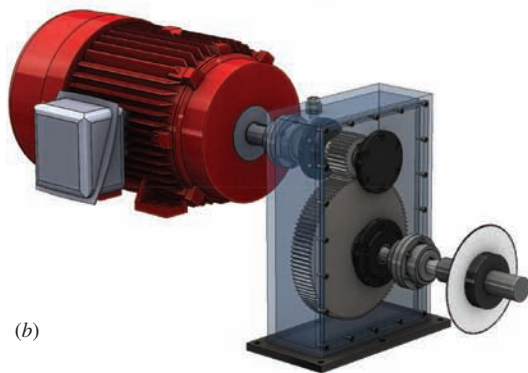
- (a) Pictorial view of the assembly of the motor and the gearbox. Note that the part on the lower right is

a schematic representation of the wall of the saw where its drive shaft connects to the gearbox.

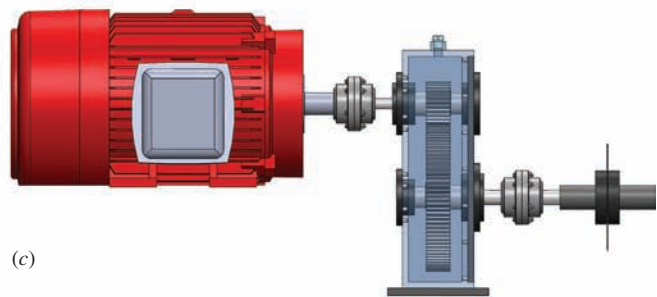
- (b) Similar view as in (a) with the walls of the gearbox made transparent.



(a)



(b)



(c)



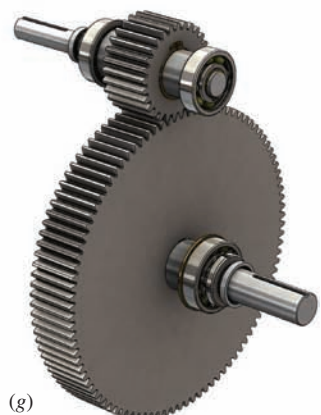
(d)



(e)



(f)



(g)

FIGURE 15–9

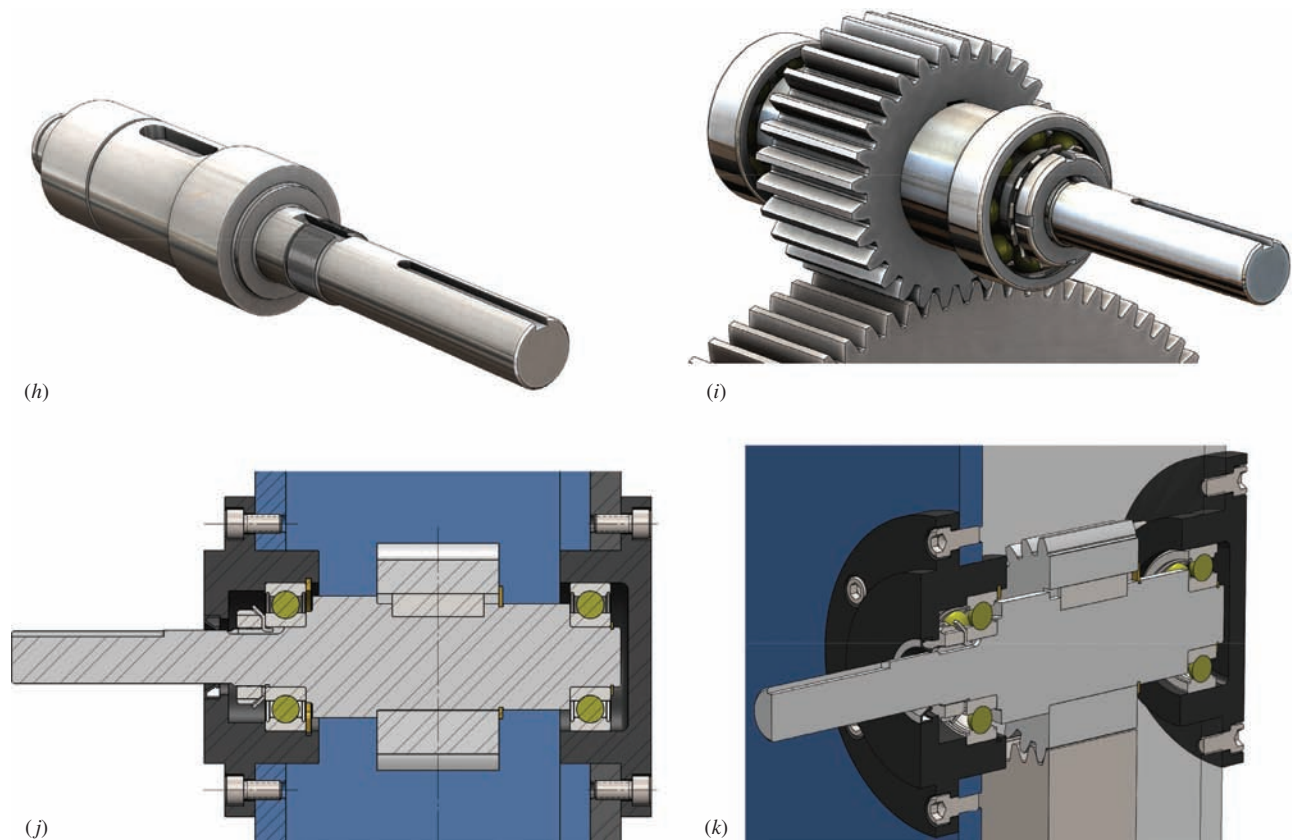


FIGURE 15-9 (continued)

- (c) Side view of the assembly with the walls of the gearbox made transparent.
- (d) Exterior view of the gear box showing the input shaft projecting from the housing along with the bearing cap for the gear shaft.
- (e) Arrangement of the pinion, gear, their shafts, and bearings from the same viewpoint as in (d).
- (f) Exterior of the gear box showing the output shaft projecting from the housing along with the bearing support for the pinion shaft.
- (g) Arrangement of the pinion, gear, their shafts, and bearings from the same viewpoint as in (f).
- (h) Pinion shaft showing keyseats, shoulders, and the threaded portion for the locknut.
- (i) Pinion and its shaft subassembly positioned in mesh with the gear.
- (j) Enlarged view of the portion of the gearbox involving the input/pinion shaft, its bearings, and bearing supports attached to the housing wall.
- (k) Similar view as in (j) shown in a pictorial view.

Critique of the Design

Figures 15-6 to 15-8 present drawings that meet the basic design requirements established at the beginning of this chapter. It is likely that refinements could be made if more details were available about the saw for which the reducer is being designed.

It appears that the length of the shafts could be made somewhat shorter. The distances between the center of the gears and the bearings was arbitrarily set at 2.50 in at the start of the design process when dimensions for any components were unknown. Now that the nominal size of the gears, bearings, and couplings are known, further iterations on the design could result in a smaller package.

You should look at other commercially available gear-type speed reducers for other features that might be built into this design. Note particularly Figures 9-29 to 9-33 in Chapter 9 and Figures 10-1, 10-2, 10-22, and 10-23 in Chapter 10.

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12. Ullman, David G. *The Mechanical Design Process*. 5th ed. New York: McGraw-Hill, 2015.
13. Ulrich, K., and S. Eppinger. *Product Design and Development*. 6th ed. New York: McGraw-Hill/Irwin, 2015.

INTERNET SITES RELATED TO TRANSMISSION DESIGN

1. **Kepner-Tregoe, Inc.** A management consulting and training firm specializing in strategic and operational decision making. The firm's work evolves from the popular book listed as Reference 5. The Kepner-Tregoe approach is applicable to problem solving and decision making in product development and manufacturing operations. An online self-study learning mode called *eLearning Problem Solving and Decision Making* is now offered.
2. **Peerless-Winsmith, Inc.** Manufacturer of a broad line of speed reducers. This company produced the book listed in Reference 9.

DESIGN DETAILS AND OTHER MACHINE ELEMENTS

OBJECTIVES AND CONTENT OF PART III

Chapters 16–23 present methods of analysis and design of several important machine elements that were not particularly pertinent to the design of a power transmission as presented in Part II of this book. These chapters can be covered in any order or can be used as reference material for general design projects.

Chapter 16: Plain Surface Bearings discusses plain surface bearings, sometimes called *journal bearings*. These bearings employ smooth-surfaced elements to support loads from shafts or other devices in which relative motion occurs. You will learn how to specify materials for the parts of the bearing, the geometry of the bearing, and the lubricant. Full-film hydrodynamic lubrication and boundary lubrication are discussed.

Chapter 17: Linear Motion Elements discusses devices that convert rotary motion to linear motion, or vice versa. They are often used in machine tools, automation equipment, and parts of construction machinery. You will learn about their geometry and how to analyze their performance.

Chapter 18: Springs discusses how to design and analyze helical compression springs, helical extension springs, and torsional springs.

Chapter 19: Fasteners describes machine screws, bolts, nuts, and set screws. You will learn about the kinds of materials used for fasteners and how to design them for safe, reliable performance. Also discussed briefly are rivets, quick-operating fasteners, welding, brazing, soldering, and adhesive bonding.

Chapter 20: Machine Frames, Bolted Connections, and Welded Joints presents the important skill of designing a frame for rigidity as well as strength. You will gain experience with analyzing the forces and stresses in bolted and welded connections that hold load-carrying members together. You will also learn how to analyze eccentric loads on connections.

Chapter 21: Electric Motors and Controls discusses the many types of AC and DC motors that are commercially available. Because a great many mechanical design projects involve the use of an electric motor as the prime mover, you will learn how to match their performance characteristics to the needs of the machine being designed and to specify motor controls.

Chapter 22: Motion Control: Clutches and Brakes discusses the many types of clutches and brakes that can be applied. Clutches provide the path to connect power from a prime mover to a driven machine. Brakes bring the moving equipment to a stop or slow it down. You will learn how to analyze their performance and to either design them or specify commercially available units.

Chapter 23: Design Projects presents several projects that you can complete yourself.

PLAIN SURFACE BEARINGS

The Big Picture

You Are the Designer

- 16–1 Objectives of This Chapter
- 16–2 The Bearing Design Task
- 16–3 Bearing Parameter, $\mu n/p$
- 16–4 Bearing Materials
- 16–5 Design of Boundary-Lubricated Bearings
- 16–6 Full-Film Hydrodynamic Bearings
- 16–7 Design of Full-Film Hydrodynamically Lubricated Bearings
- 16–8 Practical Considerations for Plain Surface Bearings
- 16–9 Hydrostatic Bearings
- 16–10 The Kugel Fountain—A Special Example of a Hydrostatic Bearing
- 16–11 Tribology: Friction, Lubrication, and Wear

THE BIG PICTURE

Plain Surface Bearings

Discussion Map

- The purpose of a bearing is to support a load while permitting relative motion between two elements of a machine. This chapter discusses *plain surface* bearings where the two parts that move relative to each other do not have rolling elements between them.
- Both rotating and linear sliding bearing cases depend on lubrication, a part of the field of *tribology*.

Discover

Look around your home and in your car for products that have plain surface bearings. Find some that experience rotational motion and some that have linear sliding contact. Consider simple items such as hinges, door locks, latches, or the wheels of a lawn mower. Most of these experience boundary lubrication.

Now see if you can find information about the crankshaft bearings in the engine of your car. Such bearings usually employ full-film hydrodynamic lubrication. What can you discover about them?

How do you think a large telescope or an astronomical radio antenna is supported to permit it to be moved easily and positioned precisely? One method is called hydrostatic bearings. What can you discover about such bearings?

This chapter will help you explore all of these kinds of bearings and complete the basic design analyses required to ensure satisfactory operation.

The purpose of a bearing is to support a load while permitting relative motion between two elements of a machine. This chapter discusses *plain surface* bearings where the two parts that move relative to each other do not have rolling elements between them. Note that rolling contact bearings were discussed in Chapter 14.

Plain surface bearings for rotating parts are, of course, cylindrical with a typical arrangement as shown in Figure 16–1. The inner member, called the *journal*, is usually that part of a shaft where any

radial reaction forces are transferred to the base of the machine. The stationary member that mates with the journal is called simply the *bearing*. Other names used for plain surface bearings are *journal bearings* and *sleeve bearings*.

Where have you seen such plain surface bearings in operation? As you did in Chapter 14 on rolling contact bearings, look for consumer products, industrial machinery, or transportation equipment (cars, trucks, bicycles, and so on—any device with rotating

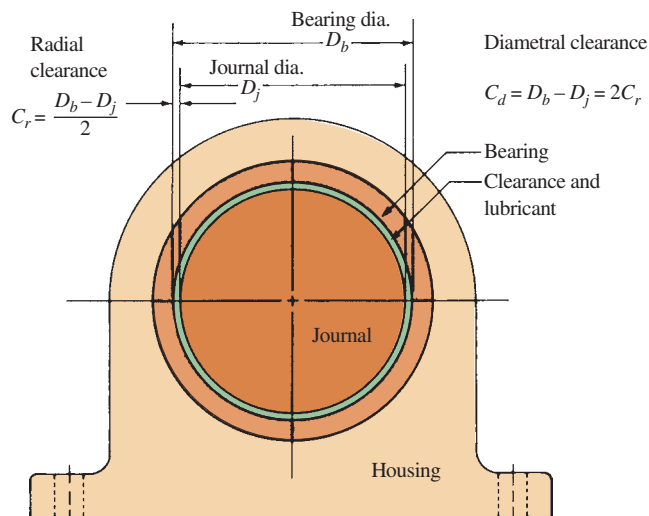


FIGURE 16-1 Bearing geometry

shafts). If the bearings are not visible from the outside (and that is often the case), you may need to partially disassemble the product to see inside.

But there are some obvious examples that should be accessible to you. The wheels of a household lawn mower, wheelbarrow, or simple cart are typically mounted directly on axles to form the plain surface bearing. Hand tools such as grass and hedge clippers, pruning shears, pliers, and adjustable or ratchet wrenches use plain surface bearings at points where one part must rotate relative to another.

Look at almost any hinge for a door. It incorporates a solid pin riding inside a cylindrical outer member. That is an example of a plain surface bearing. Garage doors often have rollers that ride in channels, but the shafts that carry the rollers are operating in plain cylindrical bearings, often with very large clearances. The cable system that connects to the counterbalance springs runs over pulleys having plain surface bearings rotating on stationary shafts. If the door has an electric door opener, several of its components probably use plain surface bearings. Get a ladder and climb up to inspect them. *But be careful of the moving parts!*

Besides the rotational motion of the kinds of applications described previously, some plain surface bearings have linear sliding motion. Look inside a printer for a computer of the ink-jet or dot-matrix-pin type. Identify the part that delivers the actual image to the paper as it traverses across the page. It typically slides on a polished rod with great precision. Many linkages contain parts that slide. Look at locking mechanisms, latches, staplers, switches, car seat adjusters, gear shifters for cars, parts of window-lift devices in cars, and coin-operated machines. Can you identify parts that have linear sliding motion?

If you visit a factory, you will likely see numerous examples of both rotational and linear sliding action.

Transfer devices, reciprocating machinery, machine tool slides, and packaging equipment employ many plain surface bearings.

Most of the examples just mentioned have intermittent and relatively slow motion between the mating parts, whether the motion is rotational or linear. In such applications, there is often a lubricant between the moving parts. Or the materials are carefully chosen to produce low friction and reliable, smooth motion. This kind of bearing experiences *boundary lubrication* as discussed in this chapter.

You may be familiar with the bearings on the crankshaft of an internal combustion engine. They are often called *journal bearings* because they have the classic configuration shown in Figure 16-1. But consider their general operation. After the engine has been started, the crank rotates at several thousand revolutions per minute, typically from about 1500 to 6000 rpm. This is one of the most important applications of plain surface bearings using *full-film hydrodynamic lubrication*. In such bearings, there is a continuous film of lubricant, usually oil, that actually lifts the rotating shaft journal off the stationary bearing. Thus, there is no metal-to-metal contact between the members. Much of this chapter is devoted to this kind of bearing and the analysis of the conditions required to maintain the supporting oil film. Engine bearings are some of the most highly engineered mechanical devices.

Can you identify other examples of bearings that use full-film hydrodynamic lubrication?

Another example of a class of plain surface bearings is called *hydrostatic*. Pumps create high pressure in a fluid such as oil that is delivered by carefully shaped pads where the pressure lifts the part to be moved. Then it can be moved easily, either slowly or rapidly. Think about a huge telescope or

radio antenna that must be positioned accurately and moved with ease. Such applications are rare but important, justifying the large engineering and technical effort of designing, installing, and maintaining them.

This chapter will help you understand boundary lubrication, full-film hydrodynamic lubrication, and hydrostatic lubrication. Also discussed are more general aspects of friction, lubrication, wear, and the lubricants that are used to minimize friction.

YOU ARE THE DESIGNER

Your company is designing a conveyor system to transport products in a shipping and receiving system of a major manufacturer. The design has progressed to the point where it has been decided to use a flexible belt conveyor driven by flat pulleys at either end. The shafts that carry the pulleys must be supported in bearings in the side frames of the conveyor. Your task is to design the bearings.

First you must decide what type of bearing to use: plain surface or journal bearings as discussed in this chapter, or rolling contact

bearings, which have been discussed in Chapter 14. For the plain surface bearings, you must determine whether full-film hydrodynamic lubrication can be achieved with its advantages of low friction and long life. Or will the shaft operate in the bearing with boundary lubrication? What materials will the bearing and the journal be made from? What dimensions will be specified for all components? What lubricant should be used? These and other questions are discussed in this chapter.

16-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Describe the three modes of operation of a *plain surface bearing* (boundary, mixed-film, and full-film hydrodynamic lubrication), and discuss the conditions under which each will normally occur.
2. Discuss the significance of the *bearing parameter*, $\mu n/p$.
3. List the decisions that a bearing designer must make to completely define a plain surface bearing system.
4. List the materials often used for journals and bearings, and describe their important properties.
5. Define the *pV factor* and use it in the design of boundary-lubricated bearings.
6. Describe the operation of full-film hydrodynamically lubricated bearings.
7. Complete the design of full-film bearings, defining the size of the journal and bearing, the diametral clearance, the bearing length, the minimum film thickness, the surface finish, the lubricant, and the resulting frictional performance of the bearing system.
8. Describe a *hydrostatic bearing system* and complete the basic design of such bearings.
9. Define *tribology* and discuss the essential characteristics of friction, lubrication, and wear as applied to machinery.
10. Describe the general nature of oils and greases and their effects on lubrication and wear.

16-2 THE BEARING DESIGN TASK

The term *plain surface bearing* refers to the kind of bearing in which two surfaces move relative to each other without the benefit of rolling contact. Thus, there is

sliding contact. The actual shape of the surfaces can be anything that permits the relative motion. The most common shapes are flat surfaces and concentric cylinders. Figure 16-1 shows the basic geometry of a cylindrical plain surface bearing.

A given bearing system can operate with any of the three types of lubrication:

Boundary lubrication: There is actual contact between the solid surfaces of the moving and the stationary parts of the bearing system, although a film of lubricant is present.

Mixed-film lubrication: There is a transition zone between boundary and full-film lubrication.

Full-film lubrication: The moving and stationary parts of the bearing system are separated by a complete film of lubricant that supports the load. The term *hydrodynamic lubrication* is often used to describe this type.

All of these types of lubrication can be encountered in a bearing without external pressurization of the bearing. If lubricant under pressure is supplied to the bearing, it is called a *hydrostatic bearing*, which is discussed separately. Running dry surfaces together is not recommended unless there is inherently good lubricity between the mating materials. Some plastics are used dry, as discussed in Sections 16-4 and 16-5.

Bearing design involves so many design decisions that it is not possible to develop a procedure that produces the single best design. Thus, several feasible designs could be proposed, and the designer must make judgments based on knowledge of the application and on the principles of bearing operation to define the final design. The following lists identify the information needed to design a bearing system and the types of design decisions that must be made. (See References 8, 13-17, and 19-21.) In this discussion, it is assumed that the bearing

will be cylindrical, like that used to support a rotating shaft. Modified lists could be made for linear sliding surfaces or some other geometry.

Bearing Requirements

- Magnitude, direction, and degree of variation of the radial load
- Magnitude and direction of the thrust load, if any
- Rotational speed of the journal (shaft)
- Frequency of starts and stops, and duration of idle periods
- Magnitude of the load when the system is stopped and when it is started
- Life expectancy of the bearing system
- Environment in which the bearing will operate

Design Decisions

- Materials for the journal and the bearing
- Diameters, including tolerances, of the journal and the bearing
- Nominal value and range of journal clearance in the bearing
- Surface finish for the journal and the bearing
- Length of the bearing
- Method of manufacturing the bearing system
- Type of lubricant to be used and the means of supplying it
- Operating temperature of the bearing system and of the lubricant
- Method of maintaining the lubricant cleanliness and temperature

Analyses Required

- Type of lubrication: boundary, mixed-film, and full-film
- Coefficient of friction
- Frictional power loss
- Minimum film thickness
- Thermal expansion
- Heat dissipation required and the means of accomplishing it
- Shaft stiffness and slope of the shaft in the bearing

16-3 BEARING PARAMETER, $\mu n/p$

The performance of a bearing differs radically, depending on which type of lubrication occurs. There is a marked decrease in the coefficient of friction when the operation changes from boundary to full-film lubrication. Wear also decreases with full-film lubrication. Thus, it

is desirable to understand the conditions under which one or the other type of lubrication occurs.

The creation of full-film lubrication, the most desirable type, is encouraged by light loads, high relative speed between the moving and stationary parts, and presence of a high-viscosity lubricant in the bearing in copious supply. For a rotating journal bearing, the combined effect of these three factors, as it relates to the friction in the bearing, can be evaluated by computing the *bearing parameter*, $\mu n/p$. The viscosity of the lubricant is indicated by μ , the rotational speed by n , and the bearing load by the pressure, p . To compute the pressure, divide the applied radial load on the bearing by the *projected area* of the bearing, that is, the product of the length times the diameter.

The bearing parameter $\mu n/p$ is dimensionless when each term is expressed in consistent units. Some unit systems that can be used are listed as follows:

Unit system	Viscosity, μ	Rotational speed, n	Pressure, p
SI metric	N·s/m ² or Pa·s	rev/s	N/m ² or Pa
English	lb·s/in ² or reyn	rev/s	lb/in ²
Old metric (obsolete)	dyne·s/cm ² or poise	rev/s	dynes/cm ²

The effect of the bearing parameter is shown in Figure 16-2, sometimes called the *Stribeck curve*, which is a plot of the coefficient of friction, f , for the bearing versus the value of $\mu n/p$. At low values of $\mu n/p$, boundary lubrication occurs, and the coefficient of friction is high. For example, with a steel shaft sliding slowly in a lubricated bronze bearing (boundary lubrication), the value of f would be approximately 0.08 to 0.14. At high values of $\mu n/p$, the full hydrodynamic film is created, and the value of f is normally in the range of 0.001 to 0.005. Note that this compares favorably with precision rolling contact bearings. Between boundary and full-film lubrication, the *mixed-film* type, which is some combination of the other two, occurs.

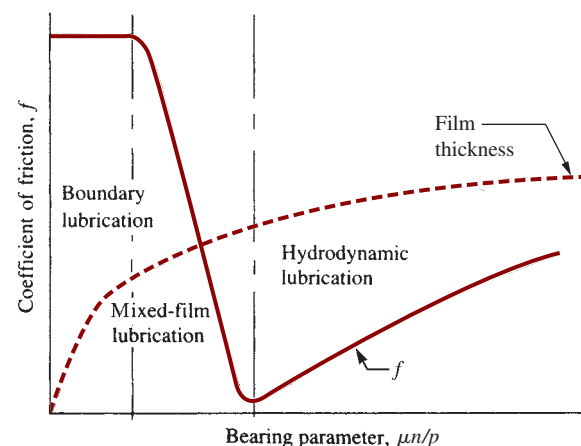


FIGURE 16-2 Bearing performance and types of lubrication related to the bearing parameter, $\mu n/p$ —the Stribeck curve

The dashed curve shows the general nature of the variation in film thickness in the bearing.

It is recommended that designers avoid the mixed-film zone because it is virtually impossible to predict the performance of the bearing system. Also notice that the curve is very steep in this zone. Thus, a small change in any of the three factors, μ , n , or p , produces a large change in f , resulting in erratic performance of the machine.

The value of $\mu n/p$ at which full-film lubrication is produced is difficult to predict. Besides the individual factors of speed, pressure (load), and viscosity (a function of the type of lubricant and its temperature), the variables affecting the production of the film include the quantity of lubricant supplied, the adhesion of the lubricant to the surfaces, the materials of the journal and the bearing, the structural rigidity of the journal and the bearing, and the surface roughness of the journal and the bearing. After completing the design process presented later in this chapter, you are advised to test the design.

In general, boundary lubrication should be expected for slow-speed operation with a surface speed less than about 10 ft/min (0.05 m/s). Reciprocating or oscillating motion or a combination of a light lubricant and high pressure would also produce boundary lubrication.

The design of bearings to produce full-film lubrication is described in Section 16–6. In general, it requires a surface speed of greater than 25 ft/min (0.13 m/s) continuously in one direction with an adequate supply of oil at a proper viscosity.

16–4 BEARING MATERIALS

In rotational applications, the journal on the shaft is frequently steel. The stationary bearing may be made of any of a wide variety of materials, including the following:

- Bronze
- Babbitt
- Aluminum
- Zinc
- Porous metals
- Plastics (nylon, TFE, PTFE, phenolic, acetal, polycarbonate, and filled polyimide)

The properties desirable for materials used for plain bearings are unique, and compromises must frequently be made. The following list discusses these properties.

1. **Strength:** The function of the bearing is to carry the applied load and deliver it to the supporting structure. At times, the loads are varied, requiring fatigue strength as well as static strength.
2. **Embeddability:** The property of embeddability relates to the ability of the material to hold contaminants in the bearing without causing damage to the rotating journal. Thus, a relatively soft material is desirable.
3. **Corrosion resistance:** The total environment of the bearing must be considered, including journal

material, lubricant, temperature, air-borne particulates, and corrosive gases or vapors.

4. **Cost:** Always an important factor, cost includes not only the material cost but also processing and installation costs.

A brief discussion of the performance of some of the bearing materials follows.

Cast Bronze

The name *bronze* refers to several alloys of copper with tin, lead, zinc, or aluminum, either singly or in combination. The leaded bronzes contain 25% to 35% lead, which gives them good embeddability and resistance to seizing under conditions of boundary lubrication. However, their strength is relatively low. The cast bearing bronze, SAE CA932, has 83% copper, 7% tin, 7% lead, and 3% zinc. It possesses a good combination of properties for such uses as pumps, machinery, and appliances. Tin and aluminum bronzes have higher strength and hardness and can carry greater loads, particularly in impact situations. But they rate lower on embeddability. See Internet sites 1, 6, 7, 9, and 10.

Babbitt

Babbitts may be lead based or tin based, having nominally 80% of the parent metal. Various alloy compositions of copper and antimony (as well as lead and tin) can tailor the properties to suit a particular application. Because of their softness, babbitts have outstanding embeddability and resistance to seizure, important properties in applications in which boundary lubrication occurs. They have rather low strength, however, and are often applied as liners in steel or cast iron housings. See Internet site 9.

Aluminum

With the highest strength of the commonly used bearing materials, aluminum is used in severe applications in engines, pumps, and aircraft. The high hardness of aluminum bearings results in poor embeddability, requiring clean lubricants.

Zinc

Bearings made from zinc alloys offer good protection to running without a continuous supply of lubricant, although they perform best if lubricated. Standard bearing greases are often used. When operating on steel journals, a thin film of the softer zinc material is transferred to the steel to protect it from wear and damage. It performs well in most atmospheric conditions except for continuously wet environments and exposure to seawater. See Internet site 8.

Porous Metals

Products of the powder metal industry, porous metals are sintered from powders of bronze, iron, and aluminum;

some are mixed with lead or copper. The sintering leaves a large number of voids in the bearing material into which lubricating oil is forced. Then, during operation, the oil migrates out of the pores, supplying the bearing. Such bearings are particularly good for slow-speed, reciprocating, or oscillating motions. See Internet sites 6, 7, and 16.

Plastics

Generally referred to as *self-lubricating materials*, plastics used in bearing applications have inherently low friction characteristics. They can be operated dry, but most improve in performance with a lubricant present. Embeddability is usually good, as is resistance to seizure. But many have low strength, limiting their load-carrying capacity. Backing with metal sleeves is frequently done to improve load-carrying capacity. Major advantages are corrosion resistance and, when operated dry, freedom from contamination. These properties are particularly important in the processing of foodstuffs and chemical products. See Internet sites 2–7, 16, and 17.

Because of the complex chemical names for plastic materials and the virtually infinite combinations of base materials, reinforcements, and fillers used, it is difficult to characterize bearing plastics. Most are composites of several components. The group known as *fluoropolymers* are popular because of the very low coefficient of friction (0.05 to 0.15) and good wear resistance. Phenolics, polycarbonates, acetals, nylons, and many other plastics are also used for bearings. Among the chemical names and abbreviations found in this field are the following:

PTFE: Polytetrafluoroethylene

PA: Polyamide

PPS: Polyphenylene sulfide

PVDF: Polyvinylidene fluoride

PEEK: Polyetheretherketone

PEI: Polyetherimide

PES: Polyethersulfone

PFA: Perfluoroalkoxy-modified tetrafluoroethylene

Reinforcements and fillers used with plastic bearing materials include glass fibers, milled glass, carbon fibers,

bronze powders, PTFE, PPS, and some solid lubricants, such as graphite and molybdenum disulfide.

16-5 DESIGN OF BOUNDARY-LUBRICATED BEARINGS

A boundary-lubricated bearing is typically a smooth hollow cylinder through which a shaft passes that rotates slowly, oscillates, or slides linearly. Figure 16–3 shows two styles of commercially available bearings. Part (a) shows a flange-mounted self-lubricating bearing that can operate in corrosive atmospheres or under water or other light liquids. It can be exposed to temperatures as low as -200°F up to $+1000^{\circ}\text{F}$ (-129°C to 538°C). Part (b) shows a bearing lined with Babbitt [for up to 130°F (54°C)] or bronze [for up to 300°F (149°C)] that is lubricated periodically with grease.

The factors to be considered when selecting materials for bearings and specifying the design details include the following:

Coefficient of friction: Both static and dynamic conditions should be considered.

Load capacity, p : Radial load divided by the projected area of the bearing (lb/in^2 or Pa).

Speed of operation, V : The relative speed between the moving and stationary components, usually in ft/min or m/s .

Temperature at operating conditions.

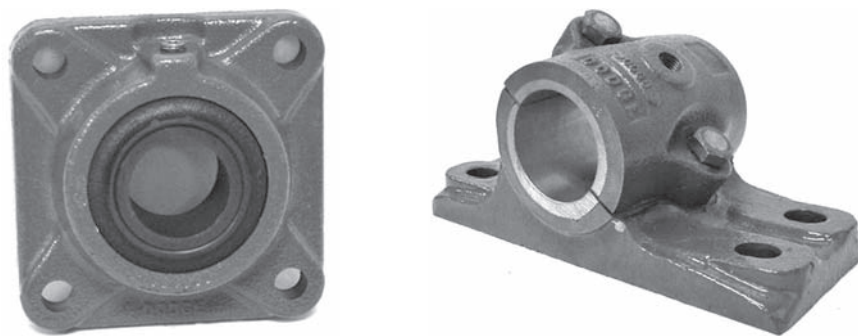
Wear limitations.

Producibility: Machining, molding, fastening, assembly, and service.

Besides specifying the material for the bearing, the designer must determine the combination of the inside diameter, D , and the length, L , for the bearing.

pV Factor

In addition to the individual consideration of load capacity, p , and speed of operation, V , the product pV is an important performance parameter for bearing design when boundary lubrication occurs. The pV value is a



(a) Self-lubricated flange-mounted bearing (b) Grease-lubricated journal bearing

FIGURE 16–3 Two styles of boundary-lubricated bearings (Baldor/Dodge, Greenville, SC)

TABLE 16-1 Typical Performance Parameters for Bearing Materials in Boundary Lubrication at Room Temperature

Material	pV		
	psi-fpm	MPa-m/s	
Vespe [®] SP-21 polyimide	300 000	10.500	Trademark of DuPont Co.
DP11 [™] , oiled	286 000	10.02	GGB Bearing Technology
Manganese bronze (C86200)	150 000	5.250	Also called SAE 430A
Aluminum bronze (C95400)	125 000	4.375	Also called SAE 68A
DX [®] 10, dry or oiled	80 000	2.800	GGB Bearing Technology
Leaded tin bronze (C93800)	75 000	2.625	Also called SAE 660
DU [®]	51 400	1.800	GGB Bearing Technology
BU dry lubricant bearing	51 000	1.785	See note 1
Porous bronze/oil impregnated	50 000	1.750	
Babbitt: high tin content (89%)	30 000	1.050	
DP11 [™] , dry	28 600	1.000	GGB Bearing Technology
Babbitt: low tin content (10%)	18 000	0.630	
Graphite/Metallized	15 000	0.525	Graphite Metallizing Corp.
Rulon [®] PTFE: 641	10 000	0.350	Food and drug applications (see note 2)
Rulon [®] PTFE: J	7500	0.263	Filled PTFE
Polyurethane: UHMW	4000	0.140	Ultra-high molecular weight
Nylon [®] 101	3000	0.105	Trademark of DuPont Co.

See Internet sites 2–7, 16, and 17.

Notes:

¹ BU Bearings consist of bonded layers of a steel backing and a porous bronze matrix overlaid with PTFE/lead bearing material. A film from the bearing material is transferred to the journal during operation.

² Rulon[®] is a registered trademark of Saint-Gobain Performance Plastics Company. Bearings are made from Rulon[®] PTFE (polytetrafluoroethylene) material in a variety of formulations and physical constructions.

measure of the ability of the bearing material to accommodate the frictional energy generated in the bearing. At the limiting pV value, the bearing will not achieve a stable temperature limit, and rapid failure will occur. A practical design value for pV is one-half of the limiting pV value, given in Table 16-1.

Units for pV . The nominal units for pV are simply the product of the units for pressure and the units for velocity. In the U.S. Customary System, this is, considering only units,

$$\begin{aligned}
 p &= F/LD = \text{lb/in}^2 = \text{psi} \\
 V &= \pi Dn/12 = \text{ft/min} = \text{fpm} \\
 &\quad \text{for continuous rotation in rpm} \\
 pV &= (\text{lb/in}^2)(\text{ft/min}) = \text{psi-fpm}
 \end{aligned}$$

Another way of looking at these units is to rearrange them into the form

$$pV = (\text{ft} \cdot \text{lb/min})/\text{in}^2$$

The numerator represents one unit for power or energy transfer per unit time. The denominator represents area. Therefore, pV can be thought of as the rate of energy input to the bearing per unit of projected area of the bearing if the coefficient of friction is 1.0. Of course, the actual coefficient of friction is normally much less than one. Then you can think of pV as a comparative measure of the bearing's ability to absorb energy without overheating.

In SI units, force, F , is in newtons (N) and bearing dimensions are in mm. The pressure is then given as follows:

➤ Bearing Pressure

$$p = F/LD = \text{N/mm}^2 = \text{MPa} \quad (16-1)$$

The linear velocity of the surface of the journal is typically computed from

$$v = \pi Dn/(60\,000) \text{ m/s}$$

with D in mm and n in rpm. Then the units for pV are

$$pV = (\text{MPa})(\text{m/s})$$

A useful conversion to the U.S. Customary System is

$$1.0 \text{ psi-fpm} = 3.50 \times 10^{-5} \text{ MPa} \cdot \text{m/s}$$

Again, the units for pV can be reformatted to reflect the rate of energy transfer per unit area:

$$pV = \text{MPa} \cdot \text{m/s} = \frac{10^6 \text{ N}}{\text{m}^2} \cdot \frac{\text{m}}{\text{s}} = \frac{10^6 \text{ W}}{\text{m}^2}$$

where $1 \text{ W} = 1 \text{ N} \cdot \text{m/s}$

Operating Temperature

Most plastics are limited to approximately 200°F (93°C). However, PTFE can operate at 500°F (260°C). Babbitt is limited to 300°F (150°C), while tin-bronze and aluminum can operate at 500°F (260°C). A major advantage

of carbon-graphite bearings is their ability to operate at up to 750°F (400°C).

Design Procedure for Continuously Rotating Shafts in Bearings

The following is one method of completing the preliminary design of boundary-lubricated plain surface bearings.

PROCEDURE FOR DESIGNING BOUNDARY-LUBRICATED PLAIN SURFACE BEARINGS ▼

Given information: Radial load on the bearing, F (lb or N); speed of rotation, n (rpm); nominal minimum shaft diameter, $D_{\min}$ (in or mm) (based on stress or deflection analysis).

Objectives of the design process: To specify the nominal diameter and the length of the bearing and a material that will have a safe value of pV .

1. Specify a trial diameter, D , for the journal and the bearing.
2. Specify a ratio of bearing length to diameter, L/D , typically in the range of 0.5 to 2.0. For nonlubricated (dry-rubbing) or oil-impregnated porous bearings, $L/D = 1$ is recommended. (See Reference 6.) For carbon-graphite bearings, $L/D = 1.5$ is recommended.
3. Compute $L = D(L/D)$, the nominal length of the bearing.

4. Specify a convenient value for L .
5. Compute the bearing pressure (lb/in² or Pa):

$$p = F/LD$$

6. Compute the linear speed of the journal surface:

$$\text{U.S. Customary units: } V = \pi Dn/12 \text{ ft/min or fpm}$$

$$\text{SI metric units: } V = \pi Dn(60\,000) \text{ m/s}$$

$$\text{Also note: } 1.0 \text{ ft/min} = 0.00508 \text{ m/s}$$

$$1.0 \text{ m/s} = 197 \text{ ft/min}$$

7. Compute pV (psi-fpm or MPa · m/s).
8. Multiply $2(pV)$ to obtain a design value for pV .
9. Specify a material from Table 16-1 with a rated value of pV equal to or greater than the design value.
10. Complete the design of the bearing system considering diametral clearance, lubricant selection, lubricant supply, surface finish specification, thermal control, and mounting considerations. Often the supplier of the bearing material provides recommendations for many of these design decisions.
11. Nominal diametral clearance: Many factors affect the final specification for clearance, such as need for precision, thermal expansion of all parts of the bearing system, load variations, shaft deflection expected, means of providing lubricant, and manufacturing capability. One long-used rule of thumb is to provide 0.001 in of clearance per inch of journal diameter. Figure 16-4

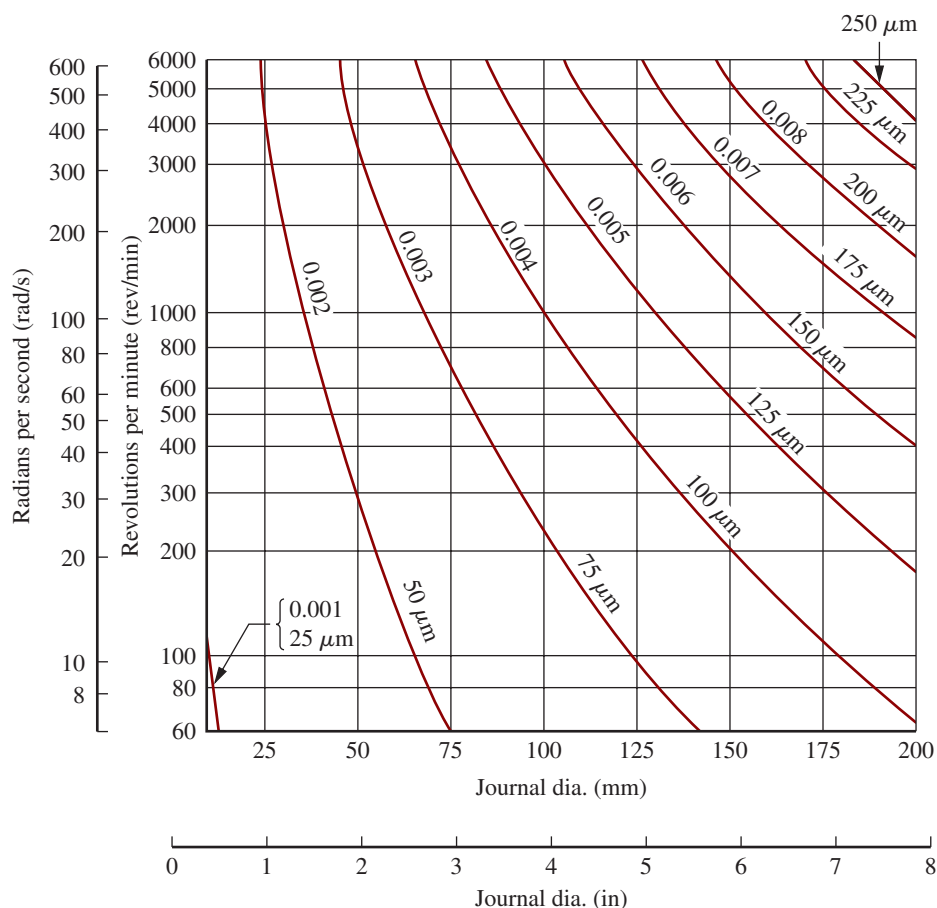


FIGURE 16-4 Minimum recommended diametral clearance for bearings considering journal diameter and rotational speed

shows recommended minimum values for clearance based on the journal diameter and the rotational speed under steady loads. These values apply to the smallest clearance under any combination of tolerances on the dimensions of the bearing system to avoid problems

of bearing heating and eventual seizure. The operating clearance will then be higher than these values because of manufacturing tolerances. Performance over the entire range of clearances should be evaluated, preferably by testing.

Example Problem 16-1

A bearing is to be designed to carry a radial load of 150 lb from a shaft having a minimum acceptable diameter of 1.50 in and rotating at 500 rpm. Design the bearing to operate under boundary-lubrication conditions.

Solution We will use the design procedure previously outlined.

Step 1. Trial diameter: Try $D = D_{\min} = 1.50$ in.

Steps 2-4. Try $L/D = 1.0$. Then $L = D = 1.50$ in.

Step 5. Bearing pressure:

$$p = F/LD = (150 \text{ lb})/(1.50 \text{ in})(1.50 \text{ in}) = 66.7 \text{ psi}$$

Step 6. Journal speed:

$$V = \pi Dn/12 = \pi(1.50)(500)/12 = 196 \text{ ft/min}$$

Step 7. pV factor:

$$pV = (66.7 \text{ psi})(196 \text{ fpm}) = 13\,100 \text{ psi-fpm}$$

Step 8. Design value of $pV = 2(13\,100) = 26\,200 \text{ psi-fpm}$.

Step 9. From Table 16-1, we could use a bearing made from high tin babbitt having a rated value of pV of 30 000 psi-fpm.

Steps 10-11. Nominal diametral clearance: From Figure 16-4, we can recommend a minimum $C_d = 0.002$ in based on $D = 1.50$ in and $n = 500$ rpm. Other design details are dependent on the system into which the bearing will be placed.

Example Problem 16-2

Design a boundary-lubricated plain surface bearing to carry a radial load of 2.50 kN from a shaft rotating at 1150 rpm. The nominal minimum diameter of the journal is 65 mm.

Solution We will use the design procedure previously outlined.

Step 1. Trial diameter: Try $D = 75$ mm.

Steps 2-4. Try $L/D = 1.0$. Then $L = D = 75$ mm.

Step 5. Bearing pressure:

$$p = F/LD = (2500 \text{ N})/(75 \text{ mm})(75 \text{ mm}) = 0.444 \text{ N/mm}^2 = 0.444 \text{ MPa}$$

Step 6. Journal speed:

$$V = \pi Dn/(60\,000) = \pi(75)(1150)/(60\,000) = 4.52 \text{ m/s}$$

Step 7. pV factor:

$$pV = (0.444 \text{ MPa})(4.52 \text{ m/s}) = 2.008 \text{ MPa} \cdot \text{m/s}$$

Step 8. Design value for $pV = 2(2.008) = 4.016 \text{ MPa} \cdot \text{m/s}$.

Step 9. From Table 16–1, we can specify aluminum bronze (C95400) having a pV rating of $4.375 \text{ MPa} \cdot \text{m/s}$.

Steps 10–11. From Figure 16–4, we can recommend a minimum $C_d = 100 \mu\text{m}$ (0.100 mm or 0.004 in) based on $D = 75 \text{ mm}$ and 1150 rpm.

Alternate design: The pV factor for the initial design, while satisfactory, is somewhat high and may require careful lubrication. Consider the following alternate design having a larger bearing diameter.

Step 1. Try $D = 150 \text{ mm}$.

Step 2. Let $L/D = 1.25$.

Step 3. Then

$$L = D(L/D) = (150 \text{ mm})(1.25) = 187.5 \text{ mm}$$

Step 4. Let's use the more convenient value of 175 mm for L .

Step 5. Bearing pressure:

$$p = F/LD = (2500 \text{ N})/(175 \text{ mm})(150 \text{ mm}) = 0.095 \text{ N/mm}^2 = 0.095 \text{ MPa}$$

Step 6. Journal speed:

$$V = \pi Dn/(60\,000) = \pi(150)(1150)/(60\,000) = 9.03 \text{ m/s}$$

Step 7. pV factor:

$$pV = (0.095 \text{ MPa})(9.03 \text{ m/s}) = 0.860 \text{ MPa} \cdot \text{m/s}$$

Step 8. Design value of $pV = 2(0.860) = 1.720 \text{ MPa} \cdot \text{m/s}$.

Step 9. From Table 16–1, we can specify an oil-impregnated porous bronze bearing having a pV rating of $1.750 \text{ MPa} \cdot \text{m/s}$ or a BU dry-lubricated bearing having a pV rating of $1.785 \text{ MPa} \cdot \text{m/s}$.

Steps 10–11. From Figure 16–4, we can recommend a minimum $C_d = 150 \mu\text{m}$ (0.150 mm or 0.006 in) based on $D = 150 \text{ mm}$ and 1150 rpm. Other design details are dependent on the system into which the bearing will be placed.

Design Procedure for Boundary-Lubricated Bearings under Oscillating Loading

Many bearings of the boundary-lubricated type oscillate on stationary shafts rather than having continuously rotating shafts within the stationary bearings. Examples are bearings on steering shafts, suspension elements for vehicles, brake and accelerator pedals, seat position adjusters, shift mechanisms, door hinges, fitness exercising machines, hospital beds, hydraulic cylinder actuators, and hatch covers.

The design procedure is modified under these conditions. We can describe the oscillation by the following data:

φ = Angle of the oscillation in one direction, degrees

n_o = Number of complete oscillation cycles in both directions per minute, cycles/min

D = Nominal inside diameter of the bearing, in or mm

L = Length of the bearing, in or mm

F = Applied radial load, lb or N

Now we can define an equivalent number of revolutions/min for the bearing motion:

$$n_{eq} = n_o(2\varphi)/360 \text{ equivalent revolutions/min} \quad (16-2)$$

We can then substitute this value in previously developed equations for the relative velocity between the bearing and the shaft as follows.

U.S. Customary units; D in inches:

$$V = \pi D n_{eq} / 12 \text{ ft/min} \quad (16-3)$$

$$\text{SI units; } D \text{ in mm: } V = \pi D n_{eq} / (60\,000) \text{ m/s} \quad (16-4)$$

The pV value is then calculated in the same manner as for rotating shafts in bearings.

**Example Problem
16–3**

A design for an industrial oven door has a 15-mm-diameter horizontal bar over the oven opening on which the door hangs. Two smooth cylindrical door hinges are lined with the graphite metalized bearing material listed in Table 16–1. The door weighs 20.40 kN and rotates 110° from vertical to 20° above horizontal as the door is opened to insert or withdraw parts to be heated. It then returns to the original position. The door is opened and closed five times per minute. Determine the required length of the hinges to produce a pV value no more than 25% of the limiting value.

Solution From the given data:

$$\varphi = 110^\circ$$

$$n_o = 5.0 \text{ cycles/min}$$

$$D = 15 \text{ mm}$$

$$L = \text{Length of the bearing, in or mm}$$

$$F = (20.40 \text{ kN})/2 = 10.2 \text{ kN} = 10\,200 \text{ N on each hinge}$$

$$pV_{\text{limit}} = 0.525 \text{ MPa} \cdot \text{m/s}$$

The allowable pV value is

$$(pV)_{\text{all}} = pV_{\text{limit}} \times 0.25 = 0.525 \text{ MPa} \cdot \text{m/s} \times 0.25 = 0.1313 \text{ MPa} \cdot \text{m/s}$$

We can now compute the equivalent number of revolutions per minute from

$$n_{\text{eq}} = n_o(2\varphi)/360 = (5.0 \text{ cycles/min})(2)(110^\circ)/360^\circ = 3.06$$

Now the equivalent sliding velocity is

$$V_e = \pi D n_{\text{eq}} / (60\,000) \text{ m/s} = V = \pi(15)(3.06) / (60\,000) \text{ m/s} = 0.00240 \text{ m/s}$$

We can now solve for the limiting bearing pressure:

$$p_{\text{all}} = (pV)_{\text{all}} / V_e = (0.1313 \text{ MPa} \cdot \text{m/s}) / (0.00240 \text{ m/s}) = 54.69 \text{ MPa} = 54.69 \text{ N/mm}^2$$

Let $p = F/LD$, we can then solve for the required length as follows:

$$L = FpD = (10\,200 \text{ N}) / (54.69 \text{ N/mm}^2)(15 \text{ mm}) = 12.43 \text{ mm}$$

A preferred value of $L = 16 \text{ mm}$ would be reasonable.

Wear Considerations for Plain Boundary-Lubricated Bearings

Wear is an important consideration in boundary-lubricated bearings because of the inherent sliding action between the shaft and the bearing. Each manufacturer provides wear data unique to its materials. See Section 16–10 for additional discussion of wear.

16–6 FULL-FILM HYDRODYNAMIC BEARINGS

In the *full-film hydrodynamic bearing*, the load on the bearing is supported on a continuous film of lubricant, usually oil, so that no contact between the bearing and the rotating journal occurs. A pressure must be developed in the oil in order to carry the load. With proper design, the motion of the journal inside the bearing creates the necessary pressure.

Figure 16–5 shows the progressive action in a plain surface bearing from start-up to the steady-state hydrodynamic operation. Observe that boundary lubrication and mixed-film lubrication precede the establishment of

full-film hydrodynamic lubrication. At start-up, the radial load applied through the journal to the bearing forces the journal off center in the direction of the load, taking up all clearance [Figure 16–5(a)]. At the initial slow rotational speeds, the friction between the journal and the bearing causes the journal to climb up the bearing wall, somewhat as shown in Figure 16–5(b). Because of the viscous shear stresses developed in the oil, the moving journal draws oil into the converging, wedge-shaped area above the region of contact. The resulting pumping action produces a pressure in the oil film; when the pressure is sufficiently high, the journal is lifted from the bearing. The frictional forces are greatly reduced under this operating condition, and the journal moves eventually to the steady-state position, as shown in Figure 16–5(c). Note that the journal is offset from the direction of the load; that there is a certain eccentricity, e , between the geometric center of the bearing and the center of the journal; and that there is a point of minimum film thickness, h_o , at the nose of the wedge-shaped pressurized zone.

Figure 16–6 illustrates the general form of the pressure distribution within a full-film hydrodynamically lubricated bearing. The clearance between the bearing and the journal

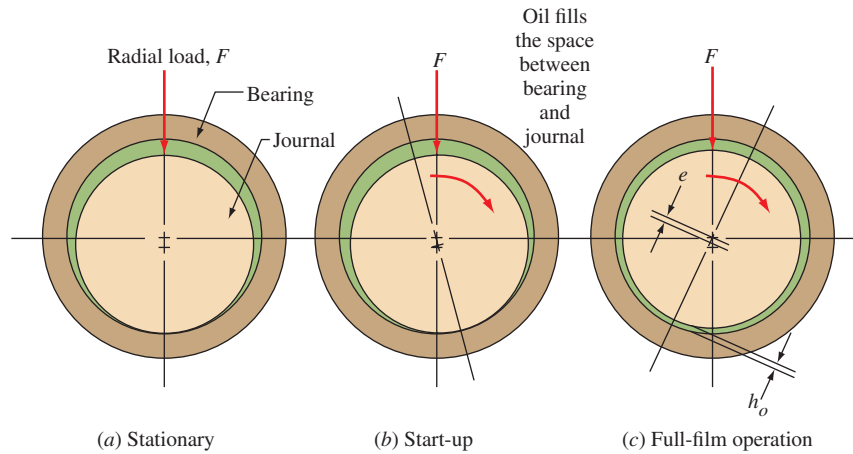


FIGURE 16-5 Position of the journal relative to the bearing as a function of mode of operation

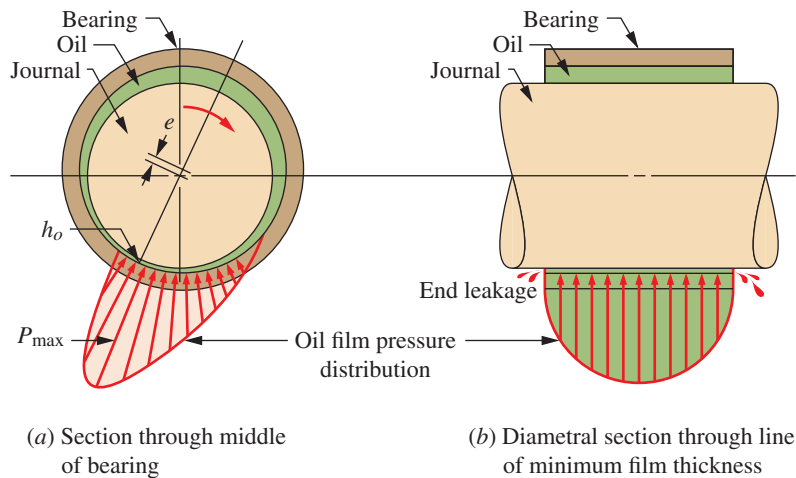


FIGURE 16-6 Pressure distribution in the oil film for hydrodynamic lubrication

is highly exaggerated. Part (a) of the figure shows the rise of pressure as the rotating shaft draws oil into the converging wedge, approaching the point of minimum film thickness. The maximum pressure occurs there and then falls rapidly to nearly zero as the space between the journal and the bearing diverges again. The integrated effect of the pressure distribution is a force sufficient to support the shaft on a film of oil without metal-to-metal contact.

Figure 16-6(b) shows the pressure distribution axially along the shaft through the line of minimum film thickness or maximum pressure. The highest pressure value occurs in the middle of the bearing length, and it falls rapidly as the ends are approached because the pressure outside the bearing is ambient pressure, typically atmospheric pressure. There is continual leakage flow from both ends of the bearing. This illustrates the importance of providing a means of continually supplying oil to the bearing to maintain the full-film operation. Without a steady, adequate supply of oil, the system would not be able to create the pressurized film to carry the shaft, and boundary lubrication would result. The significantly higher frictional forces thus created would

cause rapid heating of the bearing/journal interface, and seizure would likely occur very rapidly.

16-7 DESIGN OF FULL-FILM HYDRODYNAMICALLY LUBRICATED BEARINGS

The following discussion presents several guidelines for bearing design for typical industrial applications. The design procedure is based primarily on information in References 1–11.

Surface Roughness

A ground journal with a surface roughness average of 16 to 32 microinches (μin), or 0.40 to 0.80 μm , is recommended for bearings of good quality. The bearing should be equally smooth or made from one of the softer materials so that “wearing in” can smooth the high spots, creating a good fit between the journal and the bearing. In high-precision equipment, polishing or lapping can be used to produce a surface finish of the order of 8 to 16 μin (0.20 to 0.40 μm).

Minimum Film Thickness

The limiting acceptable value of the minimum film thickness depends on the surface roughness of the journal and the bearing because the film must be thick enough to eliminate any solid contact during expected operating conditions. The suggested design value also depends on the size of the journal. For ground journals, the following relationship can be used to estimate the design value:

$$h_o = 0.00025D \quad (16-5)$$

where D = diameter of the bearing.

Diametral Clearance

The clearance between the journal and the bearing depends on the nominal diameter of the bearing, the precision of the machine for which the bearing is being designed, the speed of rotation, and the surface roughness of the journal. The coefficient of thermal expansion for both the journal and the bearing must also be considered to ensure a satisfactory clearance under all expected operating conditions. An overall guideline of making the clearance in the range of 0.001 to 0.002 times the bearing diameter can be used. Figure 16-4 shows a graph of the recommended minimum diametral clearance as a function

of the bearing diameter and the rotational speed. Some variation above the values from the curve is permissible.

Ratio of Bearing Length to Diameter

Because the journal is a part of the shaft itself, its minimum diameter is usually limited by stress and deflection considerations of the type discussed in Chapter 12. Then the length of the bearing is specified to provide for a suitable level of bearing pressure. General-purpose industrial machinery bearings typically operate at approximately 200 to 500 psi (1.4 to 3.4 MPa) bearing pressure, based on the projected area of the bearing [$p = \text{load}/(LD)$]. The pressure may range from as low as 50 psi (0.34 MPa) for light-duty equipment to 2000 psi (13.4 MPa) for heavy machinery under varying loads, such as in internal combustion engines. The leakage of oil from the bearing is also affected by the bearing length. The typical range of length to diameter ratio (L/D) for full-film hydrodynamic bearings is from 0.35 to 1.5. But many successful bearings operate outside this range.

Lubricant Temperature

The viscosity of the oil is a critical parameter in the performance of a bearing. Figure 16-7 shows the great

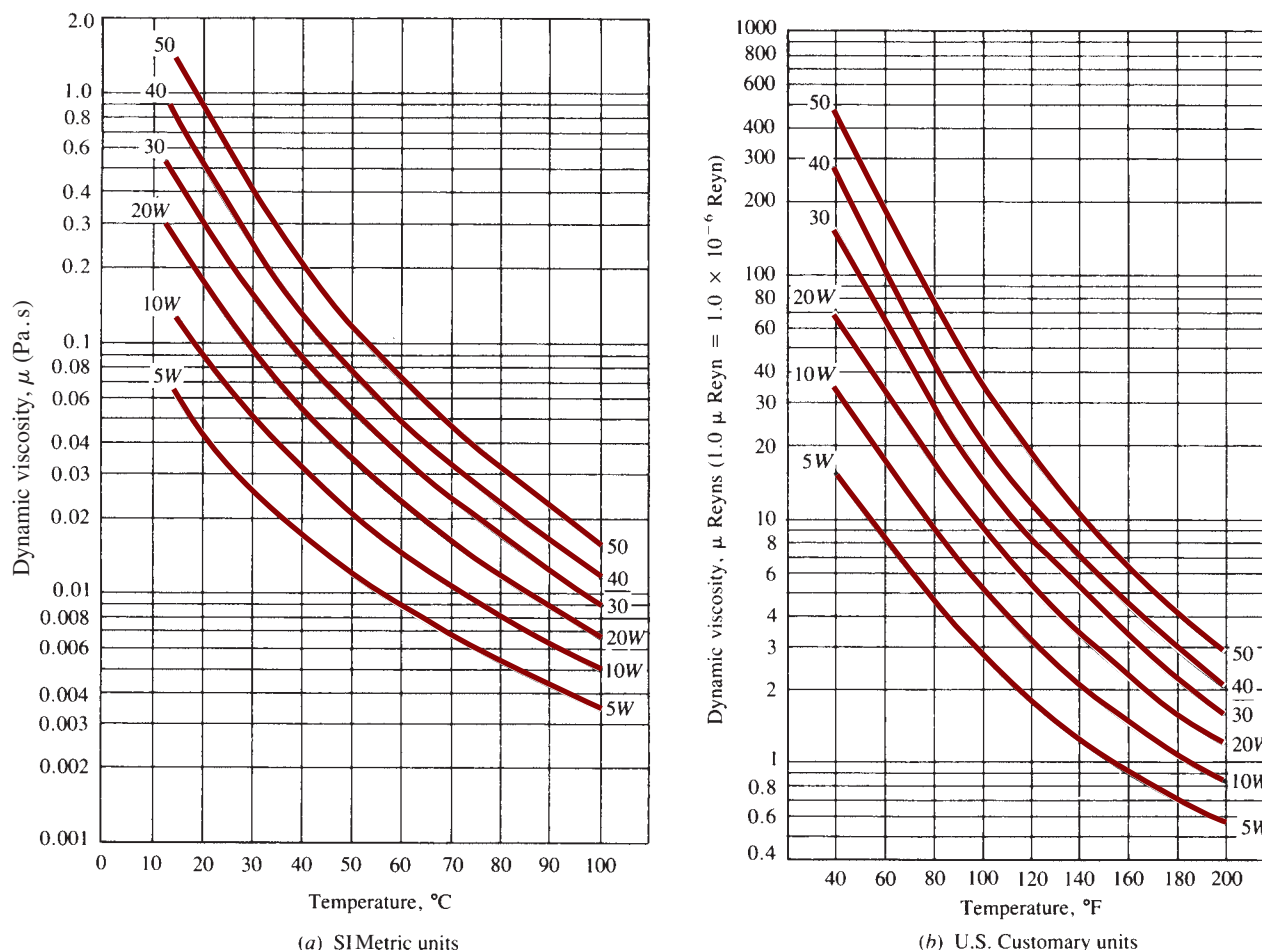


FIGURE 16-7 Viscosity versus temperature for SAE oils

variation of viscosity with temperature, which indicates that temperature control is advisable. Note that the vertical viscosity axis has a logarithmic scale. Also, most petroleum lubricating oils should be limited to approximately 160°F (70°C) in order to retard oxidation. The temperature of interest is, of course, that inside the bearing. Frictional energy or thermal energy from the equipment itself can increase the oil temperature above that in the supply reservoir. In design examples we will select the lubricant that will ensure a satisfactory viscosity at 160°F, unless otherwise noted. If the actual operating temperature is lower, the resulting film thickness will be greater than the design value—a conservative result. It is incumbent on the designer to ensure that the limiting temperature is not exceeded, using forced cooling if necessary.

Lubricant Viscosity

The specification of the lubricant for the bearing is one of the final decisions to be made in the design procedure that follows. In the calculations, it is the dynamic viscosity, μ , which is used. In the U.S. Customary Unit (lb-in-s) System, the dynamic viscosity is expressed in $\text{lb} \cdot \text{s}/\text{in}^2$, which is given the name *reyn* in honor of Osborne Reynolds, who performed much significant work in fluid flow. In SI units, the standard unit is $\text{N} \cdot \text{s}/\text{m}^2$ or $\text{Pa} \cdot \text{s}$. Some prefer the unit of poise or centipoise, derived from the dyne-cm-s metric system. Some useful conversions are

$$1.0 \text{ reyn} = 6895 \text{ Pa} \cdot \text{s}$$

$$1.0 \text{ Pa} \cdot \text{s} = 1000 \text{ centipoise}$$

Other viscosity conversions may also be useful. (See Reference 18.) Figure 16–7 shows graphs of dynamic viscosity versus temperature for both SI and U.S. Customary

units. Note in Figure 16–7(b) that common values in U.S. Customary units are very small. Values from the scale must be multiplied by 10^{-6} .

Sommerfeld Number

The combined effect of many of the variables involved in the operation of a bearing under hydrodynamic lubrication can be characterized by the dimensionless number, S , called the *Sommerfeld number*. Some refer to this as the *bearing characteristic number*, which is defined as follows:

➤ Bearing Characteristic Number

$$S = \frac{\mu n_s (R/C_r)^2}{p} \quad (16-6)$$

Note that S is similar to the bearing parameter, $\mu n/p$, discussed in Section 16–3, in that it involves the combined effect of viscosity, rotational speed, and bearing pressure. In order for S to be dimensionless, the following units should be used for the factors:

	U.S. Customary Units	SI Units
μ	$\text{lb} \cdot \text{s}/\text{in}^2$ (reyns)	$\text{Pa} \cdot \text{s}$ ($\text{N} \cdot \text{s}/\text{m}^2$)
n_s	rev/s	rev/s
p	lb/in^2 (psi)	Pa (N/m^2)
R, C_r	in	m or mm

Any consistent units can be used. Figure 16–8, adapted from Reference 4, shows the relationship between the Sommerfeld number and the film thickness

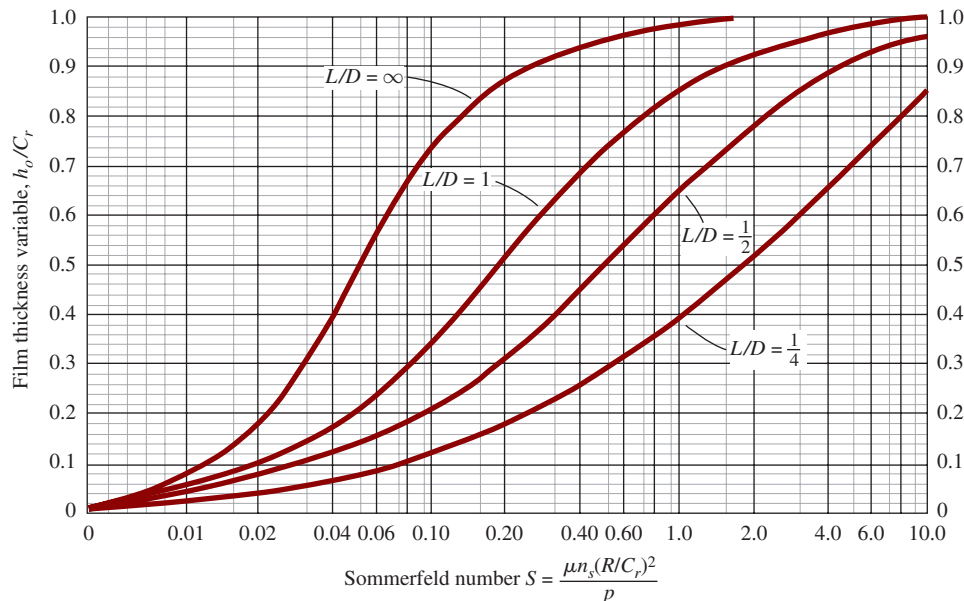


FIGURE 16–8 Film thickness variable, h_o/C_r , versus Sommerfeld number, S

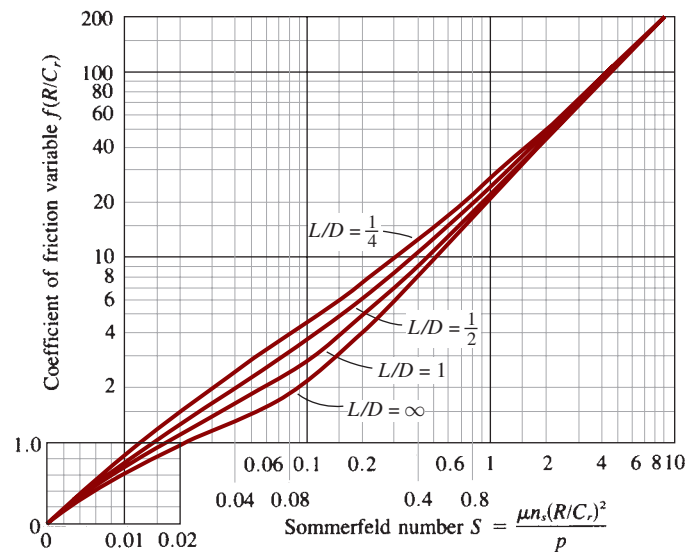


FIGURE 16-9 Coefficient of friction variable, $f(R/C_r)$, versus Sommerfeld number, S

ratio, h_0/C_r . Figure 16-9 shows the relationship between S and the coefficient of friction variable, $f(R/C_r)$. These values are used in the design procedure that follows. Because many design decisions are required, several acceptable solutions are possible.

Design Procedure

PROCEDURE FOR DESIGNING FULL-FILM HYDRODYNAMICALLY LUBRICATED BEARINGS ▼

Because the bearing design is usually done after the stress analysis of the shaft is completed, the following items are typically known:

Radial load on the bearing, F , usually in lb or N

Rotational speed, n , usually in rpm

Nominal shaft diameter at the journal, sometimes specified as a minimum acceptable diameter based on strength and stiffness

The results of the design procedure produce values for the actual journal diameter, the bearing length, the diametral clearance, the minimum film thickness of the lubricant during operation, the surface finish for the journal, the lubricant and its maximum operating temperature, the coefficient of friction, the friction torque, and the power dissipated because of friction.

1. Specify a trial value for the journal diameter, D , and the radius, $R = D/2$.
2. Specify a nominal operating bearing pressure, usually 200 to 500 psi (1.4 to 3.4 MPa), where $p = F/LD$. Solve for L :

$$L = F/pD$$

Then compute L/D . It might be desirable to redefine L/D to be a convenient value from 0.25 to 1.5 in order to use the design charts available. Finally, specify the actual design value of L/D and L , and compute the actual $p = F/LD$.

3. Referring to Figure 16-4, specify the diametral clearance, C_d , based on the values for D and n . Then compute $C_r = C_d/2$ and the ratio R/C_r .
4. Specify the desired surface finish for the journal and bearing, also based on the application. a typical value is 16 to 32 μin average (0.40 to 0.80 μm).
5. Compute the nominal minimum film thickness from Equation (16-5),

$$h_0 = 0.00025D$$

6. Compute h_0/C_r , the film thickness ratio.
7. From Figure 16-8, determine the value of the Sommerfeld number for the selected film thickness ratio and the L/D ratio. Exercise care in interpolating in this chart because of the logarithmic axis and the nonlinear spread between curves. For $L/D > 1$, only approximate data can be obtained. For $L/D = 1.5$, interpolate approximately one-fourth of the way between the curves for $L/D = 1$ and $L/D = \text{infinity}$. For $L/D = 2$, go about halfway.
8. Compute the rotational speed n_s in revolutions per second:

$$n_s = n/60$$

where n is in rpm.

9. Because each factor of the Sommerfeld number is known except the lubricant viscosity, μ , solve for the required minimum viscosity that will produce the desired minimum film thickness:

➤ Required Minimum Lubricant Viscosity

$$\mu = \frac{Sp}{n_s(R/C_r)^2} \quad (16-7)$$

10. Specify a maximum acceptable lubricant temperature, usually 160°F or 70°C. Select a lubricant from Figure 16–7 that will have at least the required viscosity at the operating temperature. If the selected lubricant has a viscosity greater than that computed in Step 9, recompute S for the new value of viscosity. The resulting value for the minimum film thickness will now be somewhat greater than the design value, a generally desirable result. You can consult Figure 16–8 again to determine the new value of the minimum film thickness, if desired.
11. From Figure 16–9, obtain the coefficient of friction variable, $f(R/C_r)$.
12. Compute $f = f(R/C_r)/(R/C_r) =$ coefficient of friction.
13. Compute the friction torque. The product of the coefficient of friction and the load F gives the frictional force

at the surface of the journal. That value multiplied by the radius gives the torque:

➤ **Friction Torque**

$$T_f = F_f R = fFR \quad (16-8)$$

14. Compute the power dissipated in the bearing from the relationship between power, torque, and speed that we have used many times:

➤ **Frictional Power Loss**

$$P_f = T_f n / 63\,000 \text{ hp} \quad (16-9)$$

This frictional power loss represents the rate of energy input to the lubricant within the bearing that can cause the temperature to rise. It is a part of the energy that must be removed from the bearing to maintain satisfactory lubricant viscosity.

Example Problem 16–4 illustrates the procedure.

Example Problem 16–4

Design a plain surface journal bearing to carry a steady radial load of 1500 lb while the shaft rotates at 850 rpm. The shaft stress analysis determines that the minimum acceptable diameter at the journal is 2.10 in. The shaft is part of a machine requiring good precision.

Solution

Step 1. Let's select $D = 2.50$ in. Then $R = 1.25$ in.

Step 2. For $p = 200$ psi, L must be

$$L = F/pD = 1500/(200)(2.50) = 3.00 \text{ in}$$

For this value of L , $L/D = 3.00/2.50 = 1.20$. In order to use one of the standard design charts, let's change L to 2.50 in so that $L/D = 1.0$. This is not essential, but it eliminates interpolation. The actual pressure is then

$$p = F/LD = 1500/(2.50)(2.50) = 240 \text{ psi}$$

This is an acceptable pressure.

Step 3. From Figure 16–4, $C_d = 0.003$ in is suitable for the diametral clearance, based on $D = 2.50$ in and $n = 850$ rpm, and $C_r = C_d/2 = 0.0015$ in. Also,

$$R/C_r = 1.25/0.0015 = 833$$

This value is used in later calculations.

Step 4. For the precision desired in this machine, use a surface finish of 16 to 32 μin , requiring a ground journal.

Step 5. Minimum film thickness (design value):

$$h_o = 0.00025D = 0.00025(2.50) = 0.0006 \text{ in (approx.)}$$

Step 6. Film thickness variable:

$$h_o/C_r = 0.0006/0.0015 = 0.40$$

Step 7. From Figure 16–8, for $h_o/C_r = 0.40$ and $L/D = 1$, we can read $S = 0.13$.

Step 8. Rotational speed in revolutions per second:

$$n_s = n/60 = 850/60 = 14.2 \text{ rev/s}$$

Step 9. Solve for the viscosity from the Sommerfeld number, S :

$$\mu = \frac{Sp}{n_s(R/C_r)^2} = \frac{(0.13)(240)}{(14.2)(833)^2} = 3.17 \times 10^{-6} \text{ reyn}$$

Step 10. From the viscosity chart, Figure 16–7, SAE 30 oil is required to ensure a sufficient viscosity at 160°F. The actual expected viscosity of SAE 30 at 160°F is approximately 3.3×10^{-6} reyn.

Step 11. For the actual viscosity, the Sommerfeld number would be

$$S = \frac{\mu n_s(R/C_r)^2}{p} = \frac{(3.3 \times 10^{-6})(14.2)(833)^2}{240} = 0.135$$

Step 12. Coefficient of friction variable (from Figure 16–9): $f(R/C_r) = 3.5$ for $S = 0.135$ and $L/D = 1$. Now, because $R/C_r = 833$,

$$f = 3.5/833 = 0.0042$$

Step 13. Friction torque:

$$T_f = fFR = (0.0042)(1500)(1.25) = 7.88 \text{ lb} \cdot \text{in}$$

Step 14. Frictional power:

$$P_f = T_f \omega / 63\,000 = (7.88)(850) / 63\,000 = 0.106 \text{ hp}$$

Comment A qualitative evaluation of the result would require more knowledge about the application. But note that a coefficient of friction of 0.0042 is quite low. It is likely that a machine requiring such a large shaft and with such high bearing forces also requires a large power to drive it. Then the 0.106-hp friction power would appear small.

Thermal considerations are also important to determine how much energy must be dissipated from the bearing. Converting the frictional power loss to thermal power gives

$$P_f = 0.106 \text{ hp} \frac{745.7 \text{ W}}{\text{hp}} = 79.0 \text{ W}$$

Expressing this in U.S. Customary units gives

$$P_f = 79.0 \text{ W} \frac{1 \text{ Btu/h}}{0.293 \text{ W}} = 270 \text{ Btu/h}$$

16-8 PRACTICAL CONSIDERATIONS FOR PLAIN SURFACE BEARINGS

The design of the bearing system must consider the method of delivery of the lubricant to the bearing, the distribution of the lubricant within the bearing, the quantity of lubricant required, the amount of heat generated in the bearing and its effect on the temperature of the lubricant, the dissipation of heat from the bearing, the maintenance of the lubricant in a clean condition, and the performance of the bearing over the complete range of operating conditions that the bearing is likely to experience.

Many of these factors are simply design details that must be worked out along with the other aspects of the machine design. But some guidelines and general recommendations will be presented here.

The lubricant can be delivered to the bearing by a pump, perhaps driven from the same source that drives the entire machine. In some gear drives, one of the gears is designed to dip into an oil sump and carry oil up to the gear mesh and to the bearings. An external oil cup can be used to supply oil by gravity if the lubricant quantity required is small.

Methods of estimating the quantity of oil required, considering the leakage of oil from the ends of the bearing, are available. (See References 1, 4, 7, 10, 11, 13, and 15.)

The delivery of oil to the bearing should always be in an area opposite the location of the hydrodynamic pressure that supports the load. Otherwise, the oil delivery hole would destroy the pressurization of the film.

Grooving is frequently used to distribute the oil along the length of the bearing. Oil would be delivered through a radial hole in the bearing at the midlength point. The groove would extend axially in both directions from the hole, but it would terminate somewhat

before the end of the bearing in order to keep the oil from leaking to the side. The rotation of the journal then carries the oil around to the area where the hydrodynamic film is generated. Figure 16–10 shows several styles of grooving in use.

Cooling of the bearing itself or of the oil in the sump that supplies the oil must always be considered. Natural convection may be sufficient to transfer heat away and maintain an acceptable bearing temperature. If not, forced convection can be used. In severe cases of heat generation, especially where the bearing system operates in a hot area such as a furnace, liquid coolant can be pumped through a jacket around the bearing. Some commercially available bearings provide this feature. Placing a heat exchanger in the oil sump or pumping the oil through an external heat exchanger can also be done.

Figure 16–11 shows one commercially available style of a plain bearing with coolant pipes to allow use of water, air, or oil as the internal coolant.

The lubricant can be cleaned by passing it through filters as it is pumped to the bearing. Magnetic plugs within the sump can be effective in attracting and holding metallic particles that can score the bearing if allowed to enter the clearance space between the journal and the bearing. Of course, frequent changing of the oil is also desirable.

The design procedure used in the preceding section was completed for one set of conditions: a given temperature, diametral clearance, load, and rotational speed. If any of these factors varies during the operation of the machine, the performance of the bearing must be evaluated under the new conditions. The testing of a prototype under a variety of conditions is also desirable. References 1–8,



FIGURE 16–10 Examples of styles of grooving for plain bearings

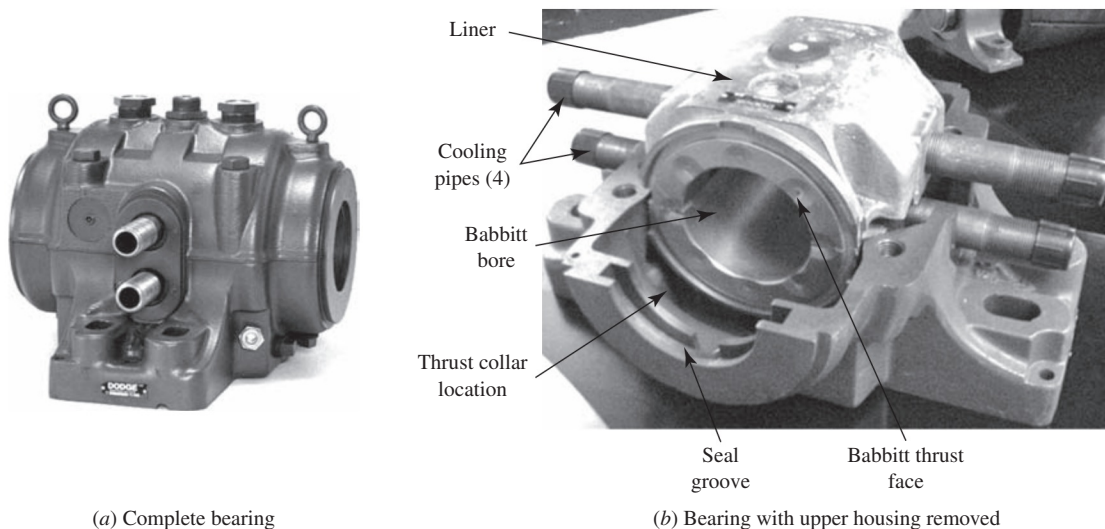


FIGURE 16–11 Sleeveoil hydrodynamically lubricated bearing with cooling pipes (Baldor/Dodge, Greenville, SC)

13–16, 19, and 20 discuss other practical considerations for the design of plain surface bearings.

16-9 HYDROSTATIC BEARINGS

Recall that hydrodynamic lubrication results from the creation of a pressurized film of oil sufficient to carry the load on the bearing, with the film being generated by the motion of the journal itself within the bearing. It was noted that a steady relative motion between the journal and the bearing is required to generate and maintain the film.

In some types of equipment, the conditions are such that a hydrodynamic film cannot be developed. Reciprocating or oscillating devices or very-slow-moving machines are examples. If the load on the bearing is very high, it may not be possible to generate a sufficiently high pressure in the film to support the load. Even in cases in which hydrodynamic lubrication can be developed during normal operation of the machine, there is still mixed-film or boundary lubrication during the start-up and shut-down cycles. This may be unacceptable.

Consider the design of the mount for a telescope or an antenna system in which rotation of the base is required to be at very slow speed and to have smooth motion. Also, low friction is desired in order to keep the drive system small and to provide fast response and accurate positioning. The mount is essentially a thrust bearing supporting the weight of the system.

In this type of application, *hydrostatic lubrication* is desirable. Lubricant is supplied to the bearing under high pressure, several hundred psi or higher, and the pressure acting on the area of the bearing literally lifts the load off the bearing, even with the equipment stationary.

Figure 16–12 shows the main elements of a hydrostatic bearing system. A positive displacement pump

draws oil from a reservoir and delivers it under pressure to a supply manifold from which several bearing pads can be supplied. At each bearing pad, the oil passes through a control element that permits the balancing of the system. The control element may be a flow-control valve, a length of small-diameter tubing, or an orifice, any of which offers a resistance to the flow of oil and permits the several bearing pads to operate at a sufficiently high pressure to lift the load on each pad. When the system is in operation, the oil enters a recess within the bearing pad. For example, Figure 16–12(b) shows a circular pad with a circular recess in its center supplied with oil through a central hole. The load initially rests on the land area, sealing the recess. As the pressure in the recess, p_r , reaches the level where the product of the pressure times the recess area equals the applied load, the load is lifted from the pad. Immediately there is a flow of oil across the land area under the lifted load, and the pressure decreases to atmospheric pressure at the outside of the pad. The flow of oil must be maintained at a level that matches the outflow from the pad. When equilibrium occurs, the integrated product of the local pressure times the area lifts the load a certain distance, h , usually in the range of 0.001 to 0.010 in (0.025 to 0.25 mm). The film thickness, h , must be large enough to ensure no solid contact over the range of operating conditions, but it should be kept as small as possible to minimize the flow of oil through each bearing and the pump power required to drive the system.

Hydrostatic Bearing Performance

Three factors characterizing the performance of a hydrostatic bearing are its load-carrying capacity, the flow of oil required, and the pumping power required, as indicated by the dimensionless coefficients a_f , q_f , and H_f . The magnitudes of the coefficients depend on the design of the pad:

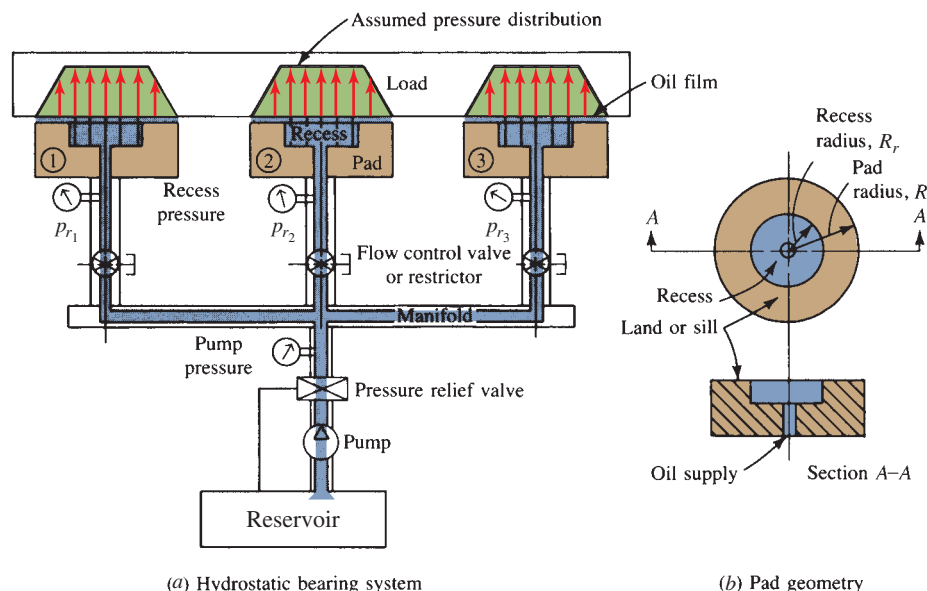


FIGURE 16–12 Main elements of a hydrostatic bearing system

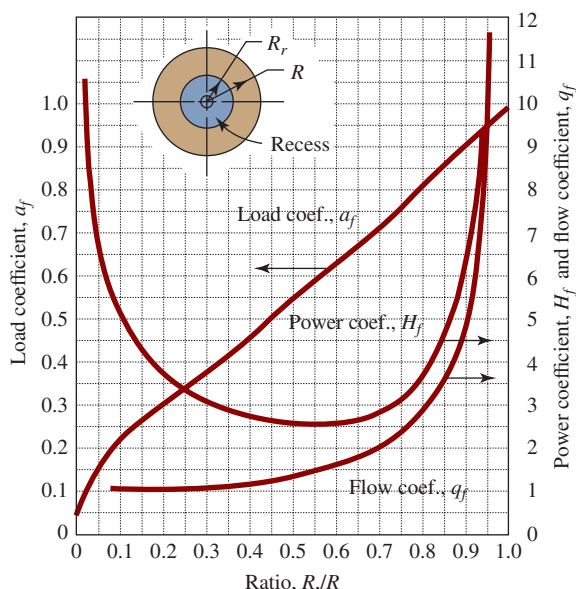


FIGURE 16-13 Dimensionless performance coefficients for circular pad hydrostatic bearing

Load-Carrying Capacity

$$F = a_f A_p p_r \quad (16-10)$$

h = film thickness, in or m

μ = dynamic viscosity of the oil, $\text{lb} \cdot \text{s}/\text{in}^2$ (reyn) or $\text{Pa} \cdot \text{s}$

Oil Flow Required

$$Q = q_f \frac{F}{A_p} \frac{h^3}{\mu} \quad (16-11)$$

Pumping Power Required

$$P = p_r Q = H_f \left(\frac{F}{A_p} \right)^2 \frac{h^3}{\mu} \quad (16-12)$$

where F = load on the bearing, lb or N

Q = volume flow rate of oil, in^3/s or m^3/s

P = pumping power, $\text{lb} \cdot \text{in}/\text{s}$ or $\text{N} \cdot \text{m}/\text{s}$ (watts)

a_f = pad load coefficient, dimensionless

q_f = pad flow coefficient, dimensionless

H_f = pad power coefficient, dimensionless
(Note: $H_f = q_f/a_f$)

A_p = pad area, in^2 or m^2

p_r = oil pressure in the recess of the pad, psi or Pa

Figure 16-13 shows the typical variation of the dimensionless coefficients as a function of the pad geometry for a circular pad with a circular recess. As the size of the recess, R_r/R , increases, the load-carrying capacity increases, as indicated by a_f . But at the same time, the flow through the bearing increases, as indicated by q_f . The increase is gradual up to a value of R_r/R of approximately 0.7, and then rapid for higher ratios. This higher flow rate requires much higher pumping power, as indicated by the rapidly increasing power coefficient. At very low ratios of R_r/R , the load coefficient decreases rapidly. The pressure in the recess would have to increase to compensate in order to lift the load. The higher pressure requires more pumping power. Therefore, the power coefficient is high either at very small ratios of R_r/R or at high ratios. The minimum power is required for ratios between 0.4 and 0.6.

These general characteristics of hydrostatic bearing pad performance are typical for many different geometries of pads. Extensive data for the performance of different pad shapes are published. (See Reference 12.)

Example Problem 16-5 illustrates the basic design procedure for hydrostatic bearings.

Example Problem 16-5

A large antenna mount weighing 12 000 lb is to be supported on three hydrostatic bearings such that each bearing pad carries 4000 lb. A positive displacement pump will be used to deliver oil at a pressure up to 500 psi. Design the hydrostatic bearings.

Solution

We will choose the circular pad design for which the performance coefficients are available in Figure 16-13. The results of the design will specify the dimensions of the pads, the required oil

pressure in the recess of each pad, the type of oil required and its temperature, the thickness of the film of oil when the bearings are supporting the load, the flow rate of oil required, and the pumping power required.

Step 1. From Figure 16–13, the minimum power required for a circular pad bearing would occur with the ratio R_f/R of approximately 0.50. For that ratio, the value of the load coefficient is $a_f = 0.55$. The pressure at the bearing recess will be somewhat below the maximum available of 500 psi because of the pressure drop in the restrictor placed between the supply manifold and the pad. Let's design for a recess pressure of approximately 400 psi. Then, from Equation (16–10),

$$A_p = \frac{F}{a_f p_r} = \frac{4000 \text{ lb}}{0.55(400 \text{ lb/in}^2)} = 18.2 \text{ in}^2$$

But $A_p = \pi D^2/4$. Then the required pad diameter is

$$D = \sqrt{4A_p/\pi} = \sqrt{4(18.2)/\pi} = 4.81 \text{ in}$$

For convenience, let's specify $D = 5.00 \text{ in}$. Then the actual pad area will be

$$A_p = \pi D^2/4 = (\pi)(5.00 \text{ in})^2/4 = 19.6 \text{ in}^2$$

The required recess pressure is then

$$p_r = \frac{F}{a_f A_p} = \frac{4000 \text{ lb}}{0.55(19.6 \text{ in}^2)} = 370 \text{ lb/in}^2$$

Also,

$$R = D/2 = 5.00 \text{ in}/2 = 2.50 \text{ in}$$

$$R_f = 0.50R = 0.50(2.50 \text{ in}) = 1.25 \text{ in}$$

Step 2. Specify the design value of the film thickness, h . It is recommended that h be between 0.001 and 0.010 in. Let's use $h = 0.005 \text{ in}$.

Step 3. Specify the lubricant and the operating temperature. Let's select SAE 30 oil and assume that the maximum oil temperature in the oil film will be 120°F. A method of estimating the actual film temperature during operation can be found by consulting Reference 12. From the viscosity/temperature curves, Figure 16–7, the viscosity is approximately $8.3 \times 10^{-6} \text{ reyn (lb} \cdot \text{s/in}^2\text{)}$.

Step 4. Compute the oil flow through the bearing from Equation (16–11). The value of $q_f = 1.4$ can be found from Figure 16–13:

$$Q = q_f \frac{F}{A_p} \frac{h^3}{\mu} = (1.4) \frac{4000 \text{ lb}}{19.6 \text{ in}^2} \frac{(0.005 \text{ in})^3}{8.3 \times 10^{-6} \text{ lb} \cdot \text{s/in}^2}$$

$$Q = 4.30 \text{ in}^3/\text{s}$$

Step 5. Compute the pumping power required from Equation (16–12). The value of $H_f = 2.6$ can be found from Figure 16–13:

$$P = p_r Q = H_f \left(\frac{F}{A_p} \right) \frac{h^3}{\mu} = 2.6 \left(\frac{4000}{19.6} \right) \frac{(0.005 \text{ in})^3}{8.3 \times 10^{-6}} = 1631 \text{ lb} \cdot \text{in/s}$$

For convenience, we can convert this to horsepower:

$$P = \frac{1631 \text{ lb} \cdot \text{in}}{\text{s}} \frac{1.0 \text{ ft}}{12 \text{ in}} \frac{1.0 \text{ hp}}{550 \text{ lb} \cdot \text{ft/s}} = 0.247 \text{ hp}$$

16-10 THE KUGEL FOUNTAIN—A SPECIAL EXAMPLE OF A HYDROSTATIC BEARING

Have you ever tried to push the Earth to make it rotate faster or tried to stop it when it was rotating? Of course, that is physically impossible for us humans. However, many simulations of this feat are on exhibit around the United States and in other parts of the world where different examples of the so-called *Kugel Fountain* are installed. Most are located at science museums or other public places.

Typically, the Kugel Fountain is a very large and heavy granite sphere placed on a shallow base with the upper part of the base made to match the spherical shape of the sphere. Water at a moderate pressure is pumped into the base and it is able to lift the heavy sphere slightly to allow a small amount of water to flow out between the sphere and the base. The sphere and the base must be accurately machined and mated because when the sphere is lifted, the clearance between it and the base is very small, not unlike the examples of hydrostatic bearings described in Section 16-9.

With the water flowing out between the sphere and the base, it is possible for visitors to push on the sphere and make it rotate in any direction. Many public examples of the Kugel Fountain have the image of Earth's land masses etched into the surface to give the visitors the feeling that they are moving the Earth!

Use an Internet search on the topic, *Kugel Fountain*, and you can find many photographs and videos from many locations. Perhaps the largest example is at the Science Museum of Virginia, located in the Virginia state capitol, Richmond. The granite sphere is over 8 ft in diameter (about 2.5 m) and it has a mass of over 59 000 lb_m (almost 27 000 kg). Yet an adult can, with quite an effort to overcome inertia, set the sphere in

motion. When the pushing force is stopped, the sphere will continue to rotate for a very long time because there is no physical contact between the sphere and its base and the moving water provides lubrication in the form of a thin pressurized film.

Figure 16-14 shows an image of a Kugel Fountain along with a schematic drawing of its features. In part (b) of the figure, the following descriptions apply:

- A. Pressurized water flows from a pump to the interface between the ball and the base. The pressure is moderate but sufficient to raise the ball above the base and to allow the water to flow outward. Water pressure decreases as it flows out to edge of the base.
- B. The water exits at atmospheric pressure and falls into an area around the base.
- C. The water returns through tubing back to the inlet of the pump.
- D. The pump increases the pressure in the fluid and delivers it to the underside of the ball to complete the circuit.

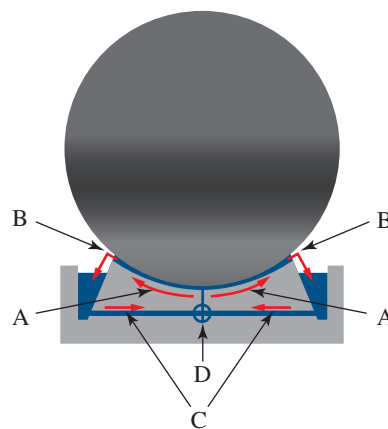
16-11 TRIBOLOGY: FRICTION, LUBRICATION, AND WEAR

The study of friction, lubrication, and wear, called *tribology*, involves many disciplines, such as solid mechanics, fluid mechanics, material science, and chemistry. It is appropriate to engage several specialists on design teams where critical bearing or general lubrication designs are completed. For study beyond the scope of this book, see References 1-3, 6-8, 13-16, 19, and 20.

This section presents some general principles of lubrication that can apply to a variety of design situations where relative motion occurs between mating machine elements. The goal is to help you recognize the many parameters that must be considered in designing machinery and in



(a) A typical Kugel Ball Fountain



(b) Cross section of a Kugel Ball

FIGURE 16-14 Example of a Kugel Ball and the fluid flow within it

analyzing failures or unsatisfactory operation of existing machines. Much of the discussion relates to minimizing or controlling *friction*, defined generally as the resistance to parallel motion of mating surfaces. *Lubrication* is used to minimize friction by introducing a film of material that inherently reduces the force required to move one component relative to its mating component. Some materials have inherently low friction coefficients and can operate satisfactorily without external lubrication. When the relative motion results in physical contact between the surfaces of the mating components, some of the surface material can be removed, resulting in *wear*.

Friction

Not all friction is undesirable. Consider the need for drive wheels to use friction to develop propulsive force against floors, rails, or roads. Clutches and brakes employ friction to engage machinery, accelerate it, decelerate it, bring it to a stop, or hold it in position. See Chapter 22. Clamps and collets use friction to hold work pieces during manufacturing operations. In such applications, large, consistent frictional forces are desirable.

Most other applications where sliding contact occurs between mating components call for minimizing friction in order to minimize the forces, torques, and power required to drive the system. We are most concerned with these applications in this section.

The primary phenomena involved in creating friction are adhesion, elastic effects such as rolling resistance, viscoelastic effects, and hydrodynamic resistance. *Adhesion* is the bonding between dissimilar materials. The strength of adhesion depends on the structure and chemistry of the mating materials. Surface characteristics also contribute, such as the height of roughness peaks and valleys, called *asperities*. Sometimes asperities on mating parts are deformed or fractured during relative motion, while for other conditions, the motion is resisted as the asperities ride up and down on one another. *Rolling resistance* is caused by the elastic deformation of the moving body or the surface on which it moves. The geometry of the members in rolling contact, the magnitude of applied forces, and the elasticity of the contacting materials all play a part in determining the amount of resistance. *Viscoelastic effects* relate to the forces caused by deformation of flexible materials, such as elastomers, during contact. *Hydrodynamic resistance*, also called the *viscous effect*, is caused by the relative motion among the molecules of fluid lubricants between the moving mating components. This is the primary form of resistance in full-film hydrodynamically lubricated bearings. All or many of these forms of friction exist simultaneously in most practical machines.

Lubricants

Some of the important functions of lubricants are to reduce friction, remove heat from bearings and other machine elements where friction occurs, and to suspend

contaminants. Several properties contribute to satisfactory lubricant performance:

1. Good lubricity or oiliness to promote low friction
2. Adequate viscosity for the application
3. Low volatility under operating conditions
4. Satisfactory flow characteristics at the temperatures encountered in use
5. Appropriate thermal conductivity and specific heat to perform the heat transfer function
6. Good chemical and thermal stability and the ability to maintain desirable characteristics for a reasonable period of use
7. Compatibility with other materials in the system such as bearings, seals, and machine parts, particularly with regard to corrosion protection and degradation
8. Environmental friendliness

Here we discuss the basic nature of oils, greases, and solid lubricants.

Oils. Vendors of lubricating oils offer a huge variety of grades. One general classification is between refined natural petroleum oil and synthetic lubricants. Typically, natural oils are lower cost and can provide satisfactory service for general-purpose lubrication. Additives are frequently compounded with the natural oil to enhance viscosity, improve viscosity index (decrease the variation of viscosity with temperature), reduce corrosion potential, retard oxidation or other forms of chemical breakdown, or increase the ability to withstand localized high pressure. Synthetic lubricants are specially designed chemical formulations that can be tailored to specific applications. While performance is better than natural oils, cost is typically higher.

Oil lubricants are offered in categories such as engine oils, gear oils, compressor oils, turbine oils, general-purpose lubricating oils, chain lubricants, bearing oils, food machinery lubricants, automatic transmission fluids, hydraulic oils, and metalworking fluids. Internet sites 11–13 provide good examples of the types of lubricants available. Properties of lubricants that can affect selection are viscosity grade, viscosity index, and corrosion protection.

Viscosity grades are typically reported using the ISO (International Organization for Standardization) rating system. The grade number is the kinematic viscosity of the oil in centistokes (mm^2/s) measured at 40°C (104°F). Common ISO grades for lubricants are 32, 46, 68, 100, 150, 220, 320, 460, 680, and 1000. Gear oils are typically grades 150 to 680, depending on environmental temperature and degree of loading. Often the viscosity will also be reported at 100°C (212°F) as an indication of the variation of viscosity with temperature. This is not a part of the ISO classification and is influenced primarily by the property of viscosity index, discussed later in this section.

The American Gear Manufacturers Association (AGMA) formerly defined lubricant numbers from 0 through 15, with numbers 3 to 8 being the most common for general power-transmission uses. The basic viscosities for the former AGMA grades 0 to 8 correlate with ISO viscosity grades as follows.

AGMA	ISO	AGMA	ISO	AGMA	ISO
0	32	3	100	6	320
1	46	4	150	7	460
2	68	5	220	8	680

Suffixes are added to the AGMA grade designations: EP for extreme pressure, CP for compounded oils, and S for synthetic oils. See References 10, 16, and 26–28 in Chapter 9. See also Section 9–15.

SAE viscosity grades are also used to report oil viscosity. Similar to the grades used in automotive engines, SAE oils are also good for general lubrication and for gear-type power transmissions. Common grades are SAE 20, 30, 40, 50, 60, 85, 90, 140, and 250. These grades must conform to limits of kinematic viscosity in centistokes measured at 100°C (212°F). The W-grades of SAE oils, such as SAE 20W, must have viscosities less than specified limits at low temperatures ranging from –5°C to –55°C (–3°F to –67°F). See Reference 18. This ensures that the lubricant can flow to critical surfaces in cold environments, particularly during equipment startup. Gear oils are typically SAE grades 80 to 250.

For details of SAE viscosity grades, see

- SAE J300 *Engine Oil Viscosity Classifications*
- SAE J306 *Automotive Gear Lubricant Viscosity Classifications*

Always use the latest version.

Viscosity index (VI) is a measure of how greatly the viscosity of a fluid changes with temperature. VI is determined by measuring the viscosity of the sample fluid at 40°C (104°F) and 100°C (212°F) and comparing these values with those of certain reference fluids that were assigned VI values of 0 and 100.

A fluid with a high VI exhibits a small change in viscosity with temperature.

A fluid with a low VI exhibits a large change in viscosity with temperature.

For most lubricants, a high VI is desirable because it would provide more reliable protection and exhibit a more uniform performance as the temperature varies. Commercially available lubricants report VI values from approximately 90 to 250. Gear lubricants typically have a VI of approximately 150. Additives, typically organic polymers, are used to improve viscosity and adjust viscosity index.

Corrosion protection can be tailored for specific applications by blending various additives with the base oil. Rust inhibitors for ferrous materials or copper and bronze corrosion protection are common. Oxidation inhibitors are used to prolong the life of oils. Extreme pressure additives help prevent scuffing in heavily loaded applications. Foam control additives prevent foaming where gears or other machine elements churn in an oil bath.

References 3, 6–8, 13–15, 19, and 20 describe recent developments in the field of lubricant technology and tribology. Reference 9 describes recent work in the analysis of wear of polymers and composites, exploring the effect of applied load and surface defects on fatigue, wear behavior, and the effects of surface cracks under different static and cyclic loading. Reference 21 emphasizes the influences of the rapidly growing field of micro- and nanoscale phenomena in tribology. Major lubricant suppliers (Internet sites 11–13, 15, and others) are developing new additives and synthetic formulations to serve the alternative energy fields of wind and solar power, with special attention to the wide range of environmental conditions in which they must operate. Some are applied at temperatures as low as –55°C (–67°F) and as high as 50°C (122°F) with related changes in humidity and exposure to corrosive elements.

Greases. Grease is a two-phase lubricant composed of a thickener dispersed in a base fluid, typically an oil. When applied to the interface between moving components, the grease tends to remain in place and adhere to the surfaces. The oil provides lubrication in a manner similar to that already discussed. As long as there is a sufficient amount of grease at the interface, continuous lubrication is provided. Some mechanisms are said to be lubricated for life. However, the designer must take great care to ensure that grease is not displaced from the critical areas that need it or to design the system for periodic reapplication of the grease. Some components such as bearings are provided with grease fittings for replenishing the supply and for discharging contaminated or oxidized grease.

Several kinds of thickeners are used in combination with natural or synthetic oils. See Internet sites 14 and 15 for examples. Thickeners are soaps formed by the reaction of animal or vegetable fats with alkaline substances such as lithium, calcium, an aluminum complex, clay, polyurea, and others. Lithium 12-hydroxy stearate is the most often used form. Soaps have a smooth buttery consistency and hold the oil in suspension until it is drawn out in the zone to be lubricated. Additives provide extreme pressure (EP) capability, rust protection, oxidation stability, and improved pumpability of the grease.

NLGI, formerly called the National Lubricating Grease Institute (Internet site 14), defines nine ranges of consistency, labeled 000 through 6, from semifluid to soft, medium hard, hard, and rigid block. Grade #2 is good for general industrial applications.

Solid Lubricants. Some applications cannot employ oils or greases because of contamination of other parts of the system, exposure to foods, excessively high or low temperatures, operation in a vacuum, or other environmental considerations. In such cases, the designer may specify solid materials that have inherently good lubricating properties or add solid lubricants on critical surfaces. Sections 16–4 and 16–5 have already discussed various formulations of PTFE (polytetrafluoroethylene) as examples of materials with good lubricity.

A solid lubricant is a thin solid film that reduces friction and wear. Some are applied in powder form by brushing, spraying, or dipping, and they then adhere to mating surfaces. Binders are often blended with the base material to facilitate application and to promote adhesion. Curing in air or by baking is required. See Reference 17.

Molybdenum disulfide (MoS_2) and graphite are two often-used types of solid lubricants. Others are lead iodide (PbI_2), silver sulfate (AgSO_4), tungsten disulfide, and stearic acid. An example of their effectiveness is the reduction of the coefficient of sliding friction for steel-on-steel from approximately 0.50 for dry clean surfaces to the range from 0.03 to 0.06.

Many solid lubricant developments are designed to enhance the efficiency and improve wear resistance of the many bearing surfaces in automotive and truck engines. Internet site 18 describes IROX™ bearings that have an overlay of polyamide-imide (PAI) polymer resin binder containing a number of additives dispersed throughout the matrix. These additives provide enhanced wear resistance, strength, thermal conductivity, and embeddability.

Wear

Wear is the gradual removal of surface material from sliding surfaces. It is a complex process with numerous variables. Only testing under real conditions of service can predict actual wear in a given system. See References 9, 15, 16, 19, and 20. Several types of wear occur:

- Pitting, galling, scuffing, or scoring that typically result from high contact stresses and fatigue of surface material during rolling or sliding contact.
- Abrasive wear, mechanical scraping, cutting, or scratching such as by hard contaminants in the interface between mating parts.
- Fretting, the cyclic sliding of very small amplitude that displaces surface material. The subsequent accumulation of the debris tends to accelerate the process. Continued operation produces a surface appearance similar to corrosion and can generate small cracks from which ultimate fatigue failure can occur. It often occurs when tight-fitting parts are subjected to oscillating loads or vibration.

- Impingement wear caused by the eroding of material due to hard moving materials hitting the surface, perhaps carried by air or fluids. High-velocity fluids, such as the discharge from high-pressure washers, can themselves cause such wear.

While it is not possible to prescribe specific approaches to reduce wear, the following are the things a designer may try. Again, testing is the only way to ensure satisfactory operation.

1. Keep contact force low between sliding surfaces.
2. Maintain low temperature at the mating surfaces.
3. Use hard contacting surfaces.
4. Produce smooth mating surfaces.
5. Maintain continuous lubrication to reduce friction.
6. Keep the relative velocity between mating surfaces low.
7. Specify materials that have inherently good wear properties.

Many material suppliers will report wear properties of their materials when operating against a similar or dissimilar material. These data are acquired by testing under carefully controlled laboratory conditions. Typically one member of a pair of materials is moved at a known velocity such as by rotation. The mating material is held stationary and under a known load. Careful measurements are made of the original weight and dimensions of the specimens of mating materials. After a sizable time of operation, the specimens are again weighed and measured to determine how much material has been removed. The results are reported as wear, computed from an equation such as,

$$K = W/FVT \quad (16-13)$$

where K = wear factor for the materials

W = wear measured as loss of weight or volume

F = applied load

V = linear relative velocity between the sliding members

T = time of operation

Comparing the K factors for a variety of materials being considered can aid the designer in material selection.

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3. **Saint-Gobain Performance Plastics**. Manufacturer of plain plastic bearings under the brand names Norglide® and Solglide®.
4. **GGB Bearing Technology**. Manufacturer of a variety of plain bearings made from a composite of PTFE plastic, porous bronze, and steel or fiberglass under the brand names DU®, DX®, Gar-Fil®, Gar-Max®, and others.
5. **Graphite Metallizing Corporation**. Manufacturer of plain bearings under the Graphalloy® brand name. Graphalloy® is a graphite-metal alloy formed from molten metal and graphite to make a uniform, solid, self-lubricating bushing material, available in over 100 grades.
6. **Beemer Precision, Inc.** Manufacturer of plain bearings made from cast bronze and porous sintered bronze impregnated with oil under the Oilite® and Excelite brands.
7. **Bunting Bearings LLC**. Manufacturer of plain bearings made from cast bronze, porous sintered bronze impregnated with oil, and plastic bearings made from Nylon, PTFE, and Vespel®. Also makes the BU bearing consisting of a steel backing strip, a porous bronze matrix impregnated and overlaid with a PTFE/lead lining for low friction coefficient.
8. **Zincaloy, Inc.** Manufacturer of continuous cast ZA-12 zinc-aluminum alloy (Zincaloy™) used for bearings in mining, construction, forestry, and off-road vehicle industries. Team Tube Ltd. is an authorized distributor of Zincaloy bearing stock.
9. **Baldor/Dodge**. Manufacturer of mounted plain self-lubricating sleeve bearings under the Solidlube® brand used in general industrial, material handling, and bulk materials industries. Also available are Bronzoil® bearings with oil-impregnated porous bronze, babbitted and bronze bushed bearings.
10. **Waukesha Bearings Corporation**. Manufacturer of fluid film hydrodynamic sleeve bearings, tilting pad thrust and journal bearings, magnetic bearings and housings for the bearings. These bearings are used for turbines, compressors, generators, gearboxes, and other types of rotating equipment.
11. **Exxon-Mobol, Inc.** Producer of a wide variety of lubricants for general industry, automotive, turbine, compressor, and engine applications. Data sheets provided under the *Industrial and speciality lubricants* page.
12. **Castrol Industrial**. Manufacturer of lubricants for vehicles and industrial applications such as bearings, gears, chains, and general lubrication applications. Search for *Industrial Lubricants* from the home page.
13. **Shell Oil Company**. Producer of a wide variety of lubricants for general industry, automotive, turbine, compressor, and engine applications. Search for *Industrial Lubricants* from the home page.
14. **NLGI, Inc. (Formerly National Lubricating Grease Institute)**. An association of companies and research organizations that develop, manufacture, distribute, and use greases. Establishes standards and provides publications and technical articles on grease.

INTERNET SITES RELATED TO PLAIN BEARINGS AND LUBRICATION

1. **Copper Development Association (CDA)**. Industry association of companies and organizations involved in the production and use of copper, including cast bronze

15. **Lubrizol Corporation.** Manufacturer of a large variety of additives for the lubrication industry including engine oils and industrial lubricants as well as for gasoline and diesel fuel.
16. **RBC Bearings.** Manufacturer of a wide array of bearings including spherical plain bearings, rod end bearings, and self-lubricated bearings. Products are used wherever pivoting, high-load-bearing applications are found in applications such as in construction equipment, agricultural machinery, hydraulic cylinder connections, vehicle suspensions, and articulated joints.
17. **Plastics International.** Supplier of a large number of types and grades of plastics. Selecting the *Material Properties* tab reveals a list of over 100 plastics with mechanical properties. Values of pV factor are given in the selectable PDF file for materials commonly used for bearings, such as Delrin, PEEK, and Rulon.
18. **Federal Mogul Corporation.** Manufacturer of numerous components and systems for automotive applications including engine bearings. Select *Powertrain* then *Bearings*, *Bushings and Washers* or *Self-lubricating Bearings*.

PROBLEMS

For Problems 1–8 and the data listed in Table 16–2, design a plain surface bearing, using the boundary-lubricated approach of Section 16–5. Use an L/D ratio for the bearing in the range of 0.50 to 1.50. Compute the pV factor, and specify a material from Table 16–1.

TABLE 16–2

Problem number	Radial load (lb)	Shaft diameter (in)	Shaft speed (rpm)
1.	225	3.00	1750
2.	100	1.50	1150
3.	200	1.25	850
4.	75	0.50	600
5.	850	4.50	625
6.	500	3.75	450
7.	800	3.00	350
8.	60	0.75	750

For Problems 9–18 and the data listed in Table 16–3, design a hydrodynamically lubricated bearing, using the method outlined in Section 16–7. Specify the journal nominal diameter, the bearing length, the diametral clearance, the minimum film

thickness of the lubricant during operation, the surface finish of the journal and bearing, the lubricant, and its maximum operating temperature. For your design, compute the coefficient of friction, the friction torque, and the power dissipated as the result of friction.

For Problems 19–28 and the data listed in Table 16–4, design a hydrostatic bearing of circular shape. Specify the pad diameter, the recess diameter, the recess pressure, the film thickness, the lubricant and its temperature, the oil flow rate, and the pumping power. The load specified is for a single bearing. You may choose to use multiple bearings. (The supply pressure is the maximum available at the pump.)

TABLE 16–3

Problem number	Radial load	Min. shaft diameter	Shaft speed (rpm)	Application
9.	1250 lb	2.60 in	1750	Electric motor
10.	2250 lb	3.50 in	850	Conveyor
11.	1875 lb	2.25 in	1150	Air compressor
12.	1250 lb	1.75 in	600	Precision spindle
13.	500 lb	1.15 in	2500	Precision spindle
14.	850 lb	1.45 in	1200	Idler sheave
15.	4200 lb	4.30 in	450	Shaft for chain drive
16.	18.7 kN	100 mm	500	Conveyor
17.	2.25 kN	25 mm	2200	Machine tool
18.	5.75 kN	65 mm	1750	Printer

TABLE 16–4

Problem number	Load	Supply pressure
19.	1250 lb	300 psi
20.	5000 lb	300 psi
21.	3500 lb	500 psi
22.	750 lb	500 psi
23.	250 lb	150 psi
24.	500 lb	150 psi
25.	22.5 kN	2.0 MPa
26.	1.20 kN	750 kPa
27.	8.25 kN	1.5 MPa
28.	12.5 kN	1.5 MPa

LINEAR MOTION ELEMENTS

The Big Picture

You Are the Designer

17-1 Objectives of This Chapter

17-2 Power Screws

17-3 Ball Screws

17-4 Application Considerations for Power Screws and Ball Screws

THE BIG PICTURE

Linear Motion Elements

Discussion Map

- Many kinds of mechanical devices produce linear motion for machines such as automation equipment, packaging systems, and machine tools.
- Power screws, jacks, and ball screws are designed to convert rotary motion to linear motion and to exert the necessary force to move a machine element along a desired path. They use the principle of a screw thread and its mating nut.

Discover

Visit a machine shop and see if you can identify power screws, ball screws, or other linear motion devices. Look in particular at the lathes and the milling machines. Manual machines likely use power screws. Those using computer numerical control should have ball screws. Describe the shape of the threads. How are the screws powered? How are they fastened to other machine parts?

Can you find other equipment that uses linear motion devices? Look in laboratories where materials are tested or where large forces must be generated.

This chapter will help you learn to analyze power screws and ball screw drives and to specify suitable sizes for a given application.

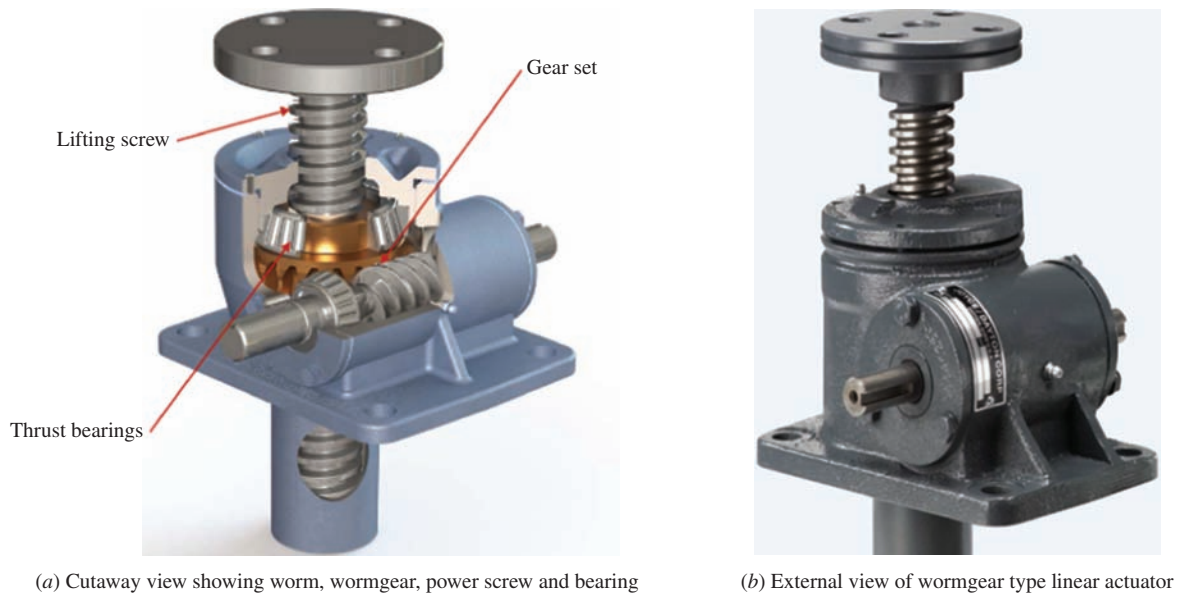
A common requirement in mechanical design is to move components in a straight line. Elevators move vertically up or down. Machine tools move cutting tools or parts to be machined in straight lines, either horizontally or vertically, to shape metal into desired forms. A precision measuring device moves a probe in straight lines to determine electronically the dimensions of a part. Assembly machines require many straight-line motions to insert components and to fasten them together. A packaging machine moves products into cartons, closes flaps, and seals the cartons.

Some examples of components and systems that facilitate linear motion are as follows:

Power screws	Jacks
Linear actuators	Ball bushings
Linear solenoids	X-Y-Z tables
Ball screws	Fluid power cylinders
Linear slides	Rack-and-pinion sets
Positioning stages	Gantry tables

Figure 17-1(a) shows a cutaway view of a jack that employs a power screw to produce linear motion. Power is delivered to the input shaft by an electric motor. The worm, machined integral with the input shaft, drives the wormgear, resulting in a reduction in the speed of rotation. The inside of the wormgear has machined threads that engage the external threads of the power screw, driving it vertically. Figure 17-1(b) shows the complete screw jack with its integral housing that constitutes the linear motion system. Limit switches, position sensors, and programmable logic controllers can be used to control the motion cycle. See Internet site 6. Other types of linear actuators can be seen on Internet sites 1-14.

Rack-and-pinion sets are discussed in Chapter 8. Fluid power actuators employ oil hydraulic or pneumatic fluid pressure to extend or retract a piston rod within a cylinder as discussed in textbooks on fluid power. Positioning stages, X-Y-Z tables, and gantry tables typically are driven by precision stepper motors or servomechanisms that allow precision location of



(a) Cutaway view showing worm, wormgear, power screw and bearing (b) External view of wormgear type linear actuator

FIGURE 17-1 Example of wormgear type linear actuator (Joyce/Dayton Corp.); (see Internet site 6)

components anywhere within their control volume. Linear solenoids are devices that cause a rodlike core to be extended or retracted as power is supplied to an electrical coil, producing rapid movement over small distances. Applications are seen in office equipment, automation devices, and packaging systems. See Internet site 11.

Linear slides and ball bushings are designed to guide mechanical components along a precise linear track. Low-friction materials or rolling contact elements are used to produce smooth motion with low power required. See Internet sites 1, 3, 5–9, and 15.

Power screws and ball screws are designed to convert rotary motion to linear motion and to exert the necessary force to move a machine element along a desired path. Power screws operate on the classic principle of the screw thread and its mating nut. If the screw is supported in bearings and rotated while the nut is restrained from rotating, the nut will translate along the screw. If the nut is made an integral part of a machine, for example, the tool holder for a lathe, the screw will drive the tool holder along the bed of the machine to take a cut. Conversely, if the nut is supported while it is rotating, the screw can be made to translate. The screw jack uses this approach.

A ball screw is similar in function to a power screw, but the configuration is different. The nut contains many small, spherical balls that make rolling contact with the threads of the screw, giving low friction and high efficiencies when compared with power screws. Modern machine tools, automation equipment, vehicle steering systems, and actuators

on aircraft use ball screws for high precision, fast response, and smooth operation.

Visit a machine shop where there are metal-cutting machine tools. Look for examples of power screws that convert rotary motion to linear motion. They are likely to be on manual lathes moving the tool holder. Or look at the table drive for a milling machine. Inspect the form of the threads of the power screw. Are they of a form similar to that of a screw thread with sloped sides? Or are the sides of the threads straight? Compare the screw threads with those shown in Figure 17-2 for square, Acme, and buttress forms.

While in the shop, do you see any type of material-testing equipment or a device called an *arbor press* that exerts large axial forces? Such machines often employ square-thread power screws to produce the axial force and motion from rotational input, through either a hand crank or an electric motor drive. If they are not in the machine shop, look for them in the metallurgy lab or another room where materials testing is done.

Now look further in the machine shop. Are there machines that use digital readouts to indicate position of the table or the tool? Are there computer numerical control machine tools? Any of these types of machines should have ball screws rather than the traditional power screws because ball screws require significantly less power and torque to drive them against a given load. They can also be moved faster and positioned more accurately than power screws. You may or may not be able to see the recirculating balls in the nut of the

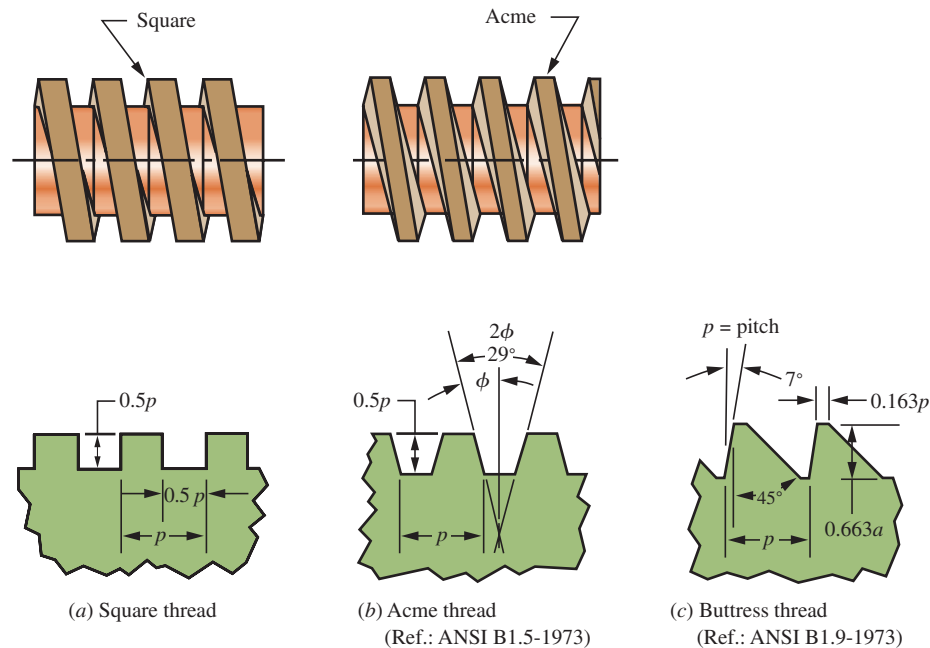


FIGURE 17-2 Three forms of power screw threads

power screw, as illustrated in Figure 17-3. But you should be able to see the different-shaped threads looking like grooves with circular bottoms in which the spherical balls roll.

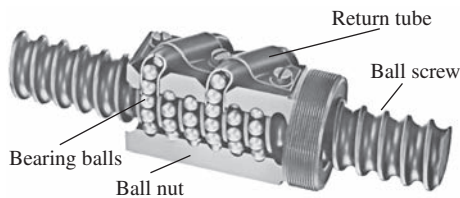


FIGURE 17-3 Ball bearing screw

Have you seen such power screws or ball screws anywhere outside a machine shop? Some garage door openers employ a screw drive, but others use chain drives. Perhaps your home has a screw jack or a scissors jack for raising the car to change a tire. Both use power screws. Have you ever sat in a seat on an airplane where you can see the mechanisms that actuate the flaps on the rear edge of the wings? Try it sometime, and observe the actuators during takeoff or landing. It is likely that you will see a ball screw in action.

This chapter will help you learn the methods of analyzing the performance of power screws and ball screws and to specify the proper size for a given application.

YOU ARE THE DESIGNER

You are a member of a plant engineering team for a large steel processing plant. One of the furnaces in the plant in which steel is heated prior to final heat treatment is installed beneath the floor, and the large ingots are lowered vertically into it. While the ingots are soaking in the furnace, a large, heavy hatch is placed over the opening to minimize the escape of heat and provide more uniform temperature. The hatch weighs 25 000 lb.

You are asked to design a system that will permit the hatch to be raised at least 15 in above the floor within 12.0 s and to lower it again within 12.0 s.

What design concept would you propose? Of course, there are many feasible concepts, but suppose that you proposed a system like that sketched in Figure 17-4. An overhead support structure

is suggested on which a worm/wormgear drive system would be mounted. One shaft would be driven directly by the gear drive while a second shaft would be driven simultaneously by a chain drive. The shafts are power screws, supported in bearings at the top and bottom. A yoke is connected to the hatch and is mounted on the screws with the nuts that mate with the screw integral with the yoke. Therefore, as the screw rotates, the nuts carry the yoke and the hatch vertically upward or downward.

As the designer of the hatch lift system, you must make several decisions. What size screw is required to ensure that it can safely raise the 25 000-lb hatch? Notice that the screws are placed in tension as they are supported on the collars on the upper support system. What diameter, thread type, and thread